FREESTONE COUNTY AMATEUR RADIO EMERGENCY SERVICE

FREESTONE COUNTY ARES TACTICAL FIELD BINDER

EMERGENCY COMMUNICATIONS | TACTICAL FIELD OPERATIONS | DEPLOYMENT REFERENCE

WHEN ALL ELSE FAILS

FIELD EDITION

REVISION 7.5

FOR OFFICIAL ARES FIELD USE

FREESTONE COUNTY, TEXAS

FREESTONE COUNTY AMATEUR RADIO EMERGENCY SERVICE FREESTONE COUNTY ARES TACTICAL FIELD BINDER

FREESTONE COUNTY ARES
TACTICAL FIELD BINDER

EMERGENCY COMMUNICATIONS • FIELD OPERATIONS • TACTICAL REFERENCE

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UPDATED: MAY 2026

This Tactical Field Binder is a quick reference and operational guide for Freestone County ARES members. It contains plans, procedures, reference information, and tools needed to support emergency communications and field operations. Use this binder as your go-to resource in the field.

★ SECTION 1 – ADMINISTRATION & COMMUNICATIONS FOUNDATION ★

SECTION	TITLE	DESCRIPTION	PAGE
1.1	Approval & Signature Page	Official approval page with signature of Freestone County ARES Emergency Coordinator.	1-1
1.2	Document Control & Revision History	Document control information and a history of all revisions made to this binder.	1-2
1.3	Communications PACE Plan	Overview of communication methods, priorities, alternates, contingencies, and escalation paths.	1-3
1.4	ICS-205 Communications Annex	Official ICS-205 Communications Plan including all frequencies and usage.	1-4
1.5	Radio Programming Standard	Standard programming guide for all ARES radios. Includes channel layout, tones, and settings.	1-5
1.6	Deployment Checklist	Pre-deployment checklist to ensure personnel and equipment are ready to deploy.	1-6
1.7	Net Control Station (NCS) Quick Guide	Quick reference guide for Net Control Stations during normal and emergency operations.	1-7
1.8	Operator Accountability Roster	Roster of ARES operators with contact information, roles, and qualifications.	1-8

IMPORTANT: Keep this binder current. Refer to the Document Control & Revision History (1.2) for the latest updates. Outdated information can impact mission success and safety.

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SECTION 2

SECTION 2 – COMMUNICATIONS

Section 2 contains all communications plans, procedures, and reference materials used by Freestone County ARES to maintain reliable communications during normal operations and emergency activations.

SECTION	TITLE	DESCRIPTION	PAGE
2.0	**Communications Divider**	Section divider page for Communications. Overview of communications scope and purpose.	2-0
2.1	**Tactical ICS-205 (Full)**	Complete ICS-205 Communications Plan. Includes all frequencies, usage, and assignments.	2-1
2.2	**Tactical Channel Matrix**	Quick reference matrix of all channels, modes, tones, offsets, and uses. Two pages.	2-5
2.3	**Net Procedures**	Standard net formats, schedules, and procedures for all ARES nets.	2-7
2.4	**Message Handling**	Procedures for handling ICS messages, traffic priority, and message logging.	2-11
2.5	**Interoperability Guide**	Interoperability frequencies, protocols, and coordination guidelines.	2-15
2.6	**Communications Failure SOP**	Standard Operating Procedure for communications failures and degradation.	2-19

KEY FOCUS: Communications are the backbone of every mission.

- Maintain clear, concise, and accurate communications.
- Follow established net and message procedures.
- Use assigned channels and report problems immediately.
- Ensure interoperability with all served agencies.

WHEN ALL ELSE FAILS – COMMUNICATION IS CRITICAL.

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SECTION 3

SECTION 3 – WEATHER OPERATIONS

Section 3 provides procedures, reference materials, and operational guidance for weather monitoring, reporting, and severe weather support. These resources help ensure accurate situational awareness and timely information to served agencies.

SECTION	TITLE	DESCRIPTION	PAGE
3.0	Weather Operations Divider	Section divider page for Weather Operations. Overview of weather support mission.	3-0
3.1	Storm Spotter Activation Levels	Defines activation levels, triggers, and operational readiness.	3-1
3.2	Reportable Weather Criteria	List of reportable conditions for storm spotters and ARES personnel.	3-3
3.3	Severe Weather Reporting Format	Standard format and examples for relaying severe weather reports.	3-6
3.4	Radar Interpretation Guide	Guide to interpreting radar imagery for situational awareness. Two pages.	3-8
3.5	Weather Operations Communications Plan	Communications plan for weather nets and severe weather operations.	3-12
3.6	Weather Net Control Responsibilities	Duties and responsibilities of the Weather Net Control Station.	3-15
3.7	Weather Net Activation Script	Sample script for opening and managing weather nets.	3-17
3.8	Storm Deployment Safety	Safety guidelines and best practices for storm operations and field deployments.	3-20

KEY FOCUS: Accurate weather information saves lives.

- Monitor conditions continuously.
- Report accurately and promptly.
- Follow net procedures and escalation protocols.
- Always prioritize safety during storm operations.

STAY INFORMED. STAY ALERT. STAY SAFE.

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SECTION 4

★ SECTION 4 – FIELD OPERATIONS ★

Section 4 provides operational guidance for deploying ARES resources in the field. It includes activation procedures, deployment checklists, accountability methods, and support coordination with served agencies.

SECTION	TITLE	DESCRIPTION	PAGE
4.0	Field Operations Divider	Section divider page for Field Operations. Overview of field mission support and procedures.	4-0
4.1	ARES Activation Procedures	Step-by-step procedures for activating ARES for emergencies and events.	4-1
4.2	Deployment Procedures	Guidelines for mobilizing personnel and equipment, travel, staging, and on-scene operations.	4-5
4.3	Check-In / Check-Out Procedures	Standard check-in and check-out process for all ARES personnel.	4-9
4.4	Go-Kit Standards	Minimum go-kit requirements, packing list, and maintenance standards. Three pages.	4-13
4.5	Operator Accountability	Procedures for tracking personnel, assignments, and duty status.	4-16
4.6	Served Agency Support	Guidance for supporting served agencies, resource requests, and coordination.	4-19

KEY FOCUS

- Be prepared and ready to deploy.
- Follow activation and deployment procedures.
- Maintain accountability at all times.
- Communicate clearly with your team and served agencies.
- Safety is everyone's responsibility.

PREPARE.
DEPLOY.
COMMUNICATE.
SUPPORT.
STAY SAFE.

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SECTION 5

SECTION 5 – APRS OPERATIONS

Section 5 provides step-by-step guides and reference material for using Automatic Packet Reporting System (APRS) to enhance situational awareness, tracking, messaging, and field operations.

SECTION	TITLE	DESCRIPTION	PAGE
5.0	APRS Operations Divider	Section divider page for APRS Operations. Overview of APRS capabilities and mission benefits.	5-0
5.1	APRS Overview and Uses	Overview of APRS, how it works, and how it supports ARES operations in the field.	5-1
5.2	APRS.fi Setup and UV-PRO Connection	Step-by-step instructions to create an APRS.fi account, configure settings, and connect a UV-PRO.	5-6
5.3	APRSdroid Setup and UV-PRO Connection	Step-by-step instructions to install and configure APRSdroid and connect a UV-PRO.	5-11
5.4	YAAC APRS Client – Overview & Uses	Overview of YAAC, system requirements, and how it supports APRS operations.	5-16
5.5	YAAC APRS Client – Download & Install	Step-by-step instructions to download and install YAAC on your computer.	5-20
5.6	YAAC APRS Client – Basic Setup	Initial configuration, setup wizard, and essential settings for YAAC.	5-25
5.7	YAAC APRS Client – Connection Methods	How to connect YAAC to APRS via TNC, audio, or network (IGate).	5-31
5.8	YAAC APRS Client – Field Use	Using YAAC in the field: mapping, messaging, tracking, beacons, and object management.	5-37
5.9	YAAC APRS Client – Troubleshooting	Common issues, error messages, and solutions for YAAC in the field.	5-44

KEY FOCUS

- Use APRS to enhance situational awareness and resource tracking.
- Ensure proper setup and connectivity before deployment.
- Monitor APRS traffic and maintain accurate position reports.
- Keep systems updated and test regularly.
- APRS data is most effective when shared and reliable.

SEE IT.
SHARE IT.
TRACK IT.
GET HOME SAFE.

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SECTIONS 6 & 7 — APRS COMMANDS / USGS TOPOGRAPHIC ATLAS

SECTION 6 — APRS COMMANDS & TACTICAL REFERENCE

SEC	DOCUMENT TITLE	OPERATIONAL SCOPE & DESCRIPTION	PAGE
6.1	APRS Command Reference Chart	SMS gateway, WXBOT, and WLNK-1 Winlink queries	6-1
6.2	APRS Command Note Page	Operator worksheet for custom command macros	6-2
6.3	MPAD Commands Reference	MPAD location queries and asset lookups	6-3
6.4	WX Data & Position Inquiry	Weather, radar, alerts and position reports	6-4

SECTION 7 — FREESTONE COUNTY USGS TOPOGRAPHIC ATLAS

SEC	QUADRANGLE MAP NAME	PAGE
7.0	County Quadrangle Index Key — 26 USGS maps	7-0

COMPLETE 26-QUADRANGLE TOPOGRAPHIC MAP INVENTORY

7.1	BUFFALO, TX	7-1	7.14	MEXIA, TX	7-14
7.2	BUTLER, TX	7-2	7.15	OAKWOOD, TX	7-15
7.3	CAYUGA, TX	7-3	7.16	ROUSTABOUT CAMP, TX	7-16
7.4	CRESLENN RANCH, TX	7-4	7.17	STEWARDS MILL, TX	7-17
7.5	DEW, TX	7-5	7.18	STREETMAN, TX	7-18
7.6	DONIE, TX	7-6	7.19	TEAGUE NORTH, TX	7-19
7.7	FAIRFIELD EAST, TX	7-7	7.20	TEAGUE SOUTH, TX	7-20
7.8	FALLON, TX	7-8	7.21	TENNESSEE COLONY, TX	7-21
7.9	FARRAR, TX	7-9	7.22	TURLINGTON, TX	7-22
7.10	KEECHI, TX	7-10	7.23	WINKLER, TX	7-23
7.11	KIRVIN, TX	7-11	7.24	WORTHAM, TX	7-24
7.12	LANELY, TX	7-12	7.25	YARD, TX	7-25
7.13	LONG LAKE, TX	7-13	7.26	YOUNG, TX	7-26

1. APPROVAL & SIGNATURE PAGE

This manual is approved for official use by Freestone County ARES for emergency communications operations, training, and incident response.

Approved By:

Brian Davis

Freestone County ARES Emergency Coordinator

Date: 11/1/2026

2. DOCUMENT CONTROL & REVISION HISTORY

This document shall be reviewed annually or after significant communications, operational, or procedural changes.

VERSION	DATE	REVISION DESCRIPTION
1.0	05/10/2026	Initial PACE Plan
1.1	05/11/2026	Added ICS-205 Communications Annex
1.2	05/11/2026	Corrected KE5DFY frequency and channel alignment
1.3	05/12/2026	Formatting and heading corrections
2.0	05/12/2026	Expanded PACE/SOP language
3.0	05/12/2026	Added roster, NCS guide, checklist
4.0	05/12/2026	Restored net/operations verbiage
5.0	05/13/2026	Professional manual formatting
6.0	05/24/2026	Visual refinement and FEMA/ICS styling
7.0	05/24/2026	Finalized and signed by EC
7.1	09/01/2026	Minor update; routine content corrections.

2. DOCUMENT CONTROL & REVISION HISTORY

This document shall be reviewed annually or after significant communications, operational, or procedural changes.

VERSION	DATE	REVISION DESCRIPTION
7.2	10/03/2026	Document control review; corrected PACE references in Sections 2.2 and 3.5; updated issue dates.
7.3	10/03/2026	Full content-control audit aligning ICS-205, PACE, and communications failure references.
7.4	10/03/2026	Corrected PACE Emergency element to "HF / GMRS / Message Delivery" throughout document; updated all related references and procedures.
7.5	10/27/2026	Finalized for publication; EC signature required for release.

3. COMMUNICATIONS PACE PLAN

Purpose: This PACE Plan establishes a structured communications framework for Freestone County ARES to ensure reliable communications during routine operations, activations, and emergency incidents.

P — PRIMARY (P) – NORMAL OPERATIONS

FF 110 Repeater – 145.110 MHz | Offset – 600 | Tone 146.2 | Channel 1

Weekly Net – Every Sunday at 2000 Hours on the FF 110 repeater.

Standard operating mode – FM Voice.

A — ALTERNATE (A) – BACKUP METHOD

WRTM 760 Repeater (KE5DFY) – 146.760 MHz | Offset – 600 | Tone 136.5 | Channel 2

Standard operating mode – FM Voice.

C — CONTINGENCY (C) – DEGRADED COMMUNICATIONS

ARES TAC – 146.580 MHz Simplex | Tone 123.0 | Channel 9

Directed simplex operations and neighboring county relays.

E — EMERGENCY (E) – LAST RESORT

HF / GMRS / Message Delivery.

4. ICS-205 COMMUNICATIONS ANNEX

CH	NAME	FREQUENCY (MHz)	TONE (CTCSS)	MODE	USE/REMARKS
1	FF 110	145.110	146.2	FM	W5NFL Fairfield VHF (Freestone ARES)
2	WRTM 760	146.760	136.5	FM	KE5DFY Wortham VHF (Alternate)
3	FF 70CM	443.900	136.5	FM	KE5DFY Fairfield UHF
4	MEX 390	145.390	146.2	FM	W5NFL Tehuacana VHF (Limestone ARES)
5	COR 290	145.290	146.2	FM	W5NFL Corsicana Nature Park (Navarro ARES)
6	FF GMRS	462.600	136.5	FM	Shared local GMRS (Fairfield Area Link)
7	MEX GMRS	462.700	146.2	FM	Tehuacana GMRS Ch 21 (Limestone Area Link)
8	COR GMRS	462.550	146.2	FM	Corsicana GMRS Ch 15 (Navarro Area Link)
9	ARES TAC	146.580	123.0	FM	Directed simplex (Contingency)
10	ARES-1	146.300	123.0	FM	VHF Simplex
11	ARES-2	146.900	123.0	FM	VHF Simplex
12	ARES-3	147.200	123.0	FM	VHF Simplex
13	ARES-4	147.500	123.0	FM	VHF Simplex
14	ARES-5	147.800	123.0	FM	VHF Simplex
15	ARES-6	443.145	123.0	FM	UHF Simplex
16	ARES-7	433.525	123.0	FM	UHF Simplex
17	ARES-8	433.775	123.0	FM	UHF Simplex
18	ARES-9	444.215	123.0	FM	UHF Simplex
19	ARES-10	444.715	123.0	FM	UHF Simplex
28	DATA-V	144.975	123.0	DIGITAL	Digital (Winlink/Packet)
29	DATA-U	438.775	123.0	DIGITAL	Digital (Winlink/Packet)
30	APRS	144.390	NONE	PACKET	Packet

5. RADIO PROGRAMMING STANDARD

GENERAL SETTINGS

- Mode: FM unless otherwise noted
- Band: 2m / 70cm / GMRS / HF / Digital as assigned
- Step: 5kHz (narrowband where applicable)
- Power: High unless otherwise directed
- Tone: As listed in ICS-205

NAMING CONVENTION

- Use channel numbers and names as shown in ICS-205
- Example: CH 2 WORTHAM 76, CH 10 ARES-1, CH 28 DATA-V

PROGRAMMING GUIDELINES

- Program in exact order as ICS-205 for consistency
- Do not skip channels
- Lock non-use channels if radio supports
- Verify tone and mode before deployment

VERIFICATION

All radios will be checked for correct programming prior to deployment.

6. DEPLOYMENT CHECKLIST

- ☐ Radio programmed per ICS-205
- ☐ Spare battery(ies) charged
- ☐ Antenna(s) installed and secure
- ☐ Coax / feedline / adapters
- ☐ Power source (battery / solar / generator)
- ☐ Notebook / ICS forms / pens
- ☐ ID / License / vest / PPE
- ☐ Go-kit / bug-out bag
- ☐ Test contact completed
- ☐ Check into net / report status

Operator Initials: _____ **Date:** _____

7. NET CONTROL STATION (NCS) QUICK GUIDE

1. Check in early and announce net is open.
2. Take roll call and note check-ins.
3. Confirm traffic procedures and priorities.
4. Handle traffic using ICS-213 message form.
5. Maintain log using ICS-214 log form.
6. Keep transmissions short and clear.
7. Monitor for emergencies / priority traffic.
8. Close net – final announcements and thanks.

NCS REMINDERS

- Be clear, calm, and concise.
- Control the net – limit side conversations.
- Confirm all traffic is logged and delivered.

8. OPERATOR ACCOUNTABILITY ROSTER

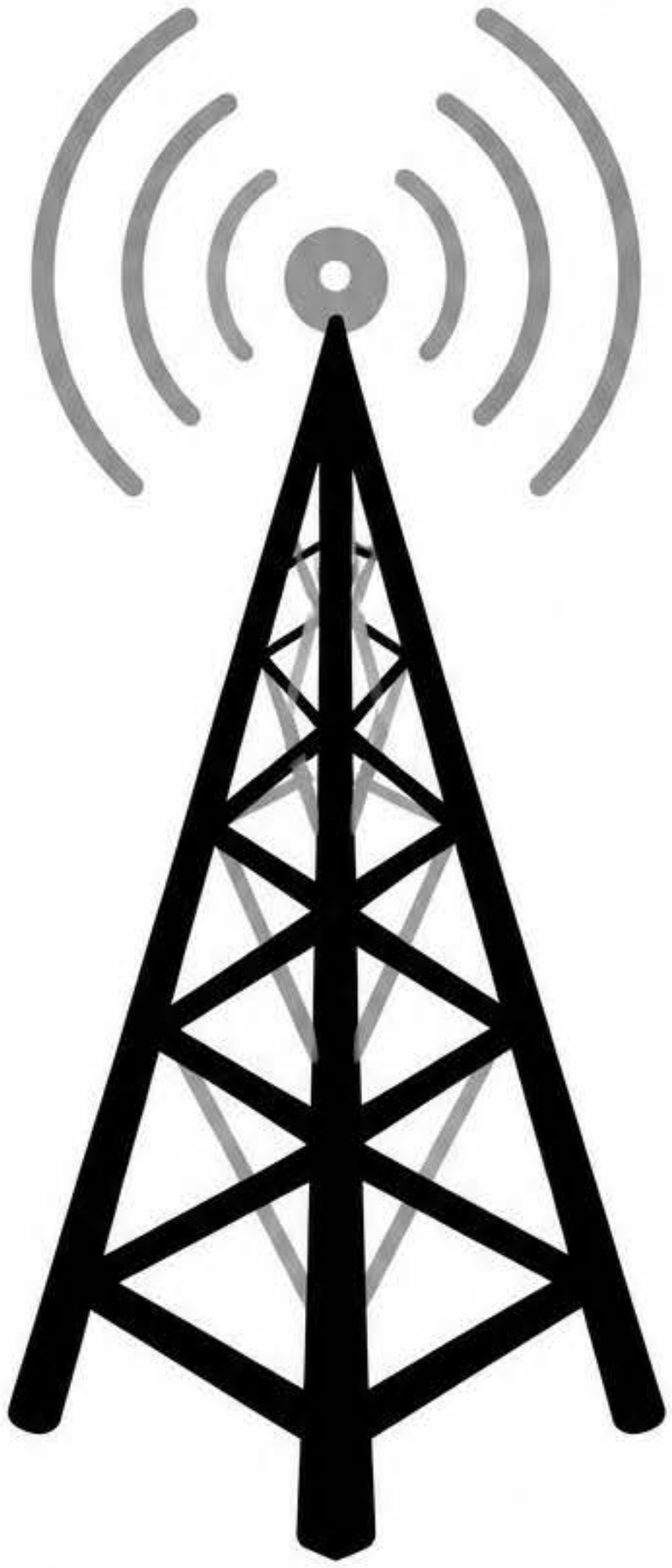
Event / Activation: _____

Date / Time: _____

NAME / CALL SIGN	POSITION / ROLE	CHECK-IN TIME	CHECK-OUT TIME

NCS Signature: _____

CHAPTER 2 – COMMUNICATIONS



COMMUNICATIONS ARE ESSENTIAL.
PLAN – PRACTICE – EXECUTE.

2.1 TACTICAL ICS-205 – COMMUNICATIONS PLAN

This ICS-205 Communications Plan establishes communications procedures and assignments for this incident. It is used for operational planning and filed in the Incident Action Plan (IAP).

COMMUNICATIONS PLAN (ICS-205)	
1. Incident Name:	2. Operational Period:
3. Date/Time From:	4. Date/Time To:
5. Incident Commander (Name and Position):	
6. Prepared By (Communications Unit Leader):	7. Date/Time Prepared:
8. Communications Objectives:	
9. Special Instructions:	

10. RESOURCE SUMMARY

Function	Channel/ System	Assignment	RX Freq (TX Freq)	Tone/NAC	Mode	Remarks
COMMAND						
TACTICAL						
AIR TO GROUND						
LOGISTICS						
PUBLIC INFO						
EMERGENCY						
CONTINGENCY						
OTHER 1						
OTHER 2						

11. ADDITIONAL INFORMATION / REMARKS:

2.2.1 TACTICAL CHANNEL MATRIX

PACE CHANNEL FLOW

PRIMARY	ALTERNATE	CONTINGENCY	EMERGENCY
CH 1 FF 110 PRIMARY OPERATIONS	CH 2 WRTM 760 ALTERNATE OPERATIONS	CH 9 ARES TAC CONTINGENCY OPERATIONS	HF / GMRS / MESSAGE DELIVERY (As Required) EMERGENCY OPTIONS

TACTICAL CHANNEL MATRIX (BY CHANNEL ICS-205) – PAGE 1

CH	CHANNEL NAME / USE	FREQUENCY (MHz)	TONE / NAC	MODE	NOTES / USE
1	FF 110	145.110	146.2	FM/Data	FM
2	WRTM 760	146.760	136.5	FM	Primary
3	FF 70C	443.900	136.5	FM	70cm
4	MEX 390	145.390	146.2	FM	Alternate
5	COR 290	145.290	146.2	FM	Navarro County
6	FF GMRS	462.600	136.5	FM	GMRS
7	MEX GMRS	462.700	146.2	FM	GMRS
8	COR GMRS	462.550	146.2	FM	GMRS
9	ARES TAC	146.580	123.0	FM	Contingency
10	ARES-1	146.300	123.0	FM	Simplex
11	ARES-2	146.900	123.0	FM	Simplex
12	ARES-3	147.200	123.0	FM	Simplex

INTEROPERATIVITY QUICK REFERENCE

CH 6	FF GMRS	462.600	136.5	FM
CH 7	MEX GMRS	462.700	146.2	FM
CH 8	COR GMRS	462.550	146.2	FM
	STATE / REGIONAL (ARES CH 28–30)			
CH 28	DATA-V	144.975	123.0	DIGITAL
CH 29	DATA-U	438.775	123.0	DIGITAL
CH 30	APRS	144.390	NONE	PACKET

SIMPLEX QUICK REFERENCE

VHF SIMPLEX — PROGRAMMED CH 10-14

146.300 / 146.900 / 147.200

147.500 / 147.800 MHz

UHF SIMPLEX — PROGRAMMED CH 15-19

433.145 / 433.525 / 433.775

444.215 / 444.715 MHz

Use listed channels only • FM • Tone 123.0

Transmit only where licensed and authorized

2.2 TACTICAL CHANNEL MATRIX

TACTICAL CHANNEL MATRIX (BY CHANNEL ICS-205) – PAGE 2					
CH	CHANNEL NAME / USE	FREQUENCY (MHz)	TONE / NAC	MODE	NOTES / USE
13	ARES-4	147.500	123.0	FM	Simplex
14	ARES-5	147.800	123.0	FM	Simplex
15	ARES-6	433.145	123.0	FM	UHF Simplex
16	ARES-7	433.525	123.0	FM	UHF Simplex
17	ARES-8	433.775	123.0	FM	UHF Simplex
18	ARES-9	444.215	123.0	FM	UHF Simplex
19	ARES-10	444.715	123.0	FM	UHF Simplex
28	DATA-V	144.975	123.0	DIGITAL	Digital (Winlink/Packet)
29	DATA-U	438.775	123.0	DIGITAL	Digital (Winlink/Packet)
30	APRS	144.390	NONE	PACKET	Packet

INTEROPERATIVITY QUICK REFERENCE				
CH	CHANNEL NAME	FREQUENCY (MHz)	TONE	MODE
6	FF GMRS	462.600	136.5	FM
7	MEX GMRS	462.700	146.2	FM
8	COR GMRS	462.550	146.2	FM
STATE / REGIONAL (ARES CH 28–30)				
CH	CHANNEL NAME	FREQUENCY (MHz)	TONE	MODE
28	DATA-V	144.975	123.0	DIGITAL
29	DATA-U	438.775	123.0	DIGITAL
30	APRS	144.390	NONE	PACKET

SIMPLEX QUICK REFERENCE

VHF SIMPLEX (CH 10–14)
146.300 – 147.800
TONE 123.0

UHF SIMPLEX (CH 15–19)
433.145 – 444.715
TONE 123.0

All Simplex Channels Use FM / TONE 123.0

2.3 NET PROCEDURES

ACTIVATION NET

- Net Control establishes net.
- Announces purpose of the net.
- Announces channel in use.
- Announces operational period.
- Requests all stations to check in.

CHECK-IN FORMAT

1	CALLSIGN
2	NAME
3	LOCATION
4	TRAFFIC / NO TRAFFIC
5	AVAILABLE RESOURCES

EXAMPLE:

```
KE5DFY
John Smith
Fairfield EOC
No Traffic
Generator Available
```

TACTICAL CALL SIGNS

Examples:

- Shelter 1
- Shelter 2
- EOC
- Damage Assessment 1
- Mobile 1

Tactical call may be used.
FCC callsign must still be given
every 10 minutes
and at end of contact.

NET PRIORITIES

1. Emergency Traffic
2. Priority Traffic
3. Welfare Traffic
4. Routine Traffic

DIRECTED NET RULES

- Wait to be called.
- No doubling.
- Keep transmissions brief.
- Use plain language.
- No 10-codes.
- No Q-signals unless necessary.

TACTICAL COMMUNICATIONS STANDARDS

✓	**GOOD** EOC from Shelter 1
✗	**BAD** Shelter 1 this is EOC are you available to receive traffic?

EMERGENCY BREAK PROCEDURE

BREAK BREAK BREAK

Used only for:

- Life safety
- Immediate danger
- Urgent operational information

2.4 MESSAGE HANDLING

MESSAGE PRECEDENCE

PRIORITY	USE
EMERGENCY	Life safety Immediate danger to life or property
PRIORITY	Incident operations Time-sensitive operational traffic
WELFARE	Health and welfare Non-life threatening needs
ROUTINE	Normal traffic Administrative / General information

STANDARD MESSAGE FORMAT

FROM: _____

TO: _____

SUBJECT: _____

MESSAGE: _____

_____

_____

_____

MESSAGE HANDLING RULES

- Write exactly what is received.
- Read back critical information.
- Use plain language.
- Avoid abbreviations and codes.
- Confirm delivery.
- Log message number.

READBACK PROCEDURE

OPERATOR A	*"Copy the following traffic."*
OPERATOR B	*"Ready to copy."*
OPERATOR A	"Message Number 24."
OPERATOR B	*"Message Number 24 copied."*
OPERATOR A	*"Go ahead."*
OPERATOR B	*"(Reads message back clearly.)"*
OPERATOR A	*"Roger. Correct."*

NOTE: *Read back **ALL** critical information and numbers.*

COMMON ERRORS

- **Assuming information**
  Never assume—always ask if unsure.
- **Correcting grammar**
  Transmit exactly as received.
- **Summarizing messages**
  Send the entire message.
- **Omitting numbers**
  Always include message numbers, times, and locations.
- **Failing to read back**
  Read back critical information for verification.

★ ACCURACY OVER SPEED ★

2.5 INTEROPERABILITY GUIDE

COMMUNICATIONS ECOSYSTEM

ARES
Amateur Radio Emergency Service

↓

EMERGENCY MANAGEMENT
County Emergency Management

↓

FIRE DEPARTMENTS
Local Fire / Rescue

↓

PUBLIC SAFETY LIAISON
Law Enforcement / Sheriff's Office

↓

STATE EOC
Texas Division of Emergency Management

WHO TALKS TO WHOM

ORGANIZATION	DIRECT CONTACT
ARES	ARES Operators
RACES	RACES Operators
Emergency Management (OEM)	EC / Net Control (NCS)
Fire Departments	Assigned Liaison
Sheriff's Office	Assigned Liaison
State EOC	Winlink / Assigned Station

GMRS INTEROPERABILITY

- ✓ Use plain language.
- ✓ Identify by FCC callsign.
- ✓ Follow GMRS regulations.
- ✓ Use interoperability channels only when needed.
- ✓ Maintain professional conduct at all times.

PUBLIC SAFETY INTEROPERABILITY

DO

- ✓ Pass information through the proper liaison.
- ✓ Use assigned tactical channels only.
- ✓ Follow Incident Command structure.
- ✓ Keep transmissions brief and concise.
- ✓ Confirm critical information.

DO NOT

- ✗ Self-deploy or self-dispatch.
- ✗ Contact dispatch directly.
- ✗ Use public safety frequencies without authorization.
- ✗ Create unnecessary radio traffic.
- ✗ Bypass the chain of command.

ESCALATION / INFORMATION FLOW PATH

ARES OPERATOR
Observes / Reports Information

→

NET CONTROL (NCS)
Logs / Verifies Information

→

EMERGENCY COORDINATOR (EC)
Evaluates / Coordinates Action

→

COUNTY EMERGENCY MANAGEMENT
Coordinates Resources

→

STATE RESOURCES
Requests / Deploys State Assets

KEY PRINCIPLE

We are the communication link — not the decision makers.
Stay in your lane. Pass information. Support the mission.

2.6 COMMUNICATIONS FAILURE SOP

LOSS OF PRIMARY REPEATER

SYMPTOMS

- ✕ Repeater not responding
- ✕ No courtesy tone
- ✕ No traffic heard

ACTIONS

1. Wait 30 seconds.
2. Retry transmission.
3. Move to Alternate Channel:

CH 2	WRTM 760 146.760 MHz Tone 136.5

Announce channel move.

LOSS OF ALL LOCAL REPEATERS

ACTIONS

Move to ARES Simplex Channels:

CH 10-14

- Establish directed simplex net.
- Use plain language.
- Make transmissions brief.

PACE FAILURE FLOW

PRIMARY
Normal Operations

↓

ALTERNATE
Backup Repeater

↓

CONTINGENCY
Direct / Simplex

↓

EMERGENCY
All Band / Digital

↓

LAST RESORT
Non-Radio Methods

LOSS OF ALTERNATE REPEATER

ACTIONS

Move to:

CH 9	ARES TAC 146.580 MHz Tone 123.0

- Establish simplex net.
- Appoint Net Control.
- Check in all stations.
- Maintain regular net discipline.

LOSS OF VHF OPERATIONS

ACTIONS

Move to:

CH 15-19

- Use UHF simplex channels.
- Increase power as needed.
- Monitor all applicable channels.
- Maintain net operations.

LOSS OF ALL VOICE COMMUNICATIONS

ACTIONS

Escalate to digital and data communications:

CH 28	DATA-V	144.975 MHz	Tone 123.0
CH 29	DATA-U	438.775 MHz	Tone 123.0
CH 30	APRS	144.390 MHz	(Packet)

- Use digital modes for traffic and check-ins.
- Use Winlink as directed.
- Log all attempts and results.

LAST RESORT COMMUNICATIONS

USE THE FOLLOWING IN ORDER:

APRS MESSAGING – Use for short text traffic.

HF RADIO – Use regional / state nets if available.

PHYSICAL MESSENGER – Send runner with written info.

WINLINK – Send/Receive messages via RMS.

TELEPHONE – Use only if infrastructure is available.

Continue attempts on all channels as conditions allow.

IMPORTANT REMINDERS

- ✓ Always announce changes to all stations.
- ✓ Use clear, plain language.
- ✓ Check in on new channel immediately.
- ✓ Log all failures and actions taken.
- ✓ Stay calm, follow the PACE plan.
- ✓ Mission first – stay in contact.

⚠ DON'T PANIC — FOLLOW THE PACE PLAN ⚠

WHEN ALL ELSE FAILS

CHAPTER
3.0
WEATHER OPERATIONS

MONITOR • REPORT • PROTECT

STORM SPOTTING

SEVERE WEATHER

RADAR AWARENESS

EFFECTIVE REPORTING

SAFETY FIRST

TEAMWORK SAVES LIVES

SITUATIONAL AWARENESS • ACCURATE REPORTS • SAVE LIVES

WEATHER OPERATIONS IS A CRITICAL MISSION

FREESTONE COUNTY ARES
TACTICAL FIELD BINDER

3.1 STORM SPOTTER ACTIVATION LEVELS

Activation levels are determined by the severity and potential impact of weather threats to Freestone County.
Levels may be upgraded or downgraded at any time by the Emergency Coordinator.

LEVEL III
MONITORING

Routine monitoring of weather conditions.
No significant threat to the county.

CRITERIA / TRIGGERS

- No severe weather outlook
- Weak or no storms expected
- General awareness level
- Normal daily nets and operations

OPERATOR ACTIONS

- ✓ Monitor weather sources
- ✓ Check in on weekly nets
- ✓ Maintain readiness
- ✓ Report significant weather if observed
- ✓ Update equipment and information

STATUS: NORMAL OPERATIONS

LEVEL II
STANDBY

Increased potential for severe weather.
Be prepared to activate.

CRITERIA / TRIGGERS

- Severe Weather Outlook (Slight / Enhanced)
- Storms possible within 48 hours
- Weather hazards possible
- Local conditions may support storms

OPERATOR ACTIONS

- ✓ Monitor weather closely
- ✓ Ensure equipment is operational
- ✓ Verify contact information
- ✓ Standby for possible activation
- ✓ Check in on scheduled standby net

STATUS: STANDBY

LEVEL I
FULL ACTIVATION

Severe weather is occurring or imminent.
Full storm spotter operations in effect.

CRITERIA / TRIGGERS

- Severe Thunderstorm Warning
- Tornado Watch or Warning
- Confirmed severe weather in county
- High confidence of severe impact

OPERATOR ACTIONS

- ✓ Check in on activation net immediately
- ✓ Deploy as directed by Net Control
- ✓ Provide timely and accurate reports
- ✓ Maintain continuous monitoring
- ✓ Follow safety procedures at all times

STATUS: FULL ACTIVATION

ACTIVATION LEVEL TRIGGER REFERENCE

WEATHER THREAT	LEVEL III	LEVEL II	LEVEL I	EXAMPLES
Tornado Potential	None / Very Low	Possible	Likely / Confirmed	Tornado Watch, Tornado Warning, Confirmed Tornado
Severe Hail	None / Very Low	Possible (≥ 1")	Likely (≥ 1"+)	Severe Thunderstorm Warning (Hail)
Damaging Winds	None / Very Low	Possible (≥ 50 mph)	Likely (≥ 58 mph)	Severe Thunderstorm Warning (Wind)
Heavy Rain / Flooding	None / Very Low	Possible	Likely / Ongoing	Flash Flood Watch / Warning
Winter Weather	None / Very Low	Possible	Likely / Ongoing	Winter Storm Watch / Warning

SITUATION AWARENESS SOURCES

NATIONAL WEATHER SERVICE weather.gov	Forecasts, watches, warnings, hazardous weather outlooks
NOAA WEATHER RADIO All Hazards	24/7 broadcasts on local weather conditions
STORM PREDICTION CENTER spc.noaa.gov	Convective outlooks and severe weather probabilities
LOCAL RADAR RadarScope, GRLevel3, etc.	Real-time radar and storm tracking

LEVEL CHANGE AUTHORITY

The Emergency Coordinator (EC) may change activation levels at any time based on the latest information.

SPOTTERS & OPERATORS → NET CONTROL (NCS) → EMERGENCY COORDINATOR (EC) → COUNTY EMERGENCY MANAGEMENT

When in doubt, increase your level of readiness.
REPORT WHAT YOU SEE • WHEN YOU SEE IT • WHERE YOU SEE IT

BE AWARE • BE PREPARED • BE SAFE

3.2 REPORTABLE WEATHER CRITERIA

If you observe any of the conditions below, report them to Net Control immediately.
When in doubt — REPORT IT.

TORNADOES / ROTATION

- ✓ Tornado
- ✓ Funnel Cloud
- ✓ Rotating Wall Cloud
- ✓ Landspout
- ✓ Debris Cloud

⚠ REPORT IMMEDIATELY

HAIL

SIZE (DIAMETER)	EXAMPLE
1.00"	Quarter
1.25"	Half Dollar
1.50"	Ping Pong Ball
1.75"	Golf Ball
2.50"+	Tennis Ball

⚠ Quarter-size hail or larger is reportable.

WIND

REPORT:

- ✓ Measured winds ≥ 50 MPH
- ✓ Trees down
- ✓ Power lines down
- ✓ Structural damage
- ✓ Roof damage
- ✓ Debris in roadway
- ✓ Dust storms

SEVERE WEATHER DAMAGE

REPORT:

- ✓ Building damage
- ✓ Large tree damage
- ✓ Utility damage (poles / lines / equipment)
- ✓ Road obstructions
- ✓ Injuries related to weather
- ✓ Crop damage
- ✓ Debris field

FLOODING

REPORT:

- ✓ Water over roadway
- ✓ Flash flooding
- ✓ Vehicles stranded
- ✓ Swift water conditions
- ✓ Flooded homes / businesses
- ✓ Dam or levee issues

WINTER WEATHER

REPORT:

- ✓ Ice accumulation
- ✓ Sleet / Freezing Rain
- ✓ Snow accumulation
- ✓ Drifting snow
- ✓ Dangerous road conditions
- ✓ Blizzard conditions
- ✓ Low visibility

REPORT FORMAT

WHO	WHAT	WHEN	WHERE
Your call sign or location.	What you are reporting.	Time observed (use 24-hour time).	Exact location (street, intersection, landmark, or GPS).

EXAMPLE REPORT

WHO:	Net Control
FROM:	Mobile 2
WHAT:	Quarter size hail
WHERE:	FM 27 and Highway 84
WHEN:	Observed 1935 hrs

⚠ REPORT WHAT YOU SEE — NOT WHAT YOU THINK ⚠

ACCURATE REPORTS SAVE LIVES • TIMELY REPORTS PROTECT PROPERTY
BE OBSERVANT • BE ACCURATE • BE PROFESSIONAL

3.3 SEVERE WEATHER REPORTING FORMAT

Accurate and timely reports help protect lives and property.
Use the Four W's — Every Time.

THE FOUR W'S

WHO
Who is making the report?

EXAMPLE: Mobile 3

WHAT
What was observed?

EXAMPLE: Quarter-size hail

WHEN
Time observed?

EXAMPLE: 1935 hrs

WHERE
Exact location?

EXAMPLE: FM 27 & Hwy 84

REPORT TEMPLATE

NET CONTROL
THIS IS _____
I HAVE A WEATHER REPORT

WHAT: _____

_____

WHERE: _____

_____

WHEN: _____

ADDITIONAL INFORMATION:

_____

_____

_____

GOOD REPORT (EXAMPLE)

- Net Control
- Mobile 2
- Quarter-size hail
- FM 27 and Highway 84
- Observed 1935 hrs

BAD REPORT (EXAMPLE)

It's hailing pretty bad over by the church.

WHY IT'S BAD

- ✕ No exact location
- ✕ No size of hail
- ✕ No time reported
- ✕ Cannot be verified
- ✕ Not useful for warning others

LOCATION REPORTING PRIORITY

1. GPS COORDINATES — Example: N 31.7372 W 96.1956
2. HIGHWAY INTERSECTION — Example: FM 27 & Hwy 84
3. STREET ADDRESS — Example: 125 County Rd 123
4. LANDMARK — Example: Near Fairfield High School

Be as precise as possible. Give enough information that others can find the location quickly.

TIME REPORTING

USE 24-HOUR TIME

- 1900 hrs
- 1935 hrs
- 2047 hrs

DO NOT USE

- About 7pm
- A little while ago
- Earlier
- Shortly before dark

Always use 24-hour time.
It eliminates confusion.

BE SPECIFIC • BE ACCURATE • BE TIMELY

ACCURATE REPORTS SAVE LIVES • TIMELY REPORTS PROTECT PROPERTY
YOUR REPORT MAKES A DIFFERENCE!

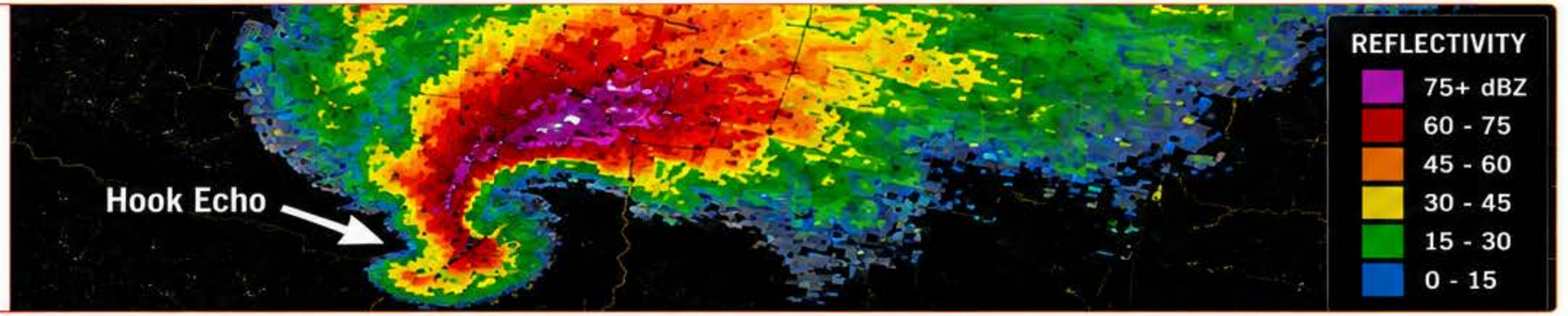
3.4 RADAR INTERPRETATION GUIDE

Common Radar Signatures Every Storm Spotter Should Recognize

1 HOOK ECHO

WHAT IT LOOKS LIKE:
Hook-shaped appendage on the southwest side of a storm cell.

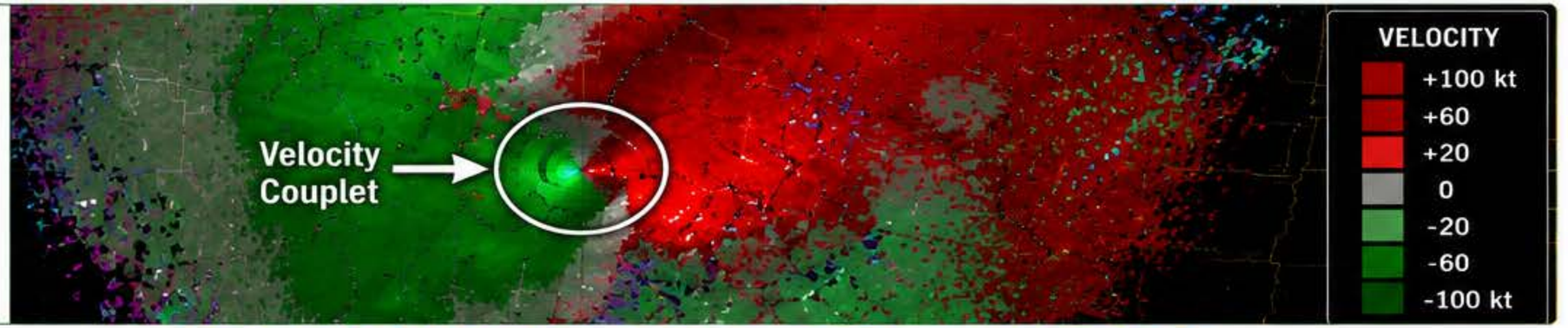
WHAT IT MEANS:
- ✔ Possible tornado
- ✔ Strong rotation
- ✔ Immediate attention required



2 VELOCITY COUPLET

WHAT IT LOOKS LIKE:
Adjacent inbound (green) and outbound (red) velocities close together.

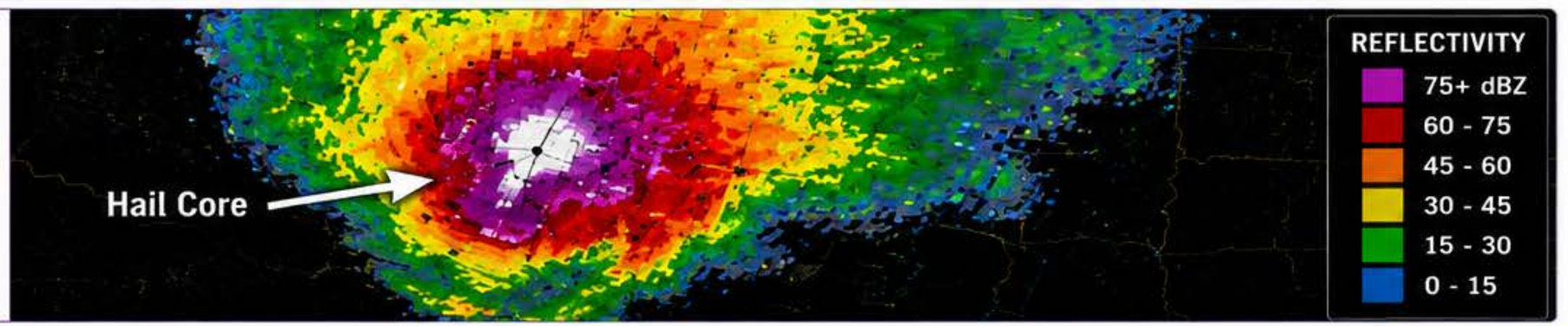
WHAT IT MEANS:
- ✔ Rotation indicated
- ✔ Potential mesocyclone
- ✔ Possible tornadic development



3 HAIL CORE

WHAT IT LOOKS LIKE:
Small, intense core of very high reflectivity (bright colors) often with a "hail spike".

WHAT IT MEANS:
- ✔ Large hail possible
- ✔ Strong updraft
- ✔ Monitor reports closely



⚠ **RADAR SUPPORTS REPORTS — IT DOES NOT REPLACE SPOTTERS** ⚠

SEE IT • REPORT IT • VERIFY IT

ACCURATE REPORTS SAVE LIVES AND PROTECT PROPERTY

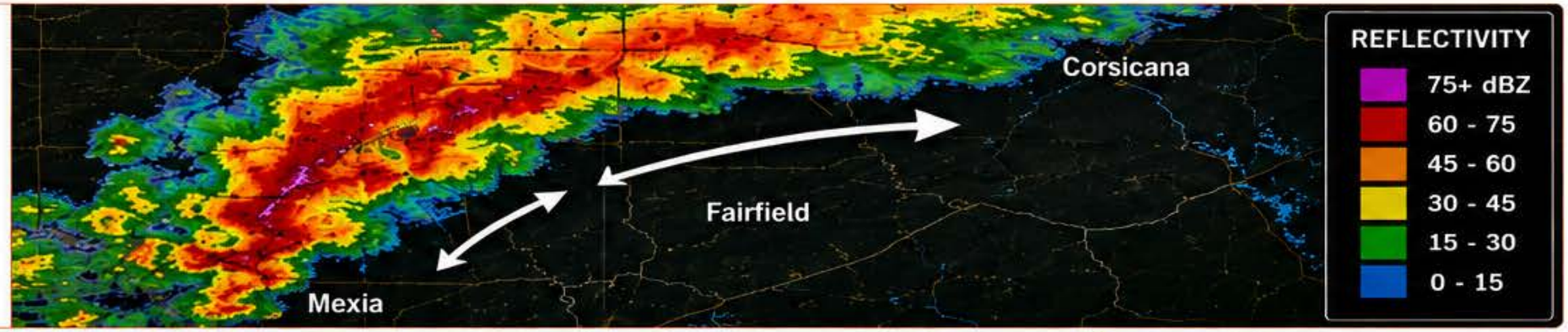
3.4 RADAR INTERPRETATION GUIDE

Common Radar Signatures Every Storm Spotter Should Recognize

4 BOW ECHO

WHAT IT LOOKS LIKE:
Arched line of storms that resembles a "bow".

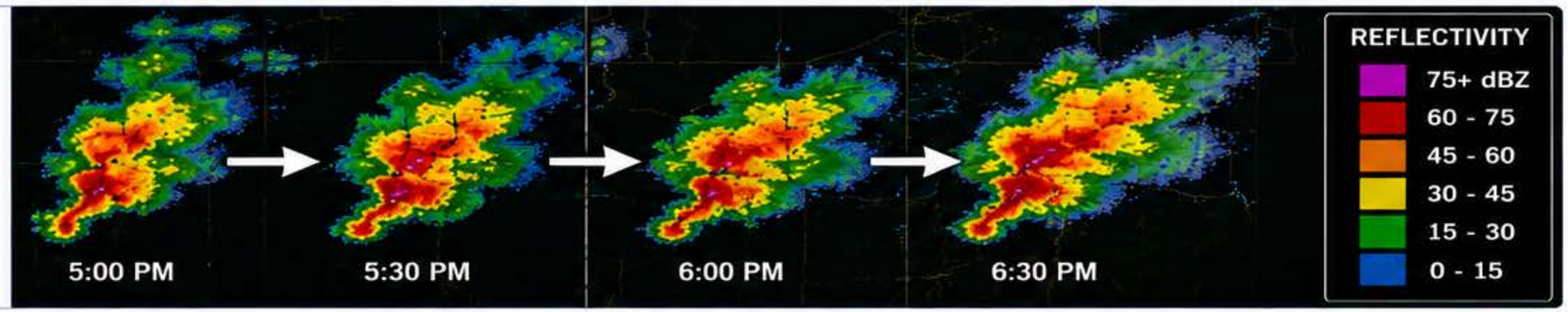
WHAT IT MEANS:
- ✓ Damaging straight-line winds
- ✓ Fast-moving storms
- ✓ Widespread damage potential



5 TRAINING CELLS

WHAT IT LOOKS LIKE:
Multiple storm cells moving over the same area repeatedly.

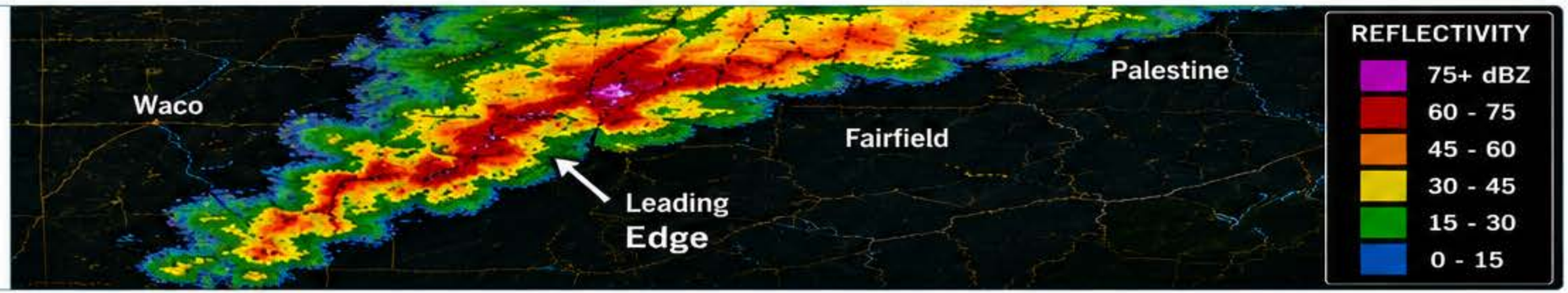
WHAT IT MEANS:
- ✓ Flash flooding risk
- ✓ Repeated rainfall
- ✓ Monitor creeks and low-water crossings



6 SQUALL LINE

WHAT IT LOOKS LIKE:
Long line of storms with leading edge.

WHAT IT MEANS:
- ✓ Strong winds
- ✓ Heavy rainfall
- ✓ Large coverage area



Radar is a powerful tool, but ground truth from spotters is critical.
Combine radar information with reports for the most accurate assessment.

SEE IT • REPORT IT • VERIFY IT

ACCURATE REPORTS SAVE LIVES AND PROTECT PROPERTY

FREESTONE COUNTY ARES TACTICAL FIELD BINDER

3.5 WEATHER OPERATIONS COMMUNICATIONS PLAN

Reliable communications save lives. Use the plan. Follow the process.

WEATHER NET PACE PLAN

PHASE	CHANNEL	FREQUENCY / TONE	PURPOSE	MODE
PRIMARY (P)	CH 1 FF 110	145.110 MHz -0.600 MHz / Tone 146.2	Weather Net Normal Operations	FM Voice
ALTERNATE (A)	CH 2 WRTM 760	146.760 MHz -0.600 MHz / Tone 136.5	Backup Net Alternate Communications	FM Voice
CONTINGENCY (C)	CH 9 ARES TAC	146.580 MHz Tone 123.0	Simplex Weather Net When repeaters fail	FM Voice
EMERGENCY (E)	HF / GMRS / MESSAGE DELIVERY	As Available (See Section 2.2)	Last Resort Communications When all else fails	HF / GMRS Message Delivery

SPOTTER CHECK-IN FORMAT

CALLSIGN	Example: Mobile 3
LOCATION	Example: FM 27 & Hwy 84
MOBILE / FIXED	Example: Mobile
TRAFFIC / NO TRAFFIC	Example: No Traffic

WEATHER NET PRIORITIES

1. Tornado Reports
2. Funnel Clouds
3. Large Hail
4. Wind Damage
5. Flooding
6. Routine Traffic

DURING ACTIVE WARNING

- Keep transmissions brief
- Use plain language
- Report only observed conditions
- Give exact locations
- Monitor assigned channel

DEPLOYMENT TYPES

MOBILE SPOTTER	Moving in vehicle. Reports observations.
FIXED SPOTTER	Stationary location. Reports local conditions.
EOC OPERATOR	Monitors nets. Relays information.
RELAY STATION	Extends coverage. Passes traffic.

SEVERE WEATHER NET FLOW

SPOTTER → Observe & Report
NET CONTROL → Receive & Verify
NWS / OEM → Evaluate & Act
PUBLIC WARNING → Issue Warnings

Accurate reports → Verified by Net Control → Relayed to NWS/OEM → Warnings issued to protect lives and property.

OBSERVE • REPORT • PROTECT

YOUR REPORT COULD BE THE ONE THAT SAVES A LIFE.
BE PREPARED • BE PROFESSIONAL • BE THE LINK

FREESTONE COUNTY ARES TACTICAL FIELD BINDER

3.6 WEATHER NET CONTROL RESPONSIBILITIES

MISSION: The Net Control Station (NCS) is responsible for managing all weather net operations, maintaining net discipline, collecting reports, and ensuring critical information reaches the National Weather Service through the Emergency Coordinator.

NET CONTROL DUTIES

- Open and close the weather net
- Maintain net discipline
- Control station check-ins
- Prioritize emergency traffic
- Assign tactical call signs if needed
- Track active spotters
- Maintain ICS-214 log
- Maintain operator accountability
- Ensure reports are complete
- **Read the NWS warnings over the air**

CHECK-IN INFORMATION

Record the following for each station:

- Callsign
- Operator Name
- Location
- Mobile / Fixed
- Available Resources

TRAFFIC PRIORITIES

PRIORITY 1
LIFE THREATENING
- Tornado
- Funnel Cloud
- Rotating Wall Cloud

IMMEDIATE RELAY TO EC

PRIORITY 2
SIGNIFICANT WEATHER
- Wind Damage
- Large Hail
- Flash Flooding

RELAY AS SOON AS PRACTICAL

PRIORITY 3
ROUTINE WEATHER
- Heavy Rain
- Reduced Visibility
- Routine Weather Reports

LOG AND MONITOR

REPORT FLOW

SPOTTER
Observe & Report
↓
NET CONTROL
Receive & Verify
↓
EC
Evaluate & Relay
↓
NATIONAL WEATHER SERVICE
Issue Warnings

WHEN A REPORT IS RECEIVED

1. Acknowledge the report
2. Read back critical details
3. Verify exact location
4. Verify time observed
5. Log the report (ICS-214)
6. Relay report to the EC
7. Monitor for updates

REPORT RELAY FORMAT (To EC)

FROM:	Net Control
SUBJECT:	Weather Report Received
WHAT:	Quarter-size hail
WHERE:	FM 27 & Hwy 84
WHEN:	1935 hrs
SOURCE:	Mobile 2

EC RESPONSIBILITIES

- Receive reports from Net Control
- Evaluate the significance
- Submit reports to NWS
- Coordinate with Emergency Management
- Maintain situational awareness
- Request additional observations as needed
- Provide feedback to Net Control

COLLECT • VERIFY • RELAY

ACCURATE INFORMATION • TIMELY RELAY • LIVES DEPEND ON IT
WE ARE THE LINK BETWEEN SPOTTER AND SCIENCE.

COMMUNICATIONS – BE PREPARED / BE PROFESSIONAL / BE THE LINK

3.7 NCS SCRIPT – ACTIVATING A WEATHER NET

Use clear, slow, and professional speech. Pause between statements. Maintain a calm and confident tone at all times.

NCS ACTIVATION CHECKLIST

- ☐ Monitor weather conditions and alerts
- ☐ Verify repeater / channel operation
- ☐ Have ICS-214, pen, and time available
- ☐ Confirm EC availability
- ☐ Announce net activation

EXAMPLE NET TITLE

Freestone County ARES Weather Net

Use this exact title when opening the net.

KEY POINTS FOR NCS

- Be clear and concise.
- Give complete information.
- Pause between statements.
- Speak slowly and clearly.
- Repeat important information when needed.
- Maintain control of the net.

NCS SCRIPT – STEP BY STEP

1. "This is [Your Call Sign], Net Control Station for the Freestone County ARES Weather Net."
2. "The reason for this net is for [whatever the severe weather warning was issued for] issued for Freestone County and surrounding areas."
3. "This net is now in session on Channel [#], [Frequency]."
4. "Net Control will now read the National Weather Service warning in its entirety."
5. "This net is for official weather observations and weather-related traffic only."
6. "All stations with weather information, damage reports, or wishing to check in, please call now."
7. "When checking in, please give: Callsign, Name, Location, and Mobile or Fixed."
8. "Priority will be given to life-threatening and time-critical reports."
9. "All reports should be factual, concise, and include: What, Where, When, and Source."
10. "We will now take check-ins. This is [Your Call Sign], Net Control."
11. "(Close) This is [Your Call Sign], closing the Freestone County ARES Weather Net. Thank you all for your participation. Net clear."

OPERATING REMINDERS

- Keep transmissions brief and to the point.
- Confirm locations and details before relaying.
- Log all traffic on ICS-214.
- Immediately relay life-threatening reports to the EC.
- Maintain professionalism and net discipline.

WEATHER NETS SAVE LIVES

ACCURATE INFO SAVES LIVES

NOTES TO NCS

- Speak clearly and slowly.
- Control the net at all times.
- Take accurate logs.
- Keep transmissions brief.
- Thank all stations for their support.

OBSERVE • REPORT • PROTECT

ACCURATE INFORMATION • TIMELY RELAY • LIVES DEPEND ON IT
WE ARE THE LINK BETWEEN SPOTTER AND SCIENCE.

3.8 STORM DEPLOYMENT SAFETY

⚠ YOUR SAFETY IS MORE IMPORTANT THAN THE REPORT ⚠

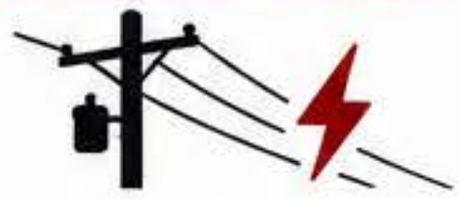
LIGHTNING SAFETY

- Stay inside a vehicle during lightning.
- Avoid open fields, hilltops, and ridges.
- Avoid trees, towers, fences, and metal structures.
- If lightning is nearby, suspend operations.
- When in doubt, **GET OUT!**

VEHICLE POSITIONING

- Position your vehicle with an escape route.
- Stay off the roadway whenever possible.
- Maintain at least 1/4 tank of fuel.
- Do not block roads or emergency access.
- Use hazard lights when stopped.

POWER LINE SAFETY



- Treat all power lines as energized.
- Stay at least 30 feet away from downed lines.
- Never drive over downed power lines.
- Report downed lines immediately to NCS and authorities.

FLOOD SAFETY

- Turn Around **Don't Drown.**
- Never drive through flooded roads.
- Avoid low water crossings and bridges.
- Do not park or stand in flood-prone areas.
- Report flooding — do not enter it.

NIGHT OPERATIONS

- Carry a flashlight and spare batteries.
- Wear a reflective safety vest.
- Maintain situational awareness at all times.
- Watch for debris, holes, and downed lines.
- Do not rely on high beams.

STORM CHASING POLICY

- Do not self-deploy into storms.
- Do not chase or pursue severe weather.
- Do not enter warned areas unnecessarily.
- Do not take unnecessary risks to obtain reports.
- Your family needs you more than we need a report.

SPOTTER SAFETY CHECKLIST

- ☐ Radio charged and functional
- ☐ Fuel topped off
- ☐ Reflective vest available
- ☐ Phone charged
- ☐ Go-kit available and ready
- ☐ Water available
- ☐ Flashlight and batteries
- ☐ GPS operational

REMEMBER

- ★ Weather can change rapidly.
- ★ Conditions can become dangerous in minutes.
- ★ Be prepared to leave the area.
- ★ Follow instructions from Net Control, the EC, and Emergency Management.
- ★ When the storm is over, **SAFELY return home.**

⚠ OBSERVE THE STORM — DON'T BECOME THE EMERGENCY ⚠

BE PREPARED • BE PROFESSIONAL • BE THE LINK

FREESTONE COUNTY ARES
TACTICAL FIELD BINDER

4.0
CHAPTER DIVIDER

EMERGENCY COMMUNICATIONS • FIELD OPERATIONS • TACTICAL REFERENCE

CHAPTER 4
FIELD OPERATIONS

This chapter provides the procedures, checklists, and operational guidance for ARES field deployments and support to served agencies.

ACTIVATE
Follow established activation procedures.

COMMUNICATE
Use proper channels and procedures.

SUPPORT
Provide reliable communications to served agencies.

OPERATE SAFELY
Follow safety guidelines and protocols.

DOCUMENT
Maintain accurate records and reports.

OUR MISSION

To provide effective, efficient, and professional communications support when it is needed most.

OBSERVE
Be aware.
Know your surroundings.

REPORT
Provide accurate information.
Be the eyes and ears.

SUPPORT
Work together.
Serve our community.
Make a difference.

PROTECT
Safety first.
Protect yourself, others, and the mission.

FREESTONE COUNTY ARES
TACTICAL FIELD BINDER

EMERGENCY COMMUNICATIONS • FIELD OPERATIONS • TACTICAL REFERENCE

4.1 ARES ACTIVATION PROCEDURES

ACTIVATION AUTHORITY

Activation Authority & Request Sources

Emergency Coordinator (EC)

County Emergency Management

National Weather Service

Served Agency

Only the EC authorizes activation.
Other agencies may request it.

ACTIVATION LEVELS

LEVEL III — MONITORING
- Situational awareness
- Monitor primary repeater
- Monitor weather conditions
- No deployment

LEVEL II — STANDBY
- Prepare equipment
- Check radios
- Fuel vehicles
- Monitor for assignment

LEVEL I — FULL ACTIVATION
- Deployment authorized
- Net established
- Operators assigned
- Incident support begins

NOTIFICATION METHODS

Phone	Voice calls to operators.
Email	Email notifications.
Amateur Radio	Directed radio notification.
Group Messaging	Text or app group alerts.

OPERATOR RESPONSIBILITIES

When notified of activation:

- [ ] Acknowledge activation.
- [ ] State availability.
- [ ] State estimated response time.
- [ ] Report equipment limitations.
- [ ] Maintain monitoring on assigned channel.

REPORTING STATUS

AVAILABLE	Ready for assignment.
DELAYED	Available at a later time.
UNAVAILABLE	Unable to participate.

IMPORTANT

Stay alert, monitor designated channels, and be prepared to respond when needed.

ACTIVATION FLOW

INCIDENT — Event occurs or threat develops.

→ **EC** — Emergency Coordinator assesses the situation.

→ **ARES ACTIVATION** — ARES is activated at the appropriate level.

→ **NET ACTIVATION** — Net Control Station is established as needed.

→ **OPERATOR NOTIFICATION** — Operators are notified and respond.

→ **DEPLOYMENT** — Operators deploy and support the mission.

FREESTONE COUNTY ARES
TACTICAL FIELD BINDER

EMERGENCY COMMUNICATIONS • FIELD OPERATIONS • TACTICAL REFERENCE

4.2 DEPLOYMENT PROCEDURES

1. PRE-DEPLOYMENT

Before leaving home:

- ☐ Confirm assignment
- ☐ Verify destination
- ☐ Notify family
- ☐ Fuel vehicle
- ☐ Charge radio batteries
- ☐ Verify go-kit
- ☐ Check weather conditions
- ☐ Confirm deployment frequency

2. EN ROUTE

While traveling:

- Drive safely.
- Monitor assigned frequency.
- Avoid unnecessary radio traffic.
- Advise NCS of delays.
- Report significant hazards.

Your safety and the safety of others is the priority. Slow down and arrive alive.

3. ARRIVAL AT ASSIGNMENT

Upon arrival:

- ✓ Check in with NCS.
- ✓ Check in with served agency representative.
- ✓ Receive assignment briefing.
- ✓ Establish communications.
- ✓ Begin ICS-214 logging.
- ✓ Verify tactical call sign if assigned.

4. WHILE ON ASSIGNMENT

Remember:

- Maintain radio monitoring.
- Log significant events.
- Work through agency chain of command.
- Do not self-assign tasks.
- Do not speak to media unless authorized.
- Report safety concerns immediately.

Safety, communication, and accountability are your primary missions.

5. DEMOBILIZATION

When released:

- ✓ Notify NCS.
- ✓ Return issued equipment.
- ✓ Submit logs / forms.
- ✓ Report mileage if required.
- ✓ Advise NCS when safely home.

Accountability does not end until you are safely home.

6. FIELD OPERATOR RULES

DO

- ✓ Be professional.
- ✓ Be on time.
- ✓ Wear identification.
- ✓ Follow directions.
- ✓ Be adaptable and flexible.

DON'T

- ✗ Freelance or self-deploy.
- ✗ Leave assignment without notification.
- ✗ Share sensitive information.
- ✗ Operate outside your training or capabilities.
- ✗ Create additional problems.

DEPLOYMENT FLOW

ASSIGNMENT → Operator is assigned by NCS or EC.

DEPLOY → Prepare and depart for assignment.

ARRIVE → Check in and receive briefing and instructions.

OPERATE → Perform duties, maintain comms, log activity.

DEMOBILIZE → Released from assignment and return equipment.

AFTER ACTION → Submit reports, provide feedback, improve readiness.

BE PREPARED • BE PROFESSIONAL • BE THE LINK

COMMUNICATIONS – BE PREPARED / BE PROFESSIONAL / BE THE LINK

FREESTONE COUNTY ARES
TACTICAL FIELD BINDER

EMERGENCY COMMUNICATIONS • FIELD OPERATIONS • TACTICAL REFERENCE

4.3 CHECK-IN / CHECK-OUT PROCEDURES

1. CHECK-IN PROCESS

Upon arrival:

- Contact NCS.
- Check in with served agency.
- Report location.
- Report tactical assignment.
- Report available capabilities.
- Begin activity log.

Always check in even if you arrive during a net break.

2. REQUIRED CHECK-IN INFORMATION

CALLSIGN: _____

NAME: _____

LOCATION: _____

ASSIGNMENT: _____

MOBILE / FIXED: _____

RESOURCES AVAILABLE:

_____

_____

TIME ON DUTY: _____

3. STATUS REPORTING

AVAILABLE
Fully available and ready for assignment.

LIMITED AVAILABILITY
Available with limitations. Specify restrictions.

UNAVAILABLE
Not available for assignment. Provide reason if able.

4. PERIODIC ACCOUNTABILITY

NCS should request accountability:

- Every 60 minutes.
- After severe weather events.
- Following major incidents.
- Prior to demobilization.

Maintain radio monitoring at all times while deployed.

5. CHECK-OUT PROCESS

Before leaving:

- Notify NCS.
- Notify served agency.
- Submit forms / logs.
- Return issued equipment.
- Report travel status.
- Advise when safely home.

You are still accountable until you advise NCS you are home.

6. LOST CONTACT PROCEDURE

If contact is lost:

1. Attempt radio contact.
2. Attempt alternate frequency.
3. Attempt telephone contact.
4. Contact EC.
5. Verify operator status.

If unable to verify status, initiate appropriate actions.

7. IMPORTANT REMINDERS

- Always check in and out.
- Provide accurate information.
- Report changes immediately.
- Maintain professional communications.
- Accountability ensures operator safety.

You are part of the team. Stay connected. Stay safe.

ACCOUNTABILITY FLOW

OPERATOR
Reports in and out. Provides status and information.

→

NCS
Maintains operator accountability and situation awareness.

→

EC
Oversees operations and ensures operator safety and support.

→

INCIDENT RECORD
All check-in/out activity is logged for operational records and review.

ACCOUNTABILITY IS OPERATIONAL READINESS

BE PREPARED • BE PROFESSIONAL • BE THE LINK

4.4 GO-KIT STANDARDS

A well-prepared operator is a force multiplier. Your go-kit should enable you to deploy quickly, operate safely, and sustain yourself during emergencies.

TIER 1 – REQUIRED ITEMS

EVERY DEPLOYABLE OPERATOR SHOULD HAVE

1. PRIMARY RADIO Programmed per county plan.	**2. SPARE BATTERIES** Fully charged or fresh.	**3. SPEAKER MIC / HEADSET** For clear and hands-free comms.	**4. CHARGING CABLES** For radio, phone, and accessories.	**5. REFLECTIVE VEST** High visibility for safety.
6. ARES IDENTIFICATION Wear at all times while deployed.	**7. NOTEBOOK & PENS** For logging and situational notes.	**8. FLASHLIGHT** LED preferred with spare bulb.	**9. SPARE AA/AAA BATTERIES** For lights, radios, and accessories.	**10. DRINKING WATER** Minimum 1 quart per operator.

TIER 2 – RECOMMENDED ITEMS

ENHANCE YOUR CAPABILITY AND COMFORT

RECOMMENDED ITEMS

1. GO-BAG (SEE CONTENTS TO THE RIGHT)
2. POWER BANK
3. COUNTY MAPS
4. RAIN GEAR
5. WORK GLOVES
6. HEADLAMP
7. EXTENSION CORD
8. BATTERY BOX
9. SNACKS
10. HYGIENE KIT

GO-BAG CONTENTS (RECOMMENDED)

- ☐ Notebook & pens / pencils
- ☐ Flashlight or headlamp
- ☐ Multi-tool or pocket knife
- ☐ Electrical tape
- ☐ Zip ties
- ☐ Small first aid kit
- ☐ Hand sanitizer / wipes
- ☐ Trash bags
- ☐ Whistle
- ☐ Emergency blanket
- ☐ Portable charger cable
- ☐ Contact list (printed)
- ☐ High energy snacks
- ☐ Cash (small bills)
- ☐ Personal medications
- ☐ Extra batteries (AA/AAA)
- ☐ 1 complete change of clothes

Tier 2 items are strongly recommended to improve self-sufficiency, safety, and mission effectiveness. Build your go-kit now, so you're ready when it matters most.

4.4 GO-KIT STANDARDS

A well-prepared operator is a force multiplier. Your go-kit should enable you to deploy quickly, operate safely, and sustain yourself during emergencies.

TIER 3 – EXTENDED OPERATIONS (24–72 HOURS)

FOR SELF-SUFFICIENCY DURING EXTENDED DEPLOYMENTS

1. SLEEPING GEAR Sleeping bag or quilt and pad.	**2. EXTRA CLOTHING** Extra shirt, pants, socks, undergarments, and layers.	**3. EXTRA FOOTWEAR** Sturdy boots or shoes plus socks.	**4. FOOD SUPPLY** Non-perishable meals and snacks for 24–72 hours.
5. ADDITIONAL WATER At least 1 gallon per day.	**6. ADDITIONAL POWER SOURCES** Batteries, power stations, or other energy sources.	**7. SHELTER EQUIPMENT** Tent, tarp, or other shelter solution.	**8. IFAK** Individual First Aid Kit (see list to the right).
9. PERSONAL HYGIENE Toothbrush, soap, deodorant, toilet paper, etc.	**10. LIGHTING** Lantern or area light plus headlamp.	**11. TOOLS** Multi-tool, duct tape, paracord, and zip ties.	**12. DOCUMENTS** ID, licenses, ICS forms, maps, and reference materials.
13. TRASH BAGS For waste management.	**14. DUCT TAPE** For repairs and multiple uses.	**15. NOTEBOOK** Field notebook for logging and notes.	**16. SUN PROTECTION** Sunglasses, hat, and sunscreen.

Be prepared to be self-sufficient. Plan ahead, pack smart, and take care of yourself so you can take care of the mission.

IFAK – INDIVIDUAL FIRST AID KIT (RECOMMENDED CONTENTS)

- ☐ Tourniquet
- ☐ Pressure dressing (4" or 6")
- ☐ Hemostatic gauze
- ☐ Compressed gauze
- ☐ Nitrile gloves (pair)
- ☐ Chest seal (vented)
- ☐ Roller gauze
- ☐ Elastic bandage (ACE wrap)
- ☐ Triangle bandage
- ☐ Medical tape
- ☐ Antiseptic wipes
- ☐ Alcohol prep pads
- ☐ Burn gel / burn dressing
- ☐ Pain reliever (e.g., ibuprofen)
- ☐ Antihistamine (e.g., Benadryl)
- ☐ Tweezers
- ☐ Trauma shears
- ☐ CPR face shield
- ☐ Safety pins
- ☐ Eye protection

Customize your IFAK based on your training, medical needs, and mission environment.

GO-KIT INSPECTION – SUMMARY (DETAILS ON NEXT PAGE)

- ☐ **EQUIPMENT** – All gear functional and clean.
- ☐ **BATTERIES** – Charged and spares available.
- ☐ **RADIOS** – Programmed per county plan.
- ☐ **LIGHTING** – Working and with spare batteries.
- ☐ **WATER & FOOD** – Stocked and within expiration.
- ☐ **DOCUMENTS** – Current and complete (forms, ID, etc.).

72-HOUR DEPLOYMENT RULE

Every operator should be prepared to operate independently for at least 72 hours whenever practical.

4.4 GO-KIT STANDARDS

A well-prepared operator is a force multiplier. Your go-kit should enable you to deploy quickly, operate safely, and sustain yourself during emergencies.

COMMUNICATIONS EQUIPMENT

MINIMUM RADIO CAPABILITY

2 METER RADIO
Programmed with county channel plan.

70 CM RADIO
Programmed with county channel plan.

COUNTY CHANNEL PLAN
Includes primary, alternate, contingency, and tactical channels.

WEATHER CHANNELS
NOAA Weather Alert and local weather channels.

SECURE / TACTICAL CHANNELS
As directed during activation.

SPEAKER MIC / HEADSET
For clear and hands-free communications.

All radios should be pre-programmed, tested, labeled with your call sign, and have fresh batteries.

DIGITAL OPERATIONS (IF CAPABLE)

LAPTOP / TABLET
For digital communications, Winlink, and documentation.

WINLINK CAPABILITY
Winlink Express or RMS Express configured and tested.

DIGITAL APPS
Helpful apps: APRS, maps, weather, logging, and ICS tools.

PORTABLE POWER
Power bank or battery pack for devices.

DATA CABLES
Spare USB cables and adapters.

CLOUD BACKUP
Secure backup for logs, documents, and forms.

Test all digital systems before deployment. Do not assume connectivity in the field.

GO-KIT INSPECTION CHECKLIST

Conduct a monthly inspection of your go-kit. Check all items and restock as needed.

- ☐ Batteries charged and spares available
- ☐ Radios programmed and functional
- ☐ Antennas and cables in good condition
- ☐ Flashlights working / bulbs updated
- ☐ Water replaced (check expiration)
- ☐ Food and snacks checked / replaced
- ☐ First aid supplies checked / restocked
- ☐ Clothing in good condition
- ☐ Forms, maps, and documents updated
- ☐ Go-bag contents complete
- ☐ Power banks charged
- ☐ Expired items removed
- ☐ Bag clean and ready to deploy

INSPECTION RECORD

Date: _____

Inspector: _____

Call Sign: _____

Result:

- ☐ All Good
- ☐ Needs Attention
- ☐ Restock Required

Notes:

_____

_____

Be ready. Stay ready. Lives may depend on it.

NOTES

72-HOUR DEPLOYMENT RULE

Every operator should be prepared to operate independently for at least 72 hours whenever practical.

4.5 OPERATOR ACCOUNTABILITY

Accountability ensures the safety of our operators and the success of the mission.
Know who is deployed, where they are, what they are doing, and their status at all times.

PERSONNEL STATUS BOARD

AVAILABLE	ASSIGNED	EN ROUTE	UNAVAILABLE	LOST CONTACT
Available for assignment.	Currently assigned to a task.	Traveling to assignment location.	Not available for duty.	Unable to establish contact.

Use this status board to track and display the condition of all deployed operators.

ACCOUNTABILITY REQUIREMENTS

- [x] Check in upon arrival at assignment.
- [x] Report location changes immediately.
- [x] Report assignment changes.
- [x] Maintain radio monitoring on assigned channel.
- [x] Check out before leaving assignment.
- [x] Report when safely home.

Failure to follow accountability procedures puts you and your team at risk.

WELFARE CHECKS

NCS should conduct regular welfare checks to ensure operator safety and well-being.

- Hourly welfare checks
- Severe weather welfare checks
- Post-incident welfare checks
- Demobilization welfare checks

If an operator cannot be contacted during a scheduled welfare check, begin Lost Contact Procedure.

RESOURCE ACCOUNTABILITY

Track and account for all resources issued or deployed.

RADIOS	BATTERIES	VEHICLES	GO KITS	DIGITAL EQUIPMENT	PORTABLE REPEATERS
Make/Model, Serial, and Assignment.	Quantity, Type, and Condition.	Unit ID, Fuel Level, Location.	Issued To, Contents, Condition.	Laptops, Tablets, Hotspots, etc.	Location, Operator, Power Source.

All resources must be accounted for at check-in and check-out.

LOST CONTACT PROCEDURE

1. Attempt contact on primary frequency.
2. Attempt contact on alternate frequency.
3. Attempt telephone contact (if available).
4. Notify Net Control Station (NCS) and advise the Emergency Coordinator (EC).
5. Escalate as needed according to the incident and safety plan.

Do not assume an operator is safe if you cannot contact them. Report lost contact immediately.

ACCOUNTABILITY CYCLE

ACTIVATED → ASSIGNED → DEPLOYED → ACCOUNTED FOR → DEMOBILIZED → HOME SAFE

ACTIVATED	ASSIGNED	DEPLOYED	ACCOUNTED FOR	DEMOBILIZED	HOME SAFE
Operator placed on activation status.	Operator assigned to mission or tasking.	Operator en route or on-scene at assignment.	Operator checked in and status confirmed.	Operator released from assignment and returning.	Operator reports safely home and off duty.

Maintain accountability from activation through safe return home.

OPERATOR ACCOUNTABILITY ROSTER (EXAMPLE)

CALL SIGN	NAME	STATUS	ASSIGNMENT	LOCATION	CHECK-IN TIME	LAST CONTACT	NOTES
N1MTR	Alex Turner	Available	Standby	Home	—	—	Monitoring Net
K7PXY	Brandon Lee	Assigned	Logistics Section	EOC	0730	0745	Supply Run
W2QRP	Carla Jenkins	En Route	Shelter Net	En Route – Fairfield	0815	0820	ETA 0830
G4ZED	Derek Morgan	Unavailable	—	—	—	—	Off Duty
M7RFS	Ethan Walsh	Available	Standby	Home	—	—	On Call
XH9LMN	Fiona Grant	En Route	Damage Assessment	En Route – Teague	0825	0830	Team Lead
Q3VHF	Gabriel Ward	Assigned	Communications Unit	Command Post	0700	0825	ICS Comms
P5YTR	Hannah Price	Lost Contact	Field Ops	Last: FM 146.760	0630	0655	See Procedure

Keep this roster current. Update statuses immediately. Accountability saves lives.

4.6 SERVED AGENCY SUPPORT

We are here to support the mission of our served agencies.
Professionalism, teamwork, and communication are the keys to successful operations.

WHO WE SUPPORT

EMERGENCY MANAGEMENT	FIRE DEPARTMENTS	LAW ENFORCEMENT	HOSPITALS	SHELTERS	RED CROSS	GOVERNMENT AGENCIES
County, regional, and state EOCs.	Fire ground, support, and standby ops.	Traffic, search, security, and investigations.	Patient transport, operations, and emergency support.	Mass care, resident tracking, and logistics.	Disaster relief, communications, and logistics.	Public works, utilities, transportation, and more.

We augment — we do not replace. We follow their direction and support their objectives.

PROFESSIONAL CONDUCT

DO

- Wear identification at all times.
- Be professional and courteous.
- Follow all directions and procedures.
- Maintain confidentiality of information.
- Be punctual and prepared.
- Ask questions when unsure.
- Keep accurate logs and records.

DON'T

- Freelance or act on your own.
- Self-deploy to an incident.
- Bypass the chain of command.
- Discuss sensitive information.
- Argue with agency personnel.
- Speak to media unless authorized.
- NO PICTURES TO BE TAKEN WITHOUT PERMISSION FROM EC OR EMC.
- Use equipment or frequencies without authorization.

CHAIN OF COMMAND

SERVED AGENCY
(Incident Commander or Designee)
↓
FREESTONE COUNTY ARES EMERGENCY COORDINATOR (EC)
↓
NET CONTROL STATION (NCS)
↓
OPERATOR (Volunteer)

Always follow the chain of command. It ensures safety, clarity, and success.

REPRESENTING ARES

Remember:

You are representing:

- Freestone County ARES
- ARRL ARES
- Amateur Radio
- Freestone County

★ ★ ★

PROFESSIONAL APPEARANCE AND CONDUCT MATTER. YOU ARE AN AMBASSADOR FOR AMATEUR RADIO.

★ ★ ★

COMMUNICATIONS RESPONSIBILITIES

- Pass traffic accurately and completely.
- Maintain message logs and records.
- Use proper radio procedures and plain language.
- Protect sensitive and confidential information.
- Relay messages exactly as received. Do not add, delete, or change.

Clear, accurate communications save time, reduce errors, and save lives.

DOCUMENTATION

Maintain the following:

- **ICS-214 Activity Log** — Record all activities and events.
- **Message Forms (ICS-213)** — Document all formal messages.
- **Resource Tracking** — Track equipment and personnel.
- **Accountability Records** — Track personnel and assignments.
- **Check-In / Check-Out Records** — Maintain accurate logs at all times.

Good documentation supports the mission and protects everyone.

CONFLICT RESOLUTION

If problems or conflicts arise:

1. Work the issue through your immediate supervisor.
2. Notify the Net Control Station (NCS).
3. Notify the Emergency Coordinator (EC).
4. Document the issue factually.
5. Remain calm, professional, and mission-focused.

Professionalism and communication resolve most issues.

KEY POINT — We are a team with our served agencies. Your support, skills, and professionalism make the mission possible and the community stronger.

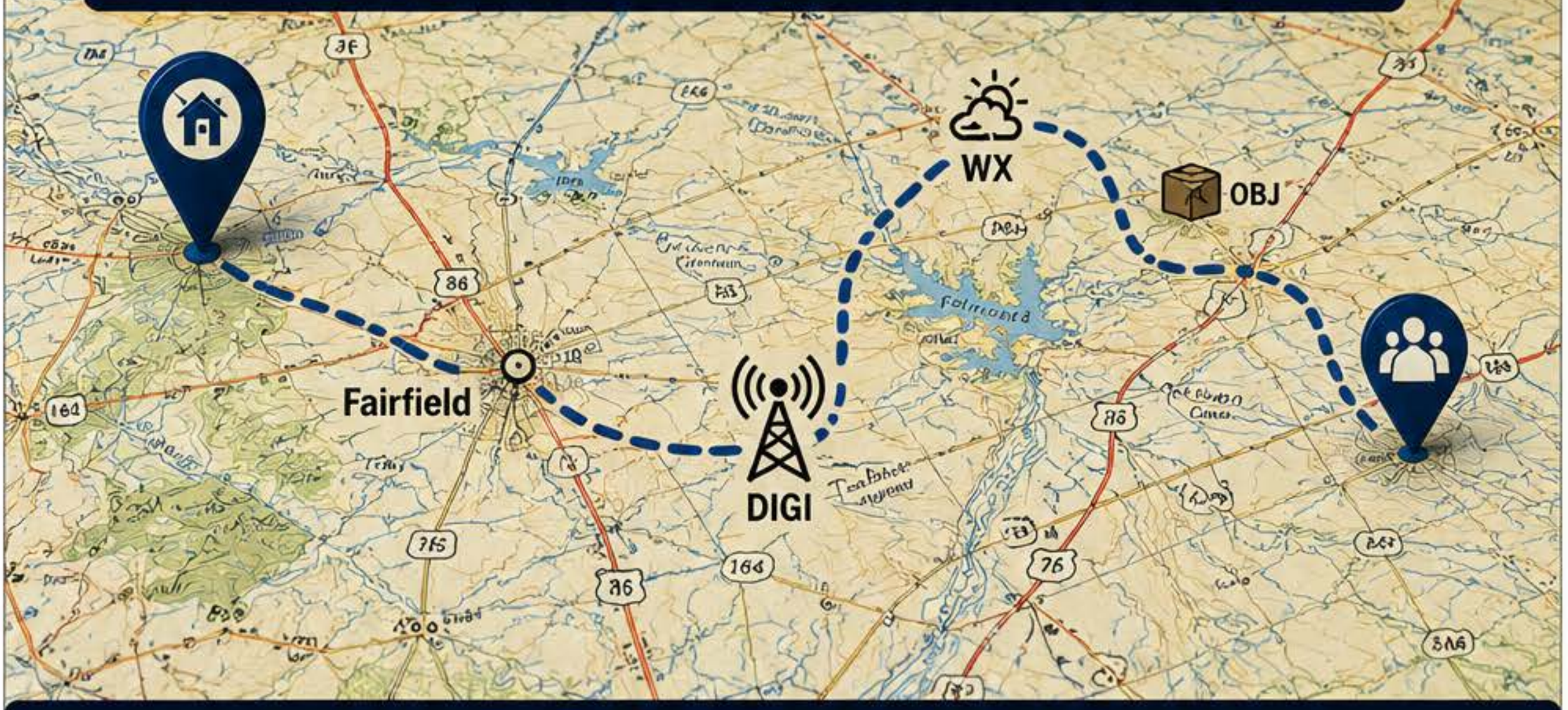
CHAPTER 5.0

APRS OPERATIONS

AUTOMATIC PACKET REPORTING SYSTEM

APRS provides real-time position reporting, short messaging, and tactical situational awareness for ARES operators.

Stay visible. Stay informed. Stay connected.



★ BETTER DATA • BETTER AWARENESS • BETTER OUTCOMES ★

APRS IS NOT PRIVATE. Do not transmit sensitive locations or information unless authorized by the served agency or Emergency Coordinator.

5.1 APRS OVERVIEW AND USES

APRS (Automatic Packet Reporting System) is a digital communication protocol used in the amateur radio service. It allows stations to send real-time data such as position, messages, objects, weather, and telemetry over RF and the internet.

WHAT APRS IS

- A real-time data communications system.
- Uses packet radio to transmit short data packets.
- Shares information between stations, digipeaters, iGates, and the internet.
- Works on VHF, UHF, and HF frequencies.
- Used worldwide by hams, emergency services, weather spotters, and public service groups.

APRS IS NOT PRIVATE

- APRS transmissions are open to everyone.
- Do NOT transmit sensitive information.
- Do NOT transmit your exact location unless necessary.
- Use good judgment and follow directives from your EC or the served agency.
- Location data can be seen by anyone using APRS websites or apps.

COMMON APRS USES

POSITION REPORTING	Share real-time location and track team members, vehicles, and assets.
SHORT MESSAGING	Send and receive short text messages between stations.
WEATHER REPORTS	Share weather observations from field stations to the network.
OBJECT TRACKING	Track assets, shelters, resources, and mobile command posts.
TELEMETRY DATA	Monitor remote sensors, battery levels, and other environmental data.

TACTICAL USES FOR ARES

- Monitor team locations during deployments.
- Track convoy movement and asset locations.
- Provide situational awareness to the EOC.
- Share field weather reports with net control.
- Mark routes, hazards, shelters, and resources.
- Send quick status updates and messages.

APRS SYMBOLS & SSIDS

identify what your station represents on the map.

EXAMPLE

KJ5ABC-7

CALLSIGN SSID

- Symbols show station type (vehicle, shelter, weather, mobile, etc.).
- SSID (-1, -7, -10, etc.) shows function or service role.
- Use the correct symbol and SSID for clarity.

COMMON APRS SSIDS

SSID	TYPICAL USE
-1	Fixed Station / Home
-2	Mobile Station
-3	Weather Station
-4	Mobile Weather
-5	Satellite / Space
-6	Boat / Maritime
-7	Emergency / ARES
-10	Digipeater / iGate
-11	Internet Gateway (iGate)

COMMON APRS FREQUENCIES

BAND	FREQUENCY	MODE
2 Meter	144.390 MHz	FM 1200
2 Meter	144.800 MHz	FM 1200
70 cm	440.775 MHz	FM 9600
70 cm	441.775 MHz	FM 9600
70 cm	442.600 MHz	FM 9600
HF (Winlink)	Various	Pactor / Winlink

Frequencies may vary by region. Always follow local band plans and directives from your EC.

GOOD APRS OPERATING PRACTICES

- Transmit only what is necessary and authorized.
- Use clear, accurate comments and status text.
- Keep your symbol and SSID updated.
- Monitor the frequency before transmitting.
- Respect other operators and network resources.
- Use appropriate beacon intervals.
- Stop tracking when not needed.
- Conserve battery power in the field.
- Know where your data is going (RF and Internet).
- Follow all ARRL and FCC rules.

SAFETY & PRIVACY WARNING

Do not transmit information that could compromise your safety, your team, or the mission.

When in doubt...
DON'T TRANSMIT IT!

5.2 APRS.FI SETUP AND UV-PRO CONNECTION

APRS.fi is an iOS application that displays live APRS data and can also connect to your radio. Follow the steps below to install, configure, and connect APRS.fi to your BTECH UV-PRO.

aprs.fi

1. DOWNLOAD APRS.FI

1. Open the App Store on your iPhone or iPad.
2. Search for "aprs.fi".
3. Install the official aprs.fi app.
4. Open the app.
5. Allow location permissions when prompted.

2. BASIC SETUP

1. Tap ☰ (Menu) > Settings.
2. Enter your Callsign (including SSID).
3. Set your Icon and Comment (optional).
4. Set Beaconing (only if authorized to transmit).
5. Adjust map and display options to your preference.
6. Verify your station appears correctly on the map.

3. PREPARE BTECH UV-PRO

1. Update radio firmware to the latest version.
2. Program APRS frequency (144.390 MHz).
3. Enable Bluetooth in the radio menu.
4. Enable KISS TNC mode: Menu > General Settings > KISS TNC > ON.
5. Set PTT Method to Bluetooth.
6. Ensure radio is on the APRS channel.

4. CONNECT APRS.FI TO UV-PRO

1 Pair Your Device with UV-PRO

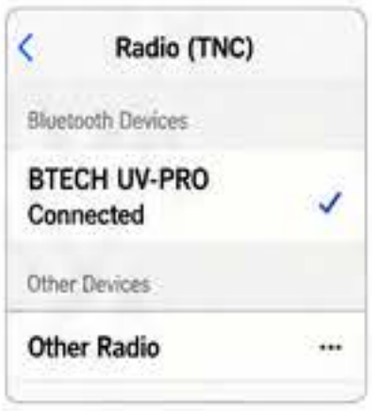
Go to iPhone Settings > Bluetooth. Select your UV-PRO from the available devices.

2 Select Radio/TNC Interface in App

In APRS.fi, go to ☰ Menu > Connection > Radio (TNC).

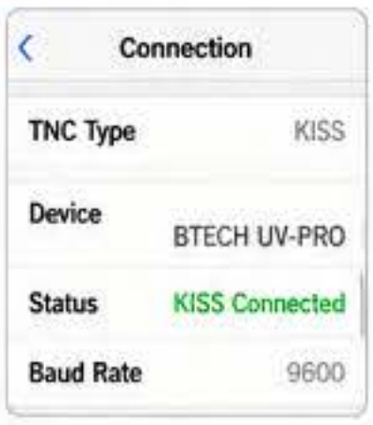
3 Choose Your Bluetooth Device

Select your UV-PRO from the device list. Ensure KISS is enabled.



4 Confirm KISS Connection

Status should show "KISS Connected" or similar.



5 Send/Receive Test Packet

Tap the transmit icon to send your position or a test message.

6 Verify on Map

Your station should appear on the APRS.fi map. Check received stations around you.

5. TROUBLESHOOTING

No Connection	Verify Bluetooth pairing and that KISS TNC is enabled.
No Packets Received	Confirm APRS frequency and squelch settings.
Cannot Transmit	Check PTT method (Bluetooth) and radio settings.
Garbage Data	Verify KISS mode, baud rate (9600), and clean audio.
No Station on Map	Confirm callsign/SSID and that you are beaconing (if authorized).
Multiple Apps Connected	Only one app can use the radio at a time via Bluetooth.

UV-PRO QUICK SETTINGS (SUMMARY)

- Frequency: 144.390 MHz (APRS)
- KISS TNC: ON (Menu > General Settings)
- PTT Method: Bluetooth
- Bluetooth: ON
- Mode: FM
- Power: As needed
- CTCSS/DCS: As programmed
- Baud Rate: 9600 (default)

Refer to your radio manual for full details on programming and menu navigation.

GOOD APRS PRACTICES

- Only transmit when authorized.
- Use appropriate beacon intervals (typically 5–10 minutes).
- Keep your comment short and meaningful.
- Use the correct symbol for your activity.
- Monitor before you transmit.
- Respect other operators and bandwidth.
- Do not transmit sensitive locations.
- Conserve battery power in the field.

APRS IS NOT PRIVATE. All APRS transmissions are public and can be viewed by anyone with an APRS receiver or through APRS websites and apps. Do not transmit sensitive information or exact locations unless authorized by your EC or the served agency.

5.3 APRSDROID SETUP AND UV-PRO CONNECTION

APRSdroid is an Android APRS application that reports your position, displays nearby stations, and supports APRS messaging. Follow the steps below to install, configure, and connect APRSdroid to your BTECH UV-PRO.

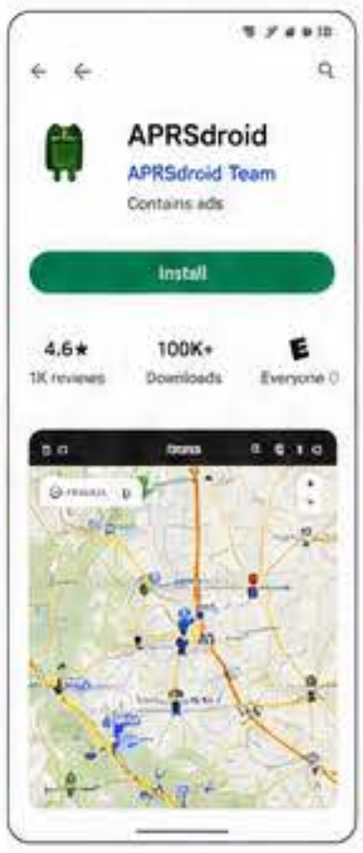
1. DOWNLOAD APRSDROID

1. Open Google Play Store on your Android device.
2. Search for "APRSdroid".
3. Install the official APRSdroid app.

— OR —

Download the latest APK from https://aprsdroid.org (APRSdroid-current.apk) if approved by your EC.

4. Open the app.
5. Grant location and storage permissions when prompted.



2. BASIC APRSDROID CONFIGURATION

1. Enter your Callsign (include SSID).
2. Enter your APRS-IS passcode (if using Internet features).
3. Set your Symbol (choose appropriate icon).
4. Set your Comment (status text).
5. Select Position Source (GPS / Network).
6. Set Beacon Interval (recommended every 5–10 min).
7. Choose Connection Protocol: KISS (TNC) for radio connection.
8. Save settings.

3. PREPARE BTECH UV-PRO

1. Update radio firmware to the latest version.
2. Program APRS frequency (144.390 MHz).
3. Enable Bluetooth in the radio menu.
4. Enable KISS TNC mode: Menu > General Settings > KISS TNC > ON.
5. Set PTT Method to Bluetooth.
6. Ensure radio is on the APRS channel.

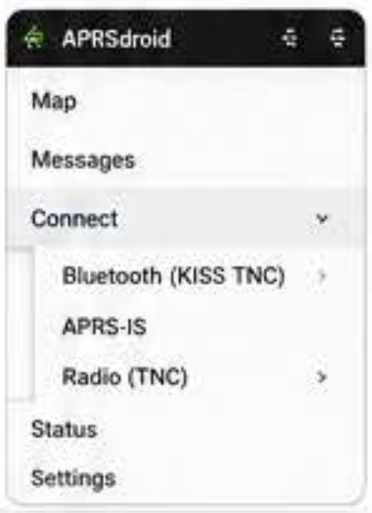
4. CONNECT APRSDROID TO UV-PRO

1 Pair Your Device with UV-PRO

Go to Android Settings > Bluetooth. Select your UV-PRO from the list of available devices.

2 Select KISS TNC Connection

In APRSdroid, tap Menu (≡) > Connect > Bluetooth (KISS TNC).



3 Choose Your Bluetooth Device

Select your UV-PRO from the device list. Ensure KISS is enabled.

4 Start Tracking

Tap the green "Start tracking" button. Status should show "Connected".

5 Verify Packets Transmit

Your position should be sent on the APRS frequency.

6 Confirm on Map or Receive Traffic

Your station should appear on APRS.fi or receive local APRS traffic.

5. TROUBLESHOOTING

No Connection	Verify Bluetooth pairing and that KISS TNC is enabled.
No Packets Received	Confirm APRS frequency, squelch, and antenna are correct.
Cannot Transmit	Check PTT method (Bluetooth) and radio settings.
No Station on Map	Confirm callsign/SSID and that you are beaconing.
Location Not Updating	Check GPS permission and position source in settings.
Multiple Apps Connected	Only one app can use the radio at a time via Bluetooth.

UV-PRO QUICK SETTINGS (SUMMARY)

- Frequency: 144.390 MHz (APRS)
- KISS TNC: ON (Menu > General Settings)
- PTT Method: Bluetooth
- Bluetooth: ON
- Mode: FM
- Power: As needed
- CTCSS/DCS: As programmed
- Baud Rate: 9600 (default)

Refer to your radio manual for full details on programming and menu navigation.

GOOD APRS PRACTICES

- Only transmit when authorized.
- Use appropriate beacon intervals (typically 5–10 minutes).
- Keep your comment short and meaningful.
- Use the correct symbol for your activity.
- Monitor before you transmit.
- Respect other operators and bandwidth.
- Do not transmit sensitive locations.
- Conserve battery power in the field.

APRS IS NOT PRIVATE. All APRS transmissions are public and can be viewed by anyone with an APRS receiver or through APRS websites and apps. Do not transmit sensitive information or exact locations unless authorized by your EC or the served agency.

5.4 YAAC APRS CLIENT – OVERVIEW AND USES

YAAC
Yet Another APRS Client

YAAC (Yet Another APRS Client) is a powerful, feature-rich APRS software application for Windows, macOS, and Linux. It allows operators to monitor APRS traffic, display stations on a map, send and receive messages, report position, and operate as an end-node or digipeater.

YAAC is a versatile tool for both field and base station APRS operations.

WHAT YAAC IS

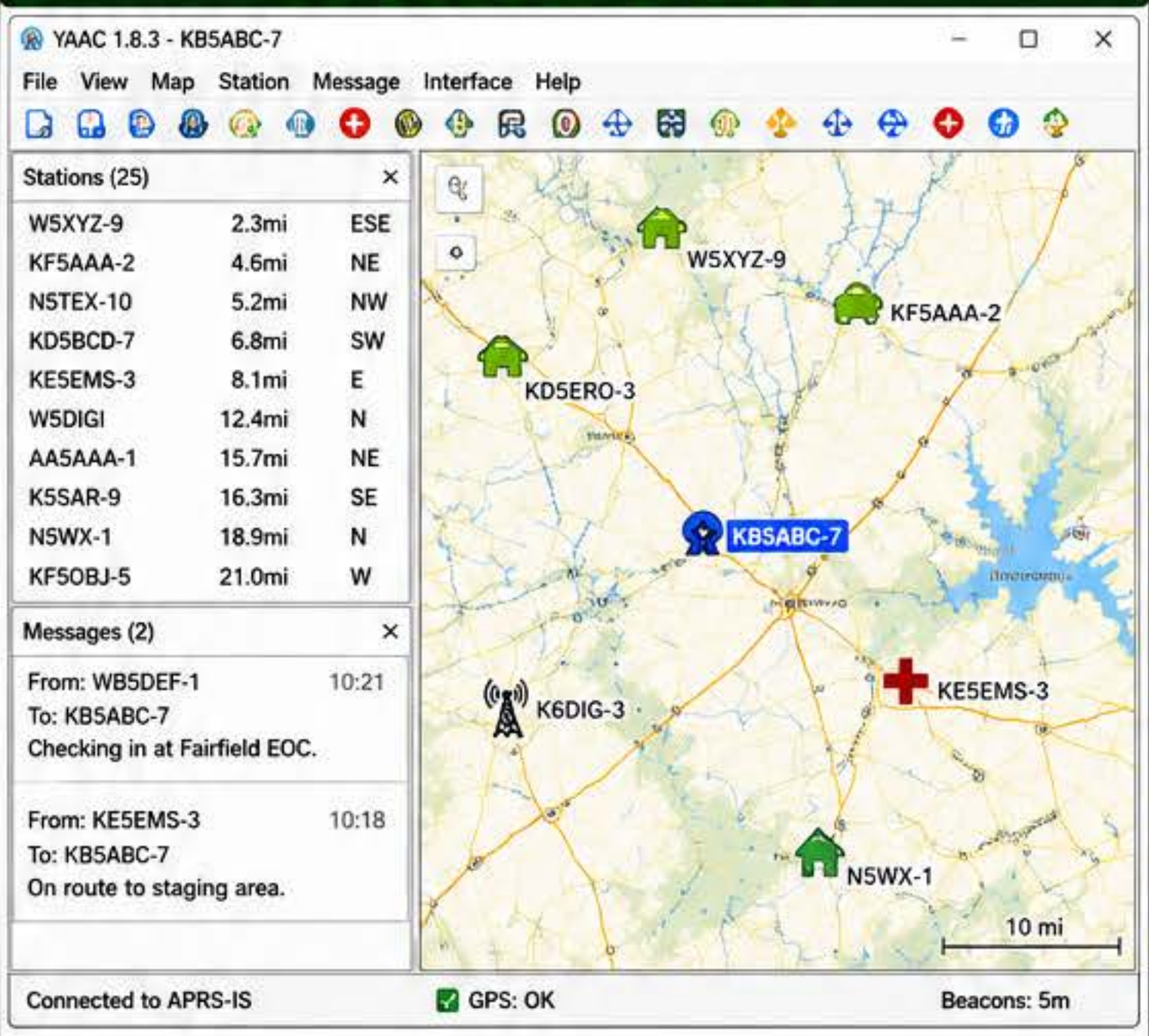
YAAC is a computer-based APRS client that connects to APRS networks via the internet (APRS-IS) or through a radio using a TNC interface.

- Free, open-source software
- Runs on Windows, macOS, and Linux
- Supports maps, messages, objects, weather, telemetry, and digipeating

WHAT YAAC IS USED FOR

- Monitor live APRS stations on a map
- Send and receive APRS messages
- Report your position and status
- Track units, assets, and teams
- View weather reports and objects
- Receive telemetry from sensors/devices
- Operate as an APRS digipeater / end-node

YAAC SCREENSHOT EXAMPLE



WHEN ARES WOULD USE YAAC

- At the EOC or command post for situational awareness
- During severe weather events or deployments
- For tracking field teams, vehicles, and assets
- Monitoring weather stations and objects
- Supporting public events and exercises
- Serving as an internet or RF digipeater

RECEIVE-ONLY VS TRANSMIT OPERATION

RECEIVE-ONLY (LISTENING)

- You only view APRS data
- No position reports transmitted
- No messages sent
- Good for monitoring situations without transmitting

TRANSMIT (ACTIVE)

- Your station transmits data
- Position, status, and messages may be sent
- Use only when authorized by your EC or agency

APRS PRIVACY AND OPSEC WARNING

- APRS transmissions are NOT private.
- Anyone can see your callsign, location, and messages on APRS networks and websites.
- Do NOT transmit sensitive locations, tactical plans, or personal information.
- Use proper judgement and follow directions from your EC or served agency.
- When in doubt... don't put it on APRS.

THINK BEFORE YOU TRANSMIT

PROTECT THE MISSION
PROTECT EACH OTHER

5.4 YAAC APRS CLIENT – SETUP AND OPERATION (2 OF 2)

1. DOWNLOAD AND INSTALL YAAC

1. Go to the official YAAC page: https://sourceforge.net/projects/yetanotheraprsclient/
2. Download the latest YAAC release for your operating system.
3. Confirm Java is installed (Java 8 or higher recommended).
4. Extract the downloaded ZIP file to a folder.
5. Run YAAC:
   - Windows: yaac.exe
   - macOS: yaac.command
   - Linux: ./yaac.sh
6. On first run, YAAC will create a configuration folder in your user profile.

2. BASIC YAAC SETUP

1. **Enter Callsign & SSID**
   Set your callsign and SSID (e.g., K5ABC-7).
2. **Set Station Symbol**
   Choose an appropriate symbol for your station.
3. **Set Station Comment**
   Enter a short comment or status text.
4. **Configure Beacons**
   Set beacon interval, path, and position source.
5. **Select Map / Display**
   Choose map type, zoom, and overlays.
6. **Click "Save" to apply your settings.**

QUICK TIP

Keep your comment short and meaningful.

Example:
"FC ARES EOC"
or "ARES Mobile 1"

Set beacon interval 5–15 minutes for field operations.

3. CONNECT YAAC TO APRS NETWORKS

YAAC can connect in multiple ways. Choose the method that best fits your equipment and mission.

APRS-IS (Internet)
Connects to APRS servers on the internet. Best for base station or internet-connected systems.

KISS TNC (Serial/USB)
Connects to a TNC (hardware or software) using KISS mode over serial or USB. Common for packet radios.

Bluetooth TNC/Radio
Connects wirelessly to a Bluetooth TNC or radio with KISS capability (e.g., BTECH UV-PRO in KISS mode).

Sound Card (AFSK)
Uses your computer's sound card to connect to a radio via audio (requires proper interface).

Go to: Connections tab > Add Connection > Choose your connection type and configure settings.

4. USING YAAC IN THE FIELD

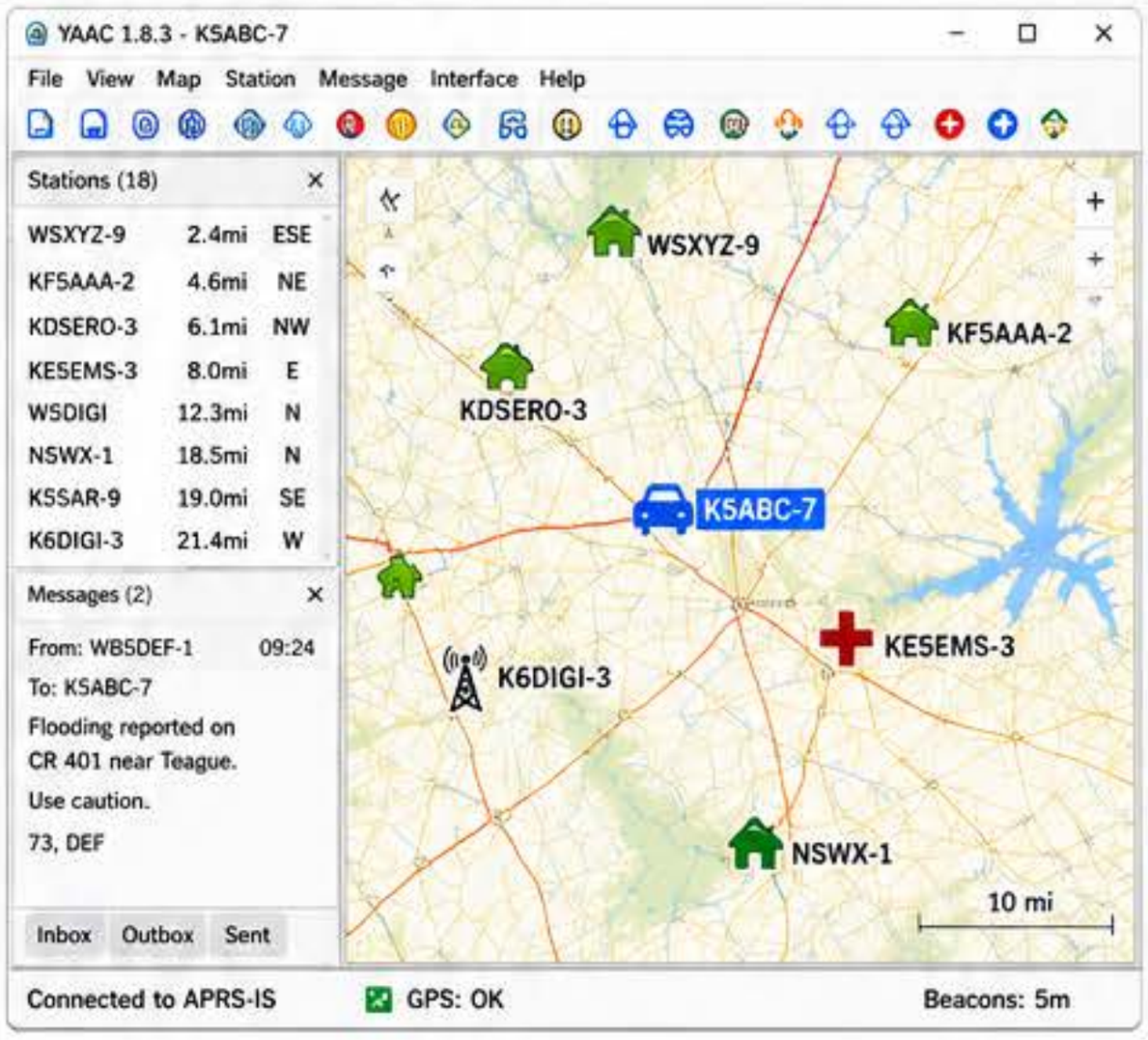
Monitor Stations
View APRS stations, vehicles, weather, digipeaters, and objects on the map.

Send Messages
Right-click a station > Send Message. Use message windows to send / receive texts.

Track & Report
Transmit your position and status. Track other team members and assets.

Objects & Weather
View weather reports, objects, and telemetry from sensors/devices.

Event / Deployment
Use YAAC at the EOC to maintain situational awareness and support field operations.



5. TROUBLESHOOTING COMMON ISSUES

ISSUE	POSSIBLE CAUSE	SOLUTION
YAAC will not start	Java not installed or outdated.	Install / update Java (8 or higher) and try again.
No map display	Map folder not set or files missing.	In Maps tab, verify map path and download maps if needed.
No stations visible	Not connected to any network.	Check connection settings (APRS-IS or TNC) and reconnect.
Cannot login to APRS-IS	Wrong callsign / passcode.	Verify callsign (no SSID) and APRS-IS passcode.
TNC not detected	COM port or device incorrect.	Check Device Manager / System Profiler and select correct port.
No packets over RF	Radio not in KISS mode / wrong frequency.	Enable KISS mode, verify APRS frequency, check antenna.
Internet connected but no packets	Firewall or router blocking APRS-IS traffic.	Allow outbound TCP port 14580 (or use different APRS-IS server).
YAAC stops transmitting	Battery saver / power settings.	Disable power saving for YAAC (especially on laptops).

HELPFUL YAAC RESOURCES

YAAC User Manual
Help > Online Manual (opens in your browser)

YAAC Project Page
https://sourceforge.net/projects/yetanotheraprsclient/

Support Forums
Ask questions and share solutions with other APRS users.

Practice at home before deploying in the field. Test your setup, maps, and connections.

APRS IS NOT PRIVATE. All APRS transmissions are public and can be viewed by anyone with an APRS receiver through APRS websites and apps. Do not transmit sensitive information or exact locations unless authorized by EC or the served agency.

5.5 YAAC APRS CLIENT – DOWNLOAD AND INSTALL YAAC

YAAC
Yet Another APRS Client

YAAC is a free, open-source APRS application that runs on Windows, macOS, and Linux. Follow the steps below to download, install, and launch YAAC on your computer. These steps apply to Windows, macOS and Linux steps are similar.

1. GO TO THE OFFICIAL YAAC PAGE

Open your web browser and go to the official YAAC SourceForge page:

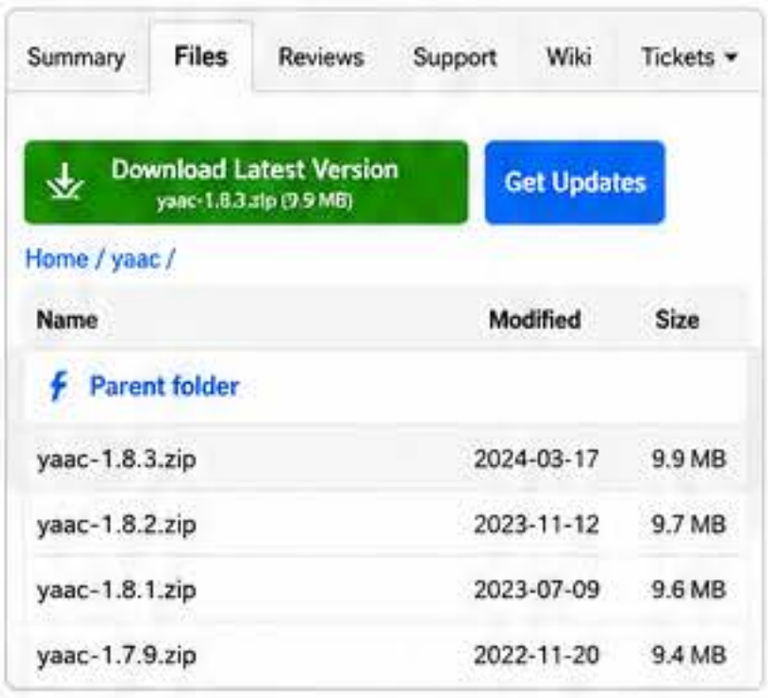
https://sourceforge.net/projects/yetanotheraprsclient/

Verify you are on the official SourceForge project page. Click the green Download button.

✓ You are now on the official YAAC SourceForge project page.

2. DOWNLOAD THE LATEST RELEASE

On the Files tab, download the latest release ZIP file for your operating system. (Windows example shown)



Click the most recent version (yaac-x.x.x.zip) to download the installer.

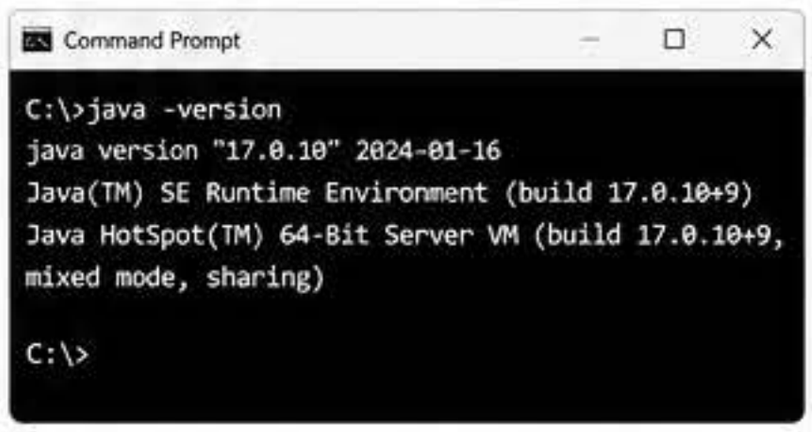
✓ The ZIP file will be downloaded to your computer.

3. CONFIRM JAVA IS INSTALLED

YAAC requires Java Runtime Environment (JRE) version 8 or higher.

To check:

- Windows: Open Command Prompt and type: `java -version`
- If Java is not installed, download it from https://www.java.com/en/download/ and install.

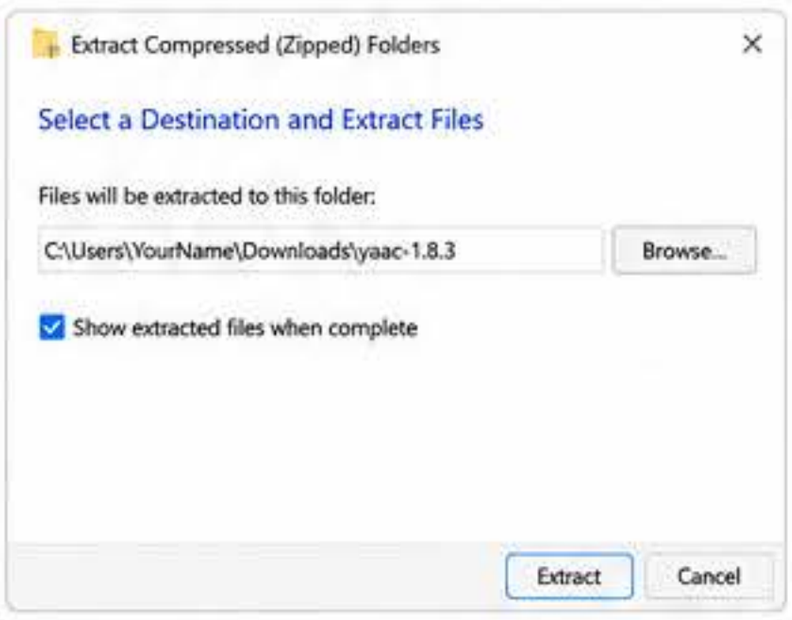


✓ You should see Java version 8 or higher. If not, install Java and reboot if needed.

4. EXTRACT THE DOWNLOAD ZIP FILE

Locate the downloaded ZIP file (e.g., yaac-1.8.3.zip) in your Downloads folder.

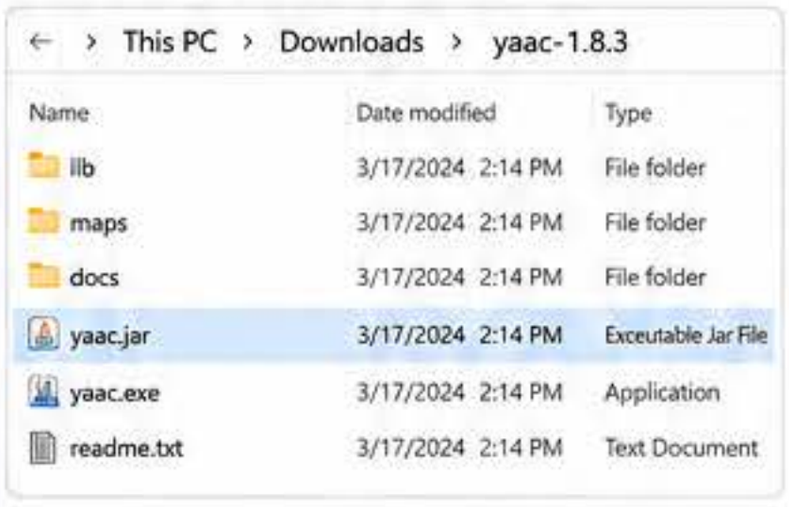
Right-click the file and select "Extract All..." Choose a destination folder and click Extract.



✓ A new folder named "yaac-1.8.3" (or similar) will be created with the YAAC program files.

5. LAUNCH YAAC

Open the extracted YAAC folder. Double-click "yaac.jar" to start YAAC.



If prompted by Windows Defender SmartScreen, click "More info" then "Run anyway". YAAC will launch.

✓ You should see the YAAC main window open with a map and toolbar.

6. VERIFY MAPS AND INTERFACE LOAD

The YAAC window should open with the map display and toolbar icons.

Make sure the map is visible and the status bar at the bottom shows "Not connected" or "Not logging in" (normal at this stage).

✓ YAAC is now installed and ready for configuration and connection.

SYSTEM REQUIREMENTS

- Windows 7/8/10/11, macOS, or Linux
- Java 8 or higher (Recommended: Java 11+)
- Internet connection (for APRS-IS)
- Serial port, USB, or Bluetooth for TNC/radio (if using RF)

mum 1 GB RAM, 200 MB disk space

INSTALL NOTES

- YAAC does not require a traditional installer. It runs from the extracted folder.
- You can create a desktop shortcut to yaac.jar for easy access.
- Keep the YAAC folder in a permanent location (e.g., C:\YAAC).

NEXT STEP

Proceed to 5.6 Basic YAAC Setup
to configure your callsign, symbol, beacons, and connection settings.

5.6 YAAC APRS CLIENT – BASIC YAAC SETUP

YAAC
Yet Another APRS Client

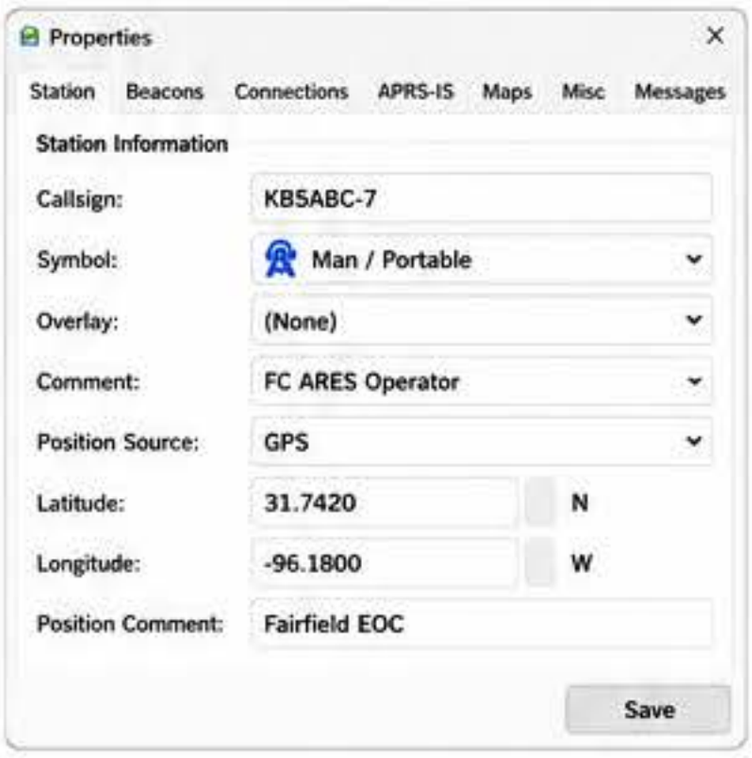
Configure YAAC with your station information, display preferences, and beacon settings. These basic settings are required before connecting to APRS networks or a radio.

Access these settings in YAAC: Station > Properties (or press Ctrl+P)

1. ENTER CALLSIGN & SSID

Enter your amateur radio callsign and SSID.

1. In the Properties window, go to the Station tab.
2. Enter your Callsign including SSID (example: KB5ABC-7).



TIP: Always include your SSID. Example: KB5ABC-7

2. SET STATION SYMBOL

Choose an appropriate APRS symbol for your station type and activity.

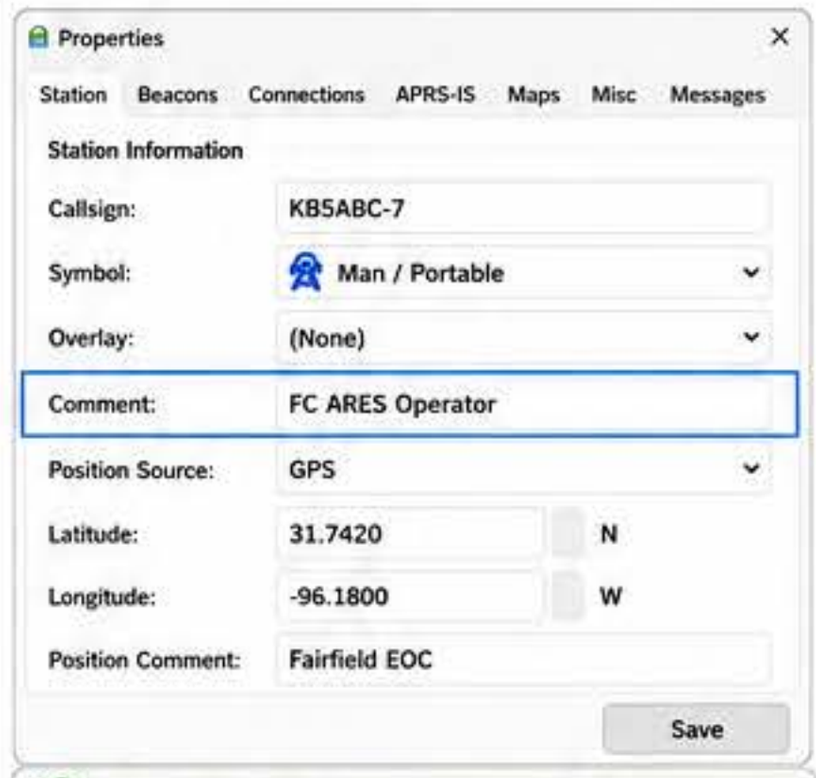
1. Click the Symbol drop-down list.
2. Select a symbol that represents your station (e.g., Man / Portable, Car, Base).

TIP: Use a symbol that clearly identifies your station and current activity.

3. SET STATION COMMENT

Enter a short comment that identifies you, your station, or your location.

1. In the Comment field, enter a short description.
2. Examples: "FC ARES Operator", "Fairfield EOC", "ARES Mobile 1".

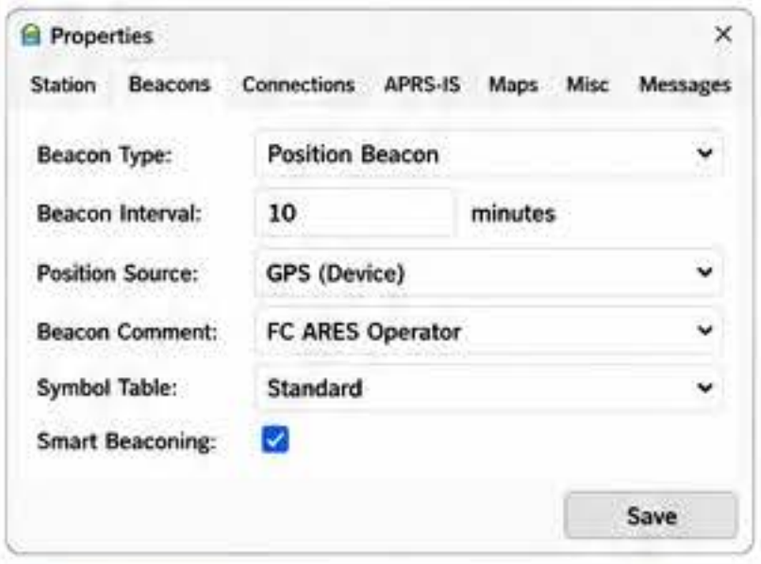


TIP: Keep it short and meaningful. This appears to all APRS users.

4. CONFIGURE BEACON SETTINGS

Set how often your position is transmitted (beaconed).

1. Go to the Beacons tab.
2. Set Beacon Interval (recommended 5–15 min).
3. Choose Position Source (GPS or Manual).

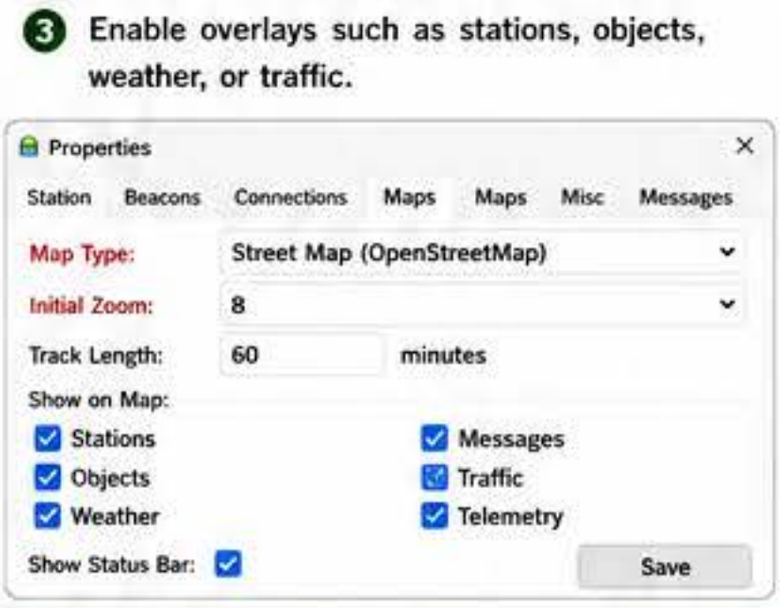


TIP: 5–15 minutes is a good beacon interval for most ARES field operations.

5. SELECT MAP / DISPLAY PREFERENCES

Customize map appearance, overlays, and object display to suit your needs.

1. Go to the Maps tab.
2. Choose your map type and zoom behavior.
3. Enable overlays such as stations, objects, weather, or traffic.



TIP: Enable "Stations" and "Objects" to see activity around you.

6. SAVE CONFIGURATION

Save your settings so YAAC will use them each time it starts.

1. Click the Save button in the Properties window.
2. Settings are stored and will be loaded automatically on startup.

Settings Saved!
Your basic YAAC setup is complete.

TIP: You can export your settings in File > Export Settings for backup.

BASIC SETUP SUMMARY

- Callsign & SSID entered
- Station symbol selected
- Station comment set
- Beacon interval configured
- Map and display preferences set
- Settings saved

IMPORTANT NOTES

- You must complete this basic setup before connecting to APRS-IS or a radio.
- You can change these settings anytime in Station > Properties.
- Accurate setup ensures your station is represented correctly on APRS.

5.7 YAAC APRS CLIENT – CONNECTING YAAC TO APRS

YAAC
Yet Another APRS Client

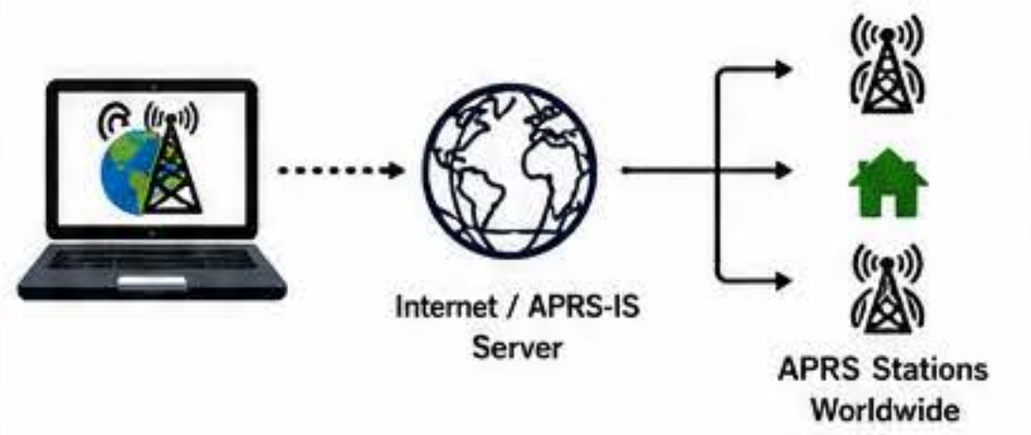
YAAC can connect to APRS networks and radios in several ways. Choose the method that best fits your equipment, mission, and operating environment.

All connection settings are configured in: Connections > Add Connection

1. APRS-IS (INTERNET CONNECTION)

Connect YAAC to the APRS-IS network to send and receive APRS data over the internet.

1. Open Connections tab and click "Add Connection".
2. Select "APRS-IS (Internet)".
3. Enter server: rotate.aprs2.net (recommended).
4. Enter your callsign-SSID and APRS passcode.
5. Check "Automatically reconnect".
6. Click Connect.

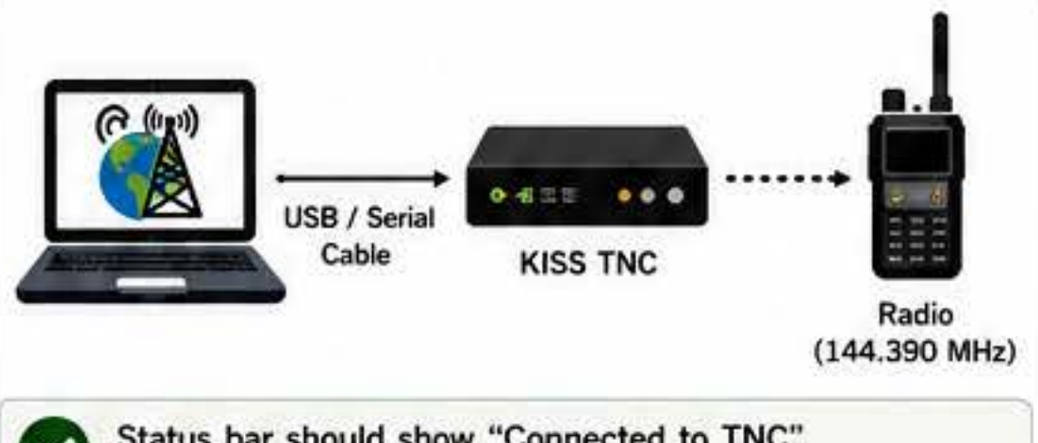


Status bar should show "Connected to APRS-IS".
You are now receiving APRS data.

2. KISS TNC (SERIAL OR USB)

Use a KISS TNC to connect YAAC to your radio for full APRS transmit and receive.

1. Connect your TNC to the computer via USB or Serial.
2. Open Connections tab and click "Add Connection".
3. Select "TNC (KISS)".
4. Choose the correct COM port.
5. Set baud rate (usually 9600 for most TNCs).
6. Click Connect.

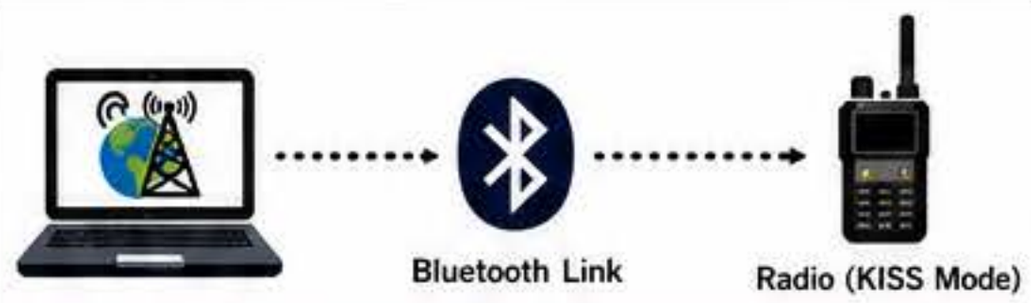


Status bar should show "Connected to TNC".
APRS packets will now flow over RF.

3. BLUETOOTH TNC / RADIO

Use a Bluetooth TNC or radio that supports KISS mode (e.g., BTECH UV-PRO in KISS).

1. Pair your Bluetooth TNC/radio with your computer.
2. Open Connections tab and click "Add Connection".
3. Select "Bluetooth (KISS)".
4. Choose your Bluetooth device.
5. Click Connect.

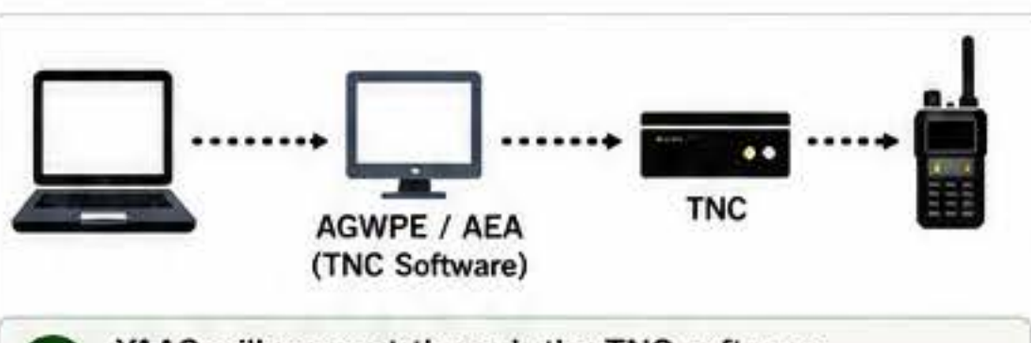


Status bar should show "Connected".
Ensure your radio is in KISS (TNC) mode.

4. SERIAL TNC (NON-KISS, AGW, OR AEA)

If using a TNC that requires a sound card or AGW/AEA connection.

1. Connect TNC audio to sound card.
2. Install and configure AGWPE or AEA TNC software.
3. In YAAC, select "AGWPE" or "AEA" connection type.
4. Enter host (127.0.0.1) and port (as configured).
5. Click Connect.



YAAC will connect through the TNC software to your radio.

5. SOUNDCARD INTERFACE (AFSK)

Use a sound card to connect YAAC to a radio using AFSK audio.

1. Connect radio audio to computer sound card.
2. Open Connections tab and click "Add Connection".
3. Select "AFSK (Sound Card)".
4. Choose input and output audio devices.
5. Click Connect.

Verify audio levels and VOX/PTT settings.
Monitor RF channel for APRS packets.

CONNECTION TIPS

- Ensure your callsign-SSID and passcode are correct for APRS-IS.
- Use Rotate Servers: rotate.aprs2.net / rotate.aprs.net.
- Enable "Reconnect" to maintain continuous connectivity.
- Monitor the status bar for connection and packet activity.
- [illegible]ceive-only (APRS-IS) to save battery.

STATUS BAR INDICATORS

Connected	—	Connection is active and working.
Connecting	—	YAAC is attempting to connect.
Disconnected	—	No active connection.
Tx / Rx	—	Packets are being transmitted / received.

SECURITY & OPSEC REMINDER

APRS is not private.
Do not transmit sensitive information, exact locations, or tactical plans on APRS.

5.8 YAAC APRS CLIENT – FIELD USE

YAAC
Yet Another APRS Client

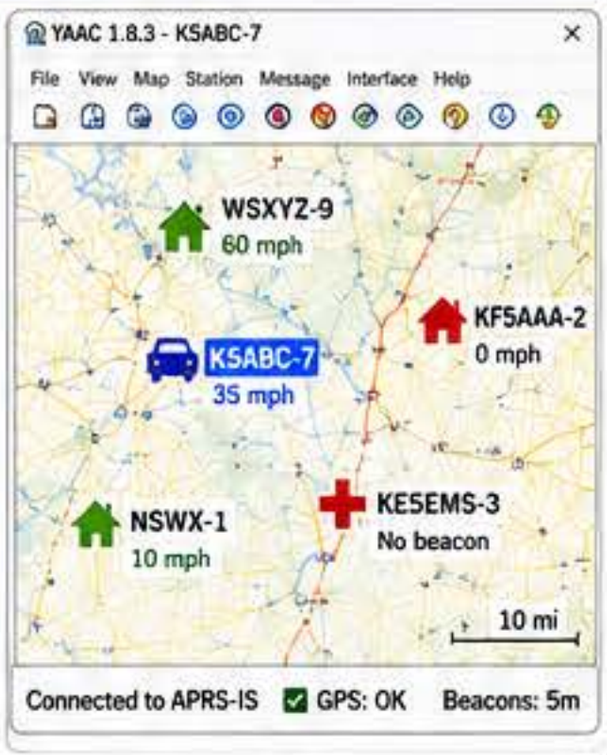
YAAC is a powerful tool for situational awareness, team tracking, messaging, and information sharing on the APRS network.

Use YAAC to support mission operations, coordination, and safety.

1. TRACK OPERATORS AND ASSETS

YAAC displays stations on a live map so you can see the location of team members, vehicles, and assets.

- View real-time position reports
- Track movement and routes
- Identify off-route or stopped units
- Monitor beacon status
- Great for events, deployments, and search operations



Colorful symbols and labels make it easy to identify stations.

2. SEND AND RECEIVE APRS MESSAGES

Exchange short text messages with other APRS stations.

- Send messages to individuals or groups
- Receive messages in real time
- Use for status updates, tasking, and coordination
- Messages are stored in the message window

Message - K5ABC-7

Inbox | Outbox | Sent

From	To	Message	Time
WSXYZ-9	K5ABC-7	Enroute to staging area	09:12
KF5AAA-2	K5ABC-7	On scene, all clear	09:18
K5ABC-7	KE5EMS-3	Need medical update	09:21
KE5EMS-3	K5ABC-7	2 patients, stable	09:23
NSWX-1	K5ABC-7	Tracking supply unit	09:27

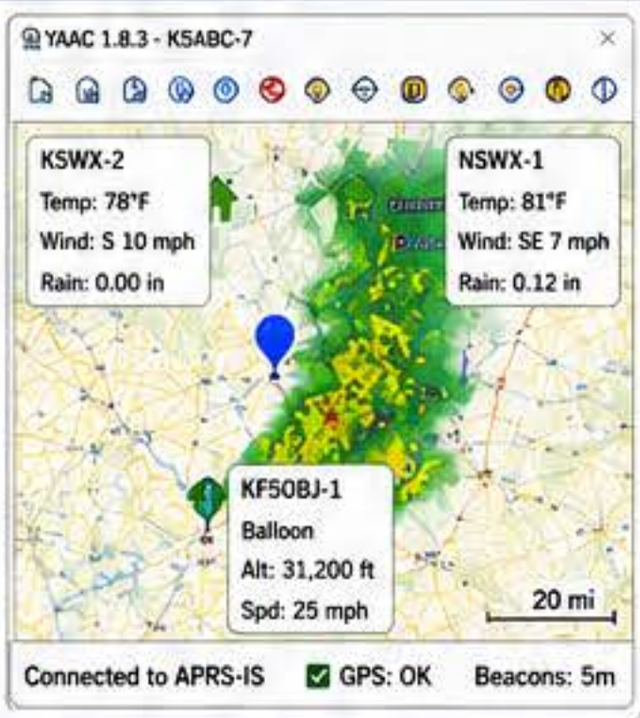
New Message | Reply | Forward | Delete

APRS messaging is ideal for quick, reliable communication when voice is not available.

3. MONITOR WEATHER AND OBJECTS

YAAC can display weather stations, objects, and other useful APRS information.

- View WX stations for real-time weather data
- Monitor NWS weather alerts (APRS)
- Track APRS objects (balloons, digipeaters, mobiles, etc.)
- Overlay information on your map



Weather and object data enhance situational awareness in the field.

4. SUPPORT EVENTS AND DEPLOYMENTS

YAAC is excellent for planned events and emergency deployments.

- Track teams, vehicles, and assets
- Monitor checkpoints and staging areas
- Share position and status information
- Improve safety and accountability

Commonly used for parades, bike rides, races, ARES nets, and disaster response.

5. SITUATIONAL AWARENESS AT THE EOC

At the EOC or command post, YAAC provides a big-picture view of the operation.

- Monitor all field units in real time
- Identify problem areas quickly
- Coordinate resources
- Share situational updates
- Provide accurate tracking to leadership

YAAC gives the EOC a clear, real-time view of the entire operation.

6. BEST PRACTICES FOR FIELD USE

- Keep your beacon interval appropriate (5–15 minutes recommended).
- Use clear station comments.
- Monitor your battery and power.
- Confirm GPS lock for accurate positions.
- Maintain radio discipline and follow net protocols.
- Protect sensitive information – APRS is not private.

★ REMEMBER

YAAC is a force multiplier for ARES operations.

Good information leads to good decisions.

Stay connected.
Stay informed.
Stay effective.

CONNECTION TIPS

- Ensure your callsign-SSID and passcode are correct for APRS-IS.
- Use Rotate Servers: rotate.aprs2.net / rotate.aprs.net.
- Enable "Reconnect" to maintain continuous connectivity.
- Monitor the status bar for connection and packet activity.
- Low power? Use receive-only (APRS-IS) to save battery.

STATUS BAR INDICATORS

- Connected — Connection is active and working.
- Connecting — YAAC is attempting to connect.
- Disconnected — No active connection.
- Tx / Rx — Packets are being transmitted / received.

SECURITY & OPSEC REMINDER

APRS is not private.

Do not transmit sensitive information, exact locations, or tactical plans on APRS.

5.9 YAAC APRS CLIENT – TROUBLESHOOTING

YAAC
Yet Another APRS Client

Even with correct setup, issues can occur. Use this guide to quickly identify and resolve common YAAC problems in the field.

Check settings, connections, and the basics first.

	PROBLEM	POSSIBLE CAUSES	SOLUTIONS	CHECK / NOTES
1	**YAAC will not start**	• Java not installed or outdated • Corrupt YAAC installation	• Install Java 8 or higher (recommended 8+) • Restart computer • Re-download and extract YAAC	Verify Java by opening Command Prompt and typing: `java -version`
2	**No map display**	• Map files not downloaded • Incorrect map path • No internet connection	• In YAAC, go to Maps tab • Set map path to correct folder • Download maps (File > Download Maps) • Verify internet connection	Street maps (OpenStreetMap) recommended for field use.
3	**No stations visible**	• Not connected to APRS-IS or TNC • Incorrect callsign/SSID • Filter settings hiding stations	• Check connection status • Verify callsign-SSID is correct • In YAAC: Interface > Filter, adjust filters or clear to default	Status bar should show "Connected" or "Connected to APRS-IS/TNC".
4	**Cannot login to APRS-IS**	• Wrong callsign or SSID • Incorrect passcode • Server down or unreachable	• Verify callsign-SSID (no spaces) • Verify passcode (not your callsign!) • Check server status at: https://aprs-is.net • Try alternate server if needed	Passcode format: CALLSIGN-10 Example: KB5ABC-10 Not: KB5ABC (wrong)
5	**TNC not detected**	• Wrong COM port • Driver not installed • TNC not powered on	• Check Device Manager for COM port • Select correct port in YAAC • Install correct drivers • Power cycle TNC	Use a known-good cable and verify LEDs on TNC.
6	**No packets over RF (KISS TNC)**	• KISS mode not enabled • Incorrect baud rate • Radio frequency wrong • No RF path or antenna issue	• Enable KISS mode in TNC • Set baud rate to 9600 (usually) • Verify radio frequency matches • Check antenna/SWR and repeater access	Most KISS TNCs: Speed: 9600 Data bits: 8 Stop bits: 1 Parity: None
7	**Internet connected but not to APRS-IS**	• Firewall blocking connection • Wrong port (14580) • Proxy or network restriction	• Allow outbound TCP port 14580 • Verify port in connection settings • Check with network admin if needed	APRS-IS uses TCP port 14580. Outbound connections only.
8	**Low battery or power issues**	• Laptop battery saver mode • Radio not getting enough power • USB power limitations	• Disable battery saver for YAAC • Use external power for radio/TNC • Use powered USB hub if needed	Field operations require reliable power. Monitor voltage regularly.
9	**Messages not received or sent**	• Outbox not connected • Message filter blocking • TNC/Internet not connected	• Verify connection is active • Check Inbox/Outbox tabs • Clear or adjust filters • Send test message to yourself	Outbox messages will queue until connection is restored.
10	**No GPS position**	• GPS not connected • Wrong position source • GPS no fix	• Check GPS connection and port • Select GPS as position source • Wait for GPS lock (outside if needed)	Status bar should show: GPS: OK or GPS: No Fix

QUICK DIAGNOSTIC CHECKLIST

- [x] 1. Power on and check all cables/connections.
- [x] 2. Verify correct callsign-SSID and passcode.
- [x] 3. Check connection status in YAAC.
- [x] 4. Confirm map display and GPS status.
- [x] 5. Monitor APRS packets on RF or Internet.
- [x] 6. Send/receive a test message.
- [ ] Save working configuration.

USEFUL YAAC TOOLS

Connections Tab
Manage all connection types (APRS-IS, TNC, Bluetooth, etc.)

Maps Tab
Download maps and configure display options.

Messages Tab
Send, receive, and manage APRS messages.

Properties (Ctrl+P)
Access all station settings quickly.

GENERAL NOTES

- Start with the basics: power, cables, and correct settings.
- Work changes one step at a time.
- Keep a backup of your working configuration (File > Export Settings).
- When in doubt, test with another station or known-good setup.

★ If you cannot resolve an issue in the field, contact your Net Control or experienced operator for assistance.

FREESTONE COUNTY ARES

APRS COMMANDS

Action	To:	Message
Send msg to SMS	SMS	@<number> <message>
Send SMS to APRS	Gtwy #	@<callsign> <message>
Add new alias	SMS	#alias #add <alias> <number>
Remove alias	SMS	#alias #remove alias <number>

Receive weather	WXBOT	[where]	[when]	[what]
		HAM callsign AIS callsign City,ST 5-dig zip code CWOP station ID 4/6-dig maidenhead Lat/lon (decimal degs)	today tonight tomorrow Monday, Tuesday, etc. morning, afternoon, night (i.e. Monday night, Friday afternoon, etc.)	full brief current metar taf
		4-dig ICAO	METAR ONLY	full
		3-dig IATA	METAR ONLY	brief

Query Winlink msgs	WLNK-1	
Query specific msg		R[message number]
Reply to specific msg		Y[message number]
Delete a specific msg		K[message number]
Forward a specific msg		F[message number] [email address]
Send long msg		SP [email, callsign, alias] [subject] *send* [text] *send* [text] [repeat as needed] /EX
Send one-line msg		SMS [callsign] [text]
Add alias		A [alias]=[email for alias]
Delete alias		A [alias]=[leave blank]
List aliases		AL
Query next RMS packet gateway		G[number] (where number is amount of nearby stations)
Show all available WLNK-1 commands		H
Help with specific cmd		?[cmd]

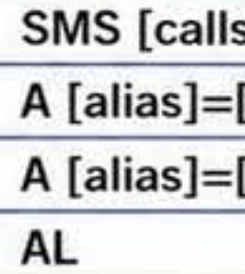
BEST PRACTICES

- Keep messages short and clear.
- Use standard addresses and formats.
- Verify service availability with your gateway operator.
- Follow FCC rules and local net procedures.

EXAMPLE

TO: MPAD

bakery



NOTE

Service availability varies. Check with your gateway operator.

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USE RESPONSIBLY. THINK BEFORE YOU TRANSMIT.

APRS COMMANDS

FAVORITE/NEW/CUSTOM APRS COMMANDS

Action	To:	Message

Notes:

TIP:

Use this section to build a personal list of favorite, new, or custom APRS commands.

EXAMPLE:

TO: MPAD

bakery

NOTE:

Service availability varies. Check with your gateway operator.

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APRS COMMANDS

APRS COMMANDS

Act	Message	TO: MPAD			
Open Street Maps data	OSM [category]	aerodrome	Airport	hostel	Hostel
		aerodrome	Airport	hostel	Hostel
		alpine_hut	Alpine Hut	hotel	Hotel
		ambulance_station	Ambulance	information	Information
		atm	ATM	laundry	Laundry
		bakery	Bakery	mall	Shop mall
		bank	Bank	motel	Motel
		butcher	Butcher	motorcycle	Motorcycle
		car_rental	Car Rental	optician	Optician
		car_repair	Car Repair	pharmacy	Pharmacy
		charging_station	Charge Stn	phone	Phone
		chemist	Pharmacy	photographer	Photographer
		clinic	Clinic/hospital	police	Police Office
		college	College	post_box	Post Box
		deli	Deli	post_office	Post Office
		dentist	Dentist	pub	Pub / Bar
		department_store	Dept Store	shoes	Shoes
		doctors	Doctors	subway	Subway Stn
		drinking_water	Drinking Wtr	supermarket	Supermarket
		dry_cleaning	Dry Cleaning	taxi	Taxi
		electronics	Electr Store	telephone	Telephone
		fire_station	Fire Station	tobacco	Tobacco
		fuel	Gas Station	toilets	Toilets
		guest_house	Guest house	train_station	Train Station
		hairdresser	Hairdresser	veterinary	Veterinary
		hospital	Hospital	university	University

Action	Message	Add'l Info
Sunrise/sunset Moonrise/moonset	riseset (based on pos) riseset [callsign][-ssid] riseset [date/daytime var]	sunrs – sunrise/sunset mn_sr – moonrise/moonset
Satellite passes	satpass [satellite name] vispass [satellite name]	+ [daytime setting]
Satellite freqs	satfreq [satellite name]	
Repeater data	repeater [band] [mode] +top2/3/4/5 (provides nearest 2/3/4/5)	70cm, 2m, etc. fm, dstar, d-star, dmr, c4fm, ysf, tetra, atv
Send msg to DAPNET	dapnet [user] [text]	
Send pos via email	posmsg [email]	
Radiosonde landing	sonde [callsign]	(includes lat, long, alt, launch time, asct rt, desct rt, burst alt)
General help	Info, help	
Set metric/imperial	metric, imperial	
Set language	Lang [2-dig language]	(enter any iso936-1 code, i.e. en, fr, hi, ja)
Fortuneteller	fortuneteller	

BEST PRACTICES

- Keep messages short and clear.
- Use standard addresses and formats.
- Verify service availability with your gateway operator.
- Follow FCC rules and local net procedures.

EXAMPLE

TO: MPAD

bakery

NOTE

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USE RESPONSIBLY. THINK BEFORE YOU TRANSMIT.

FREESTONE COUNTY ARES

APRS COMMANDS

WX DATA INQUIRY

Command	Description
WX	Summary of all available weather data
WXTEMP	Temperatures
WXWINDS	Winds
WXHUM	Humidity
WXRAIN	Rainfall
WXUV	UV Index
WXSOL	Solar Radiation
WXTHW	Temp/Humid/Wind summary
WXALERTS	Weather Alerts
WXBULLETINS	NOAA Weather Bulletins
WXNWS	NWS Forecast (your area)
WXCWOP	CWOP station list
WXSTAT	Station statistics
WXHELP	Weather command help

POSITION INQUIRY

Command	Description
P	Your current position
P<call sign>	Position of a specific station
P<SSID>	Position of a specific SSID
P<digipeater>	Position of a digipeater
PBEACONS	List of active beacons
PNEAR	Stations near you
PHERE	Who is hearing me?
PTHERE	Who am I hearing?
PINFO	Detailed station info
PREPORT	Your last position report
PHELP	Position command help

OTHER USEFUL QUERIES

TIME	Current local time
DATE	Current local date
VER	APRS software/version info
NODES	List of known nodes
PORTS	List of known ports
MHEARD	Most heard stations
H	List of available commands
HELP	General help information

MPAD WEATHER QUERIES

Command	Description	Example
wx <location>	Current conditions	wx freestone county
wx <location> forecast	Forecast	wx fairfield forecast
wx <location> alerts	Active alerts	wx freestone alerts
wx <location> radar	Radar	wx freestone radar
wx <location> satellite	Satellite imagery	wx texas satellite
wx <location> winds	Wind conditions	wx fairfield winds
wx <location> temp	Temperature	wx freestone temp
wx <location> rain	Rainfall	wx fairfield rain
wx <location> humidity	Humidity	wx freestone humidity
wx <location> uv	UV index	wx fairfield uv
wx <location> fire	Fire weather	wx freestone fire
wx <location> marine	Marine forecast	wx lake palestine marine
wx help	MPAD weather help	wx help

BEST PRACTICES

- Keep messages short and clear.
- Use standard addresses and formats.
- Verify service availability with your gateway operator.
- Follow FCC rules and local net procedures.

EXAMPLE

TO: MPAD

wx freestone county

NOTE

Service availability varies. Check with your gateway operator.

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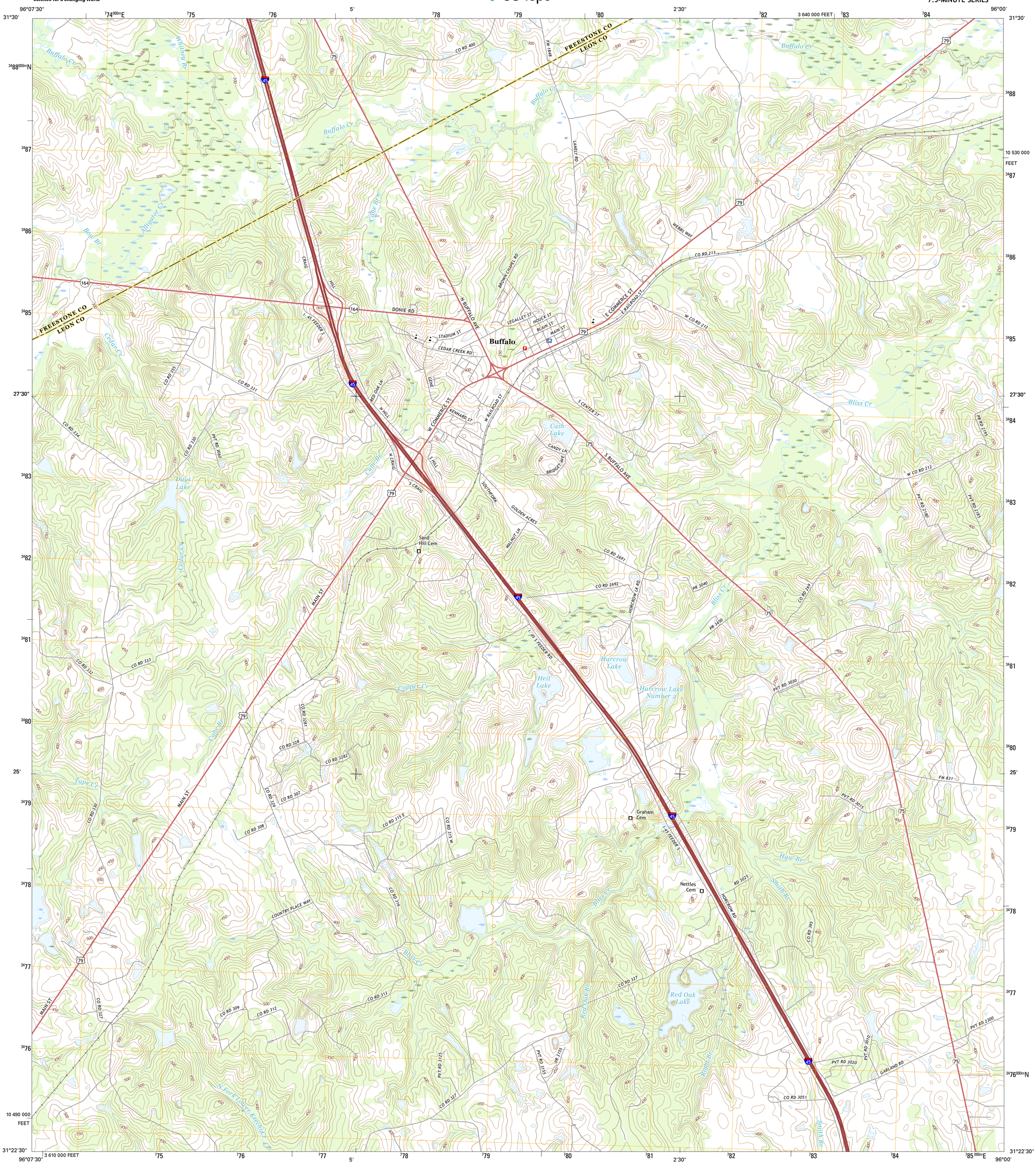
USE RESPONSIBLY. THINK BEFORE YOU TRANSMIT.

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

BUFFALO QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 5' 55 MILS
GN 1° 32' 27 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QV
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Dew
2 Lanely
3 Keechi
4 Donie
5 Flo
6 Jewett
7 Robbins
8 Centerville
ADJOINING QUADRANGLES

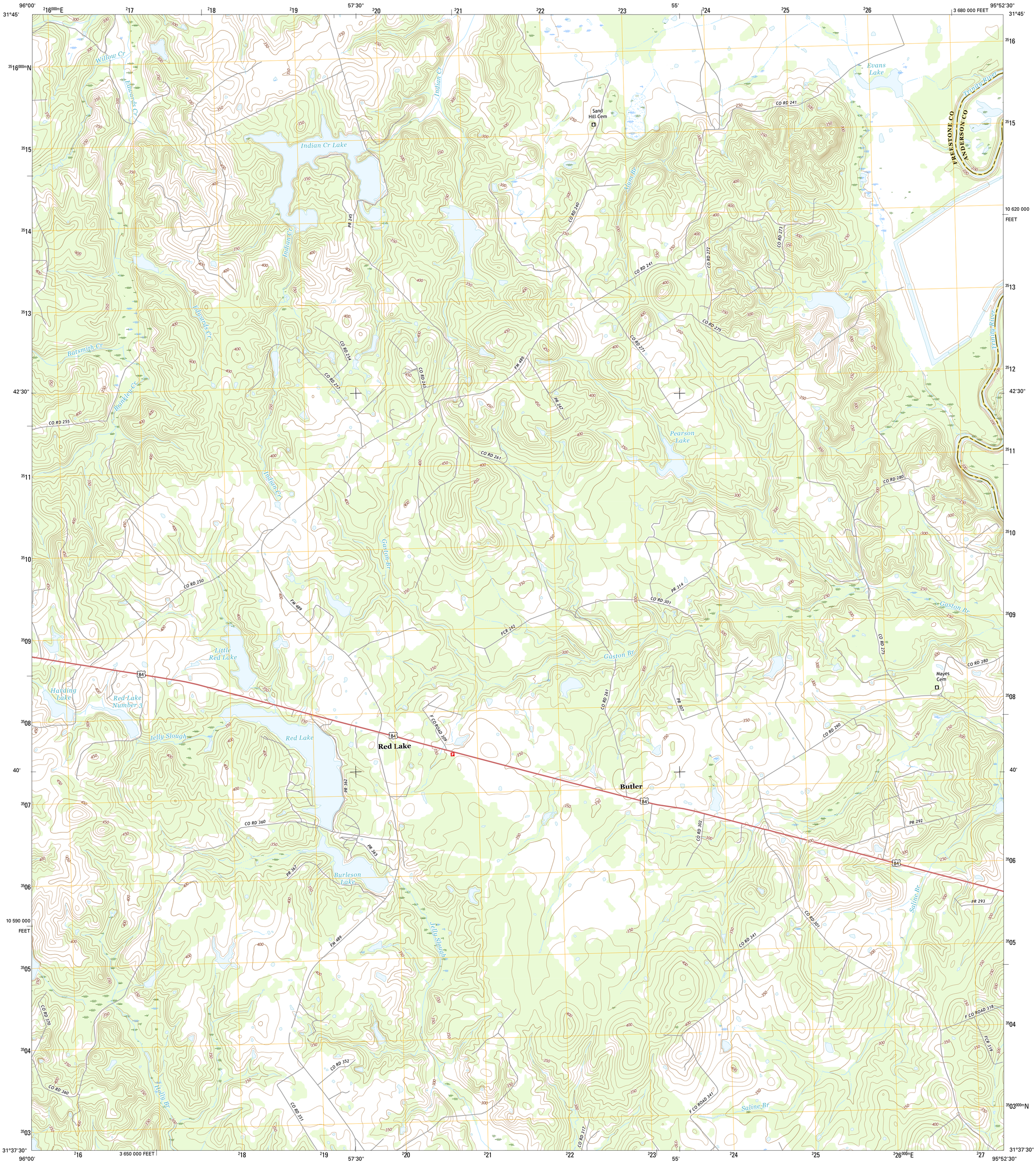
BUFFALO, TX
2016

NSN 7643015216395279
NGA REF NO. USGSX24K6013

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

BUTLER QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

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Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR
Grid Zone Designation
15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway	Local Connector
Secondary Hwy	Local Road
Ramp	4WD

Interstate Route · US Route · State Route

1	2	3
4		5
6	7	8

1 Young
2 Yard
3 Tennessee Colony
4 Turlington
5 Long Lake
6 Lanely
7 Keechi
8 Oakwood

ADJOINING QUADRANGLES

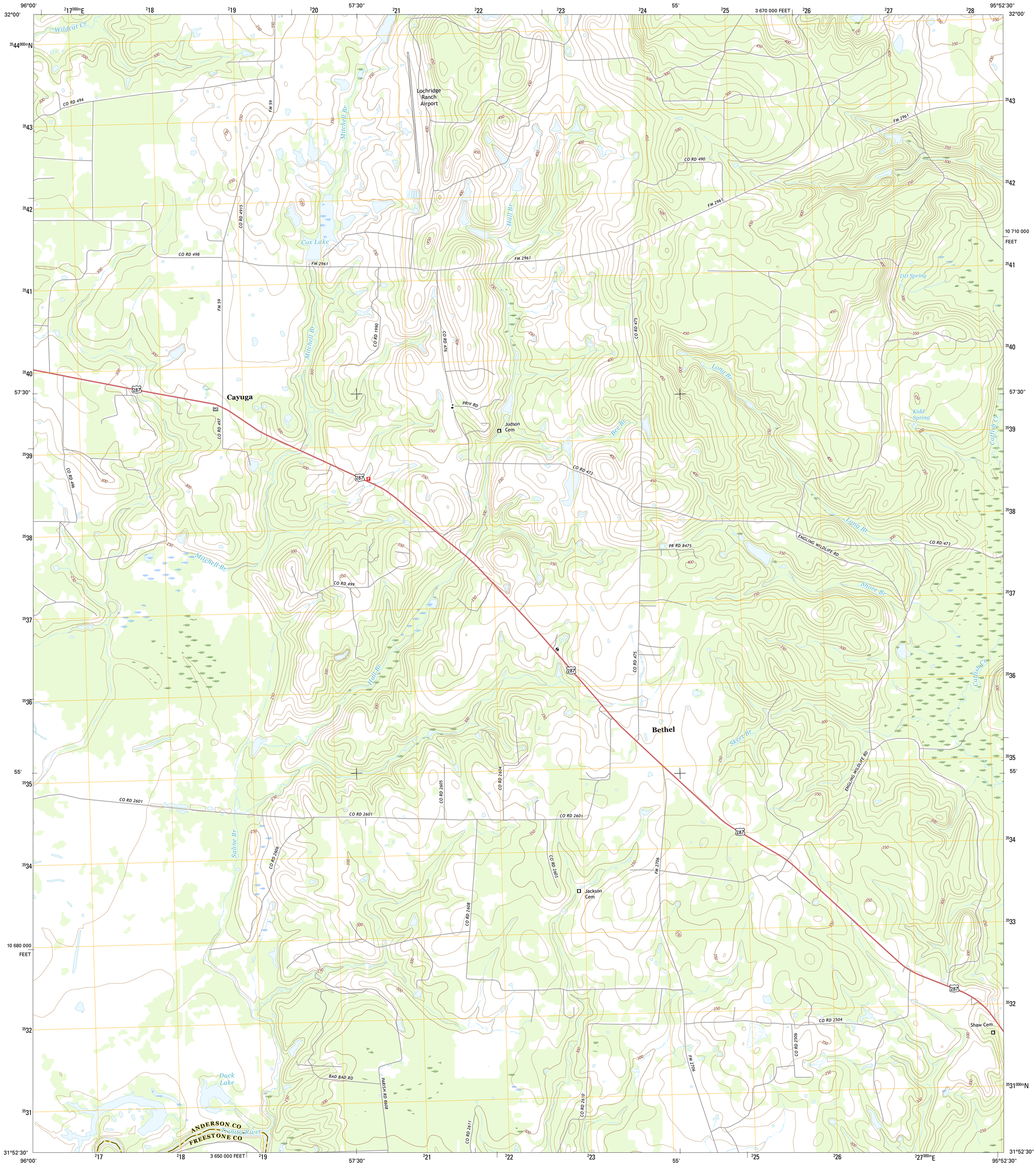
BUTLER, TX
2016

NSN 7643016395321
NGA REF NO. USGSX24K6451

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

CAYUGA QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

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Imagery.....NAIP, July 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
2° 60' 53 MILS
1° 33' 28 MILS

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR
Grid Zone Designation
15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway	Local Connector
Secondary Hwy	Local Road
Ramp	4WD
Interstate Route	US Route
State Route	

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES

1 Creslenn Ranch
2 Cross Roads
3 Coon Creek Lake
4 Roustabout Camp
5 Blackfoot
6 Young
7 Yard
8 Tennessee Colony

CAYUGA, TX
2016

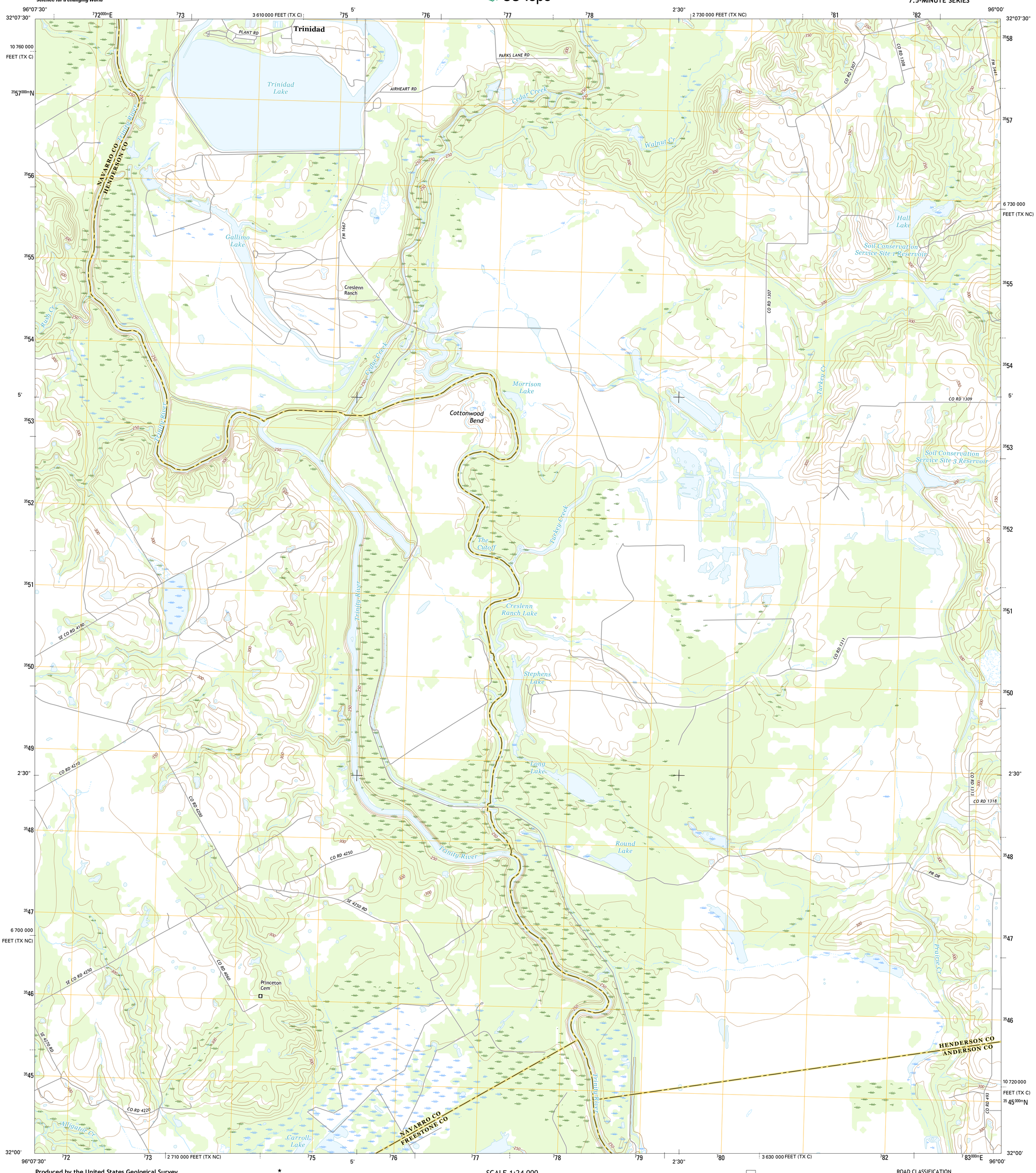
NSN. 7643016395452
NGA REF NO. USGSX24K7742

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

USGS science for a changing world

The National Map US Topo

CRESLENN RANCH QUADRANGLE
TEXAS
7.5-MINUTE SERIES



96°07'30" 32°07'30" 10 760 000 FEET (TX C) 3 610 000 FEET (TX C) 2 730 000 FEET (TX NC) 96°00' 32°07'30" 6 730 000 FEET (TX NC) 6 700 000 FEET (TX NC) 2 710 000 FEET (TX NC) 3 630 000 FEET (TX C) 32°00' 10 720 000 FEET (TX C)

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14S
10 000-foot ticks: Texas Coordinate System of 1983 (north central and central zones)

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Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
3° 4' 55 MILS
1° 33' 28 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation 14S

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

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A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Kerens
2 Malakoff
3 Mallard Hill
4 Goodlow Park
5 Cross Roads
6 Winkler
7 Roustabout Camp
8 Cayuga

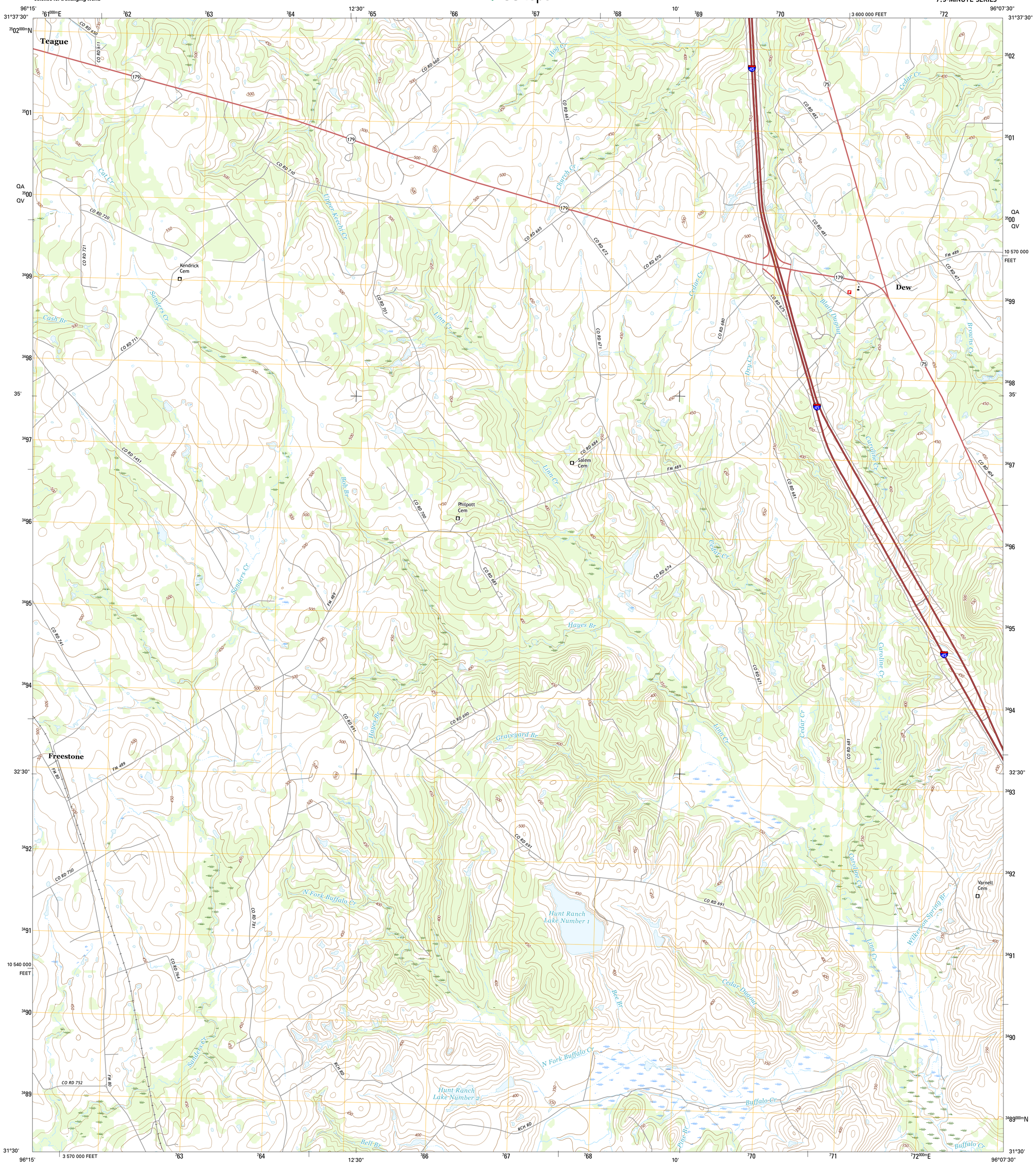
CRESLENN RANCH, TX
2016

NSN. 7643016395712
NGA REF NO. USGSX24K10693

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

DEW QUADRANGLE
TEXAS-FREESTONE CO.
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
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Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

MN 3° 9' 56 MILS
GN 1° 28' 26 MILS

U.S. National Grid
100,000-m Square ID
QA / QV (20 00)
Grid Zone Designation 14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Teague North
2 Fairfield
3 Turlington
4 Teague South
5 Lanely
6 Farrar
7 Donie
8 Buffalo

ADJOINING QUADRANGLES

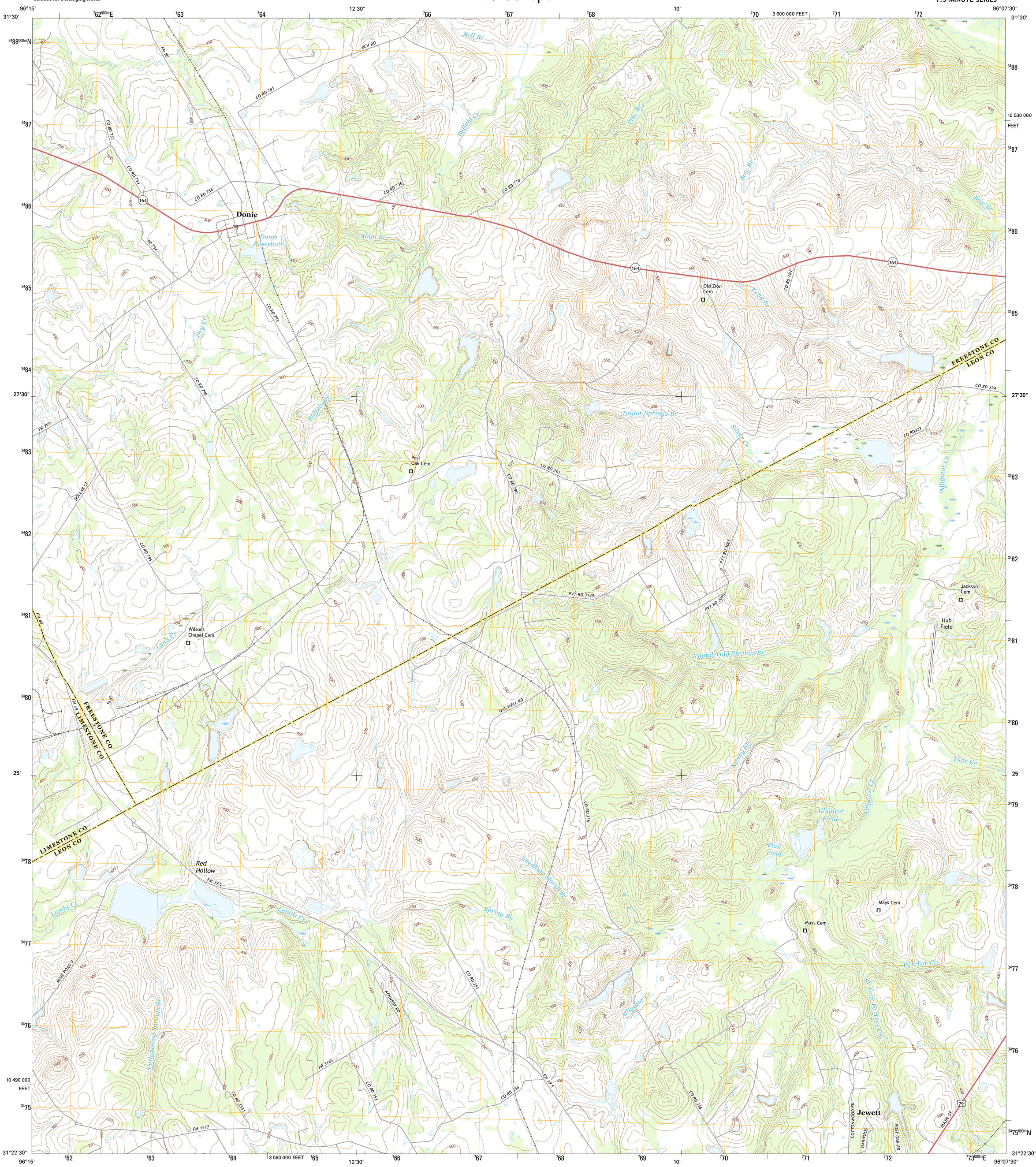
DEW, TX
2016

NSN 7643016395848
NGA REF NO. USGSX24K69875

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

DONIE QUADRANGLE
TEXAS
7.5-MINUTE SERIES



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Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET
MN 3° 9' 56 MILS
GN 1° 28' 26 MILS

U.S. National Grid
100,000-m Square ID
QV
Grid Zone Designation 14R

SCALE 1:24 000
1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Teague South
2 Dew
3 Lanely
4 Farrar
5 Buffalo
6 Round Prairie
7 Jewett
8 Robbins

ADJOINING QUADRANGLES

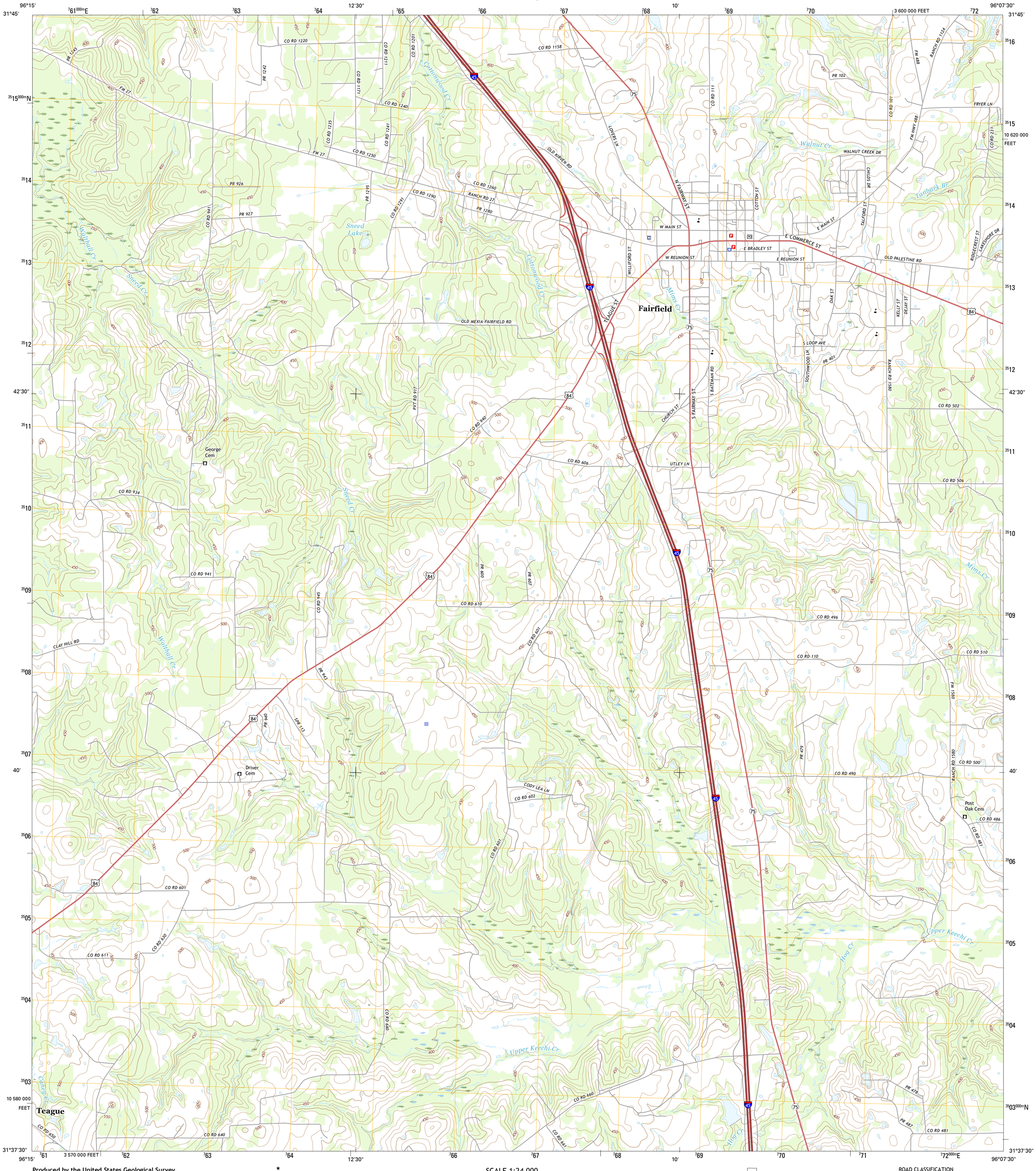
DONIE, TX
2016

NSN 7643016395907
NGA REF NO. USGSX24K12528

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

FAIRFIELD QUADRANGLE
TEXAS-FREESTONE CO.
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

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Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 9' 56 MILS
GN 1° 29' 26 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Kirvin
2 Stewards Mill
3 Young
4 Teague North
5 Turlington
6 Teague South
7 Dew
8 Lanely

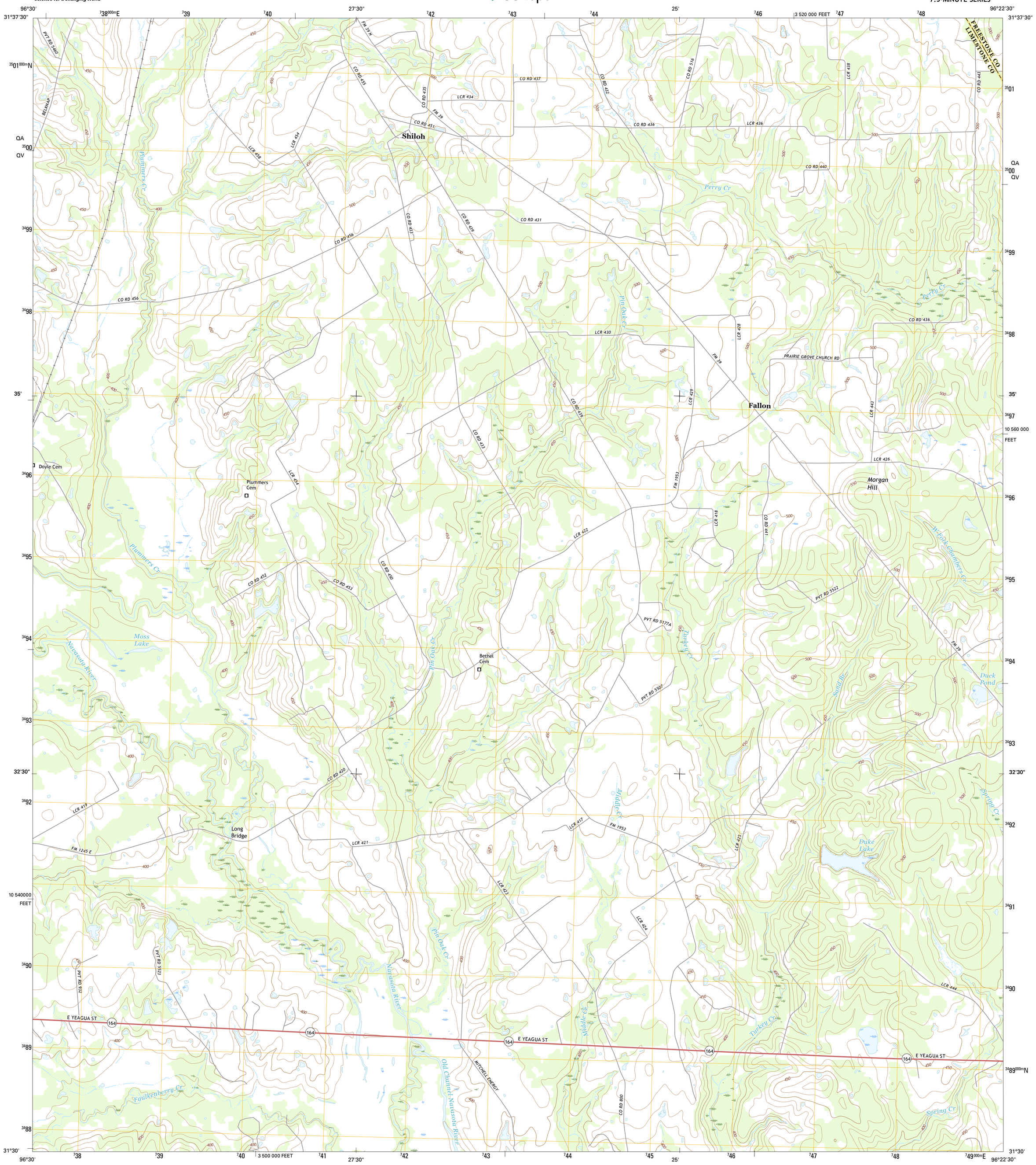
FAIRFIELD, TX
2016

NSN 7643016396099
NGA REF NO. USGSX24K14818

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

FALLON QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 18' 59 MILS
GN 1° 20' 24 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA / QV
3500
Grid Zone Designation
14R

SCALE 1:24 000
1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1 MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000 FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Tehuacana
2 Mexia
3 Teague North
4 Groesbeck
5 Teague South
6 Thornton
7 Box Church
8 Farrar
ADJOINING QUADRANGLES

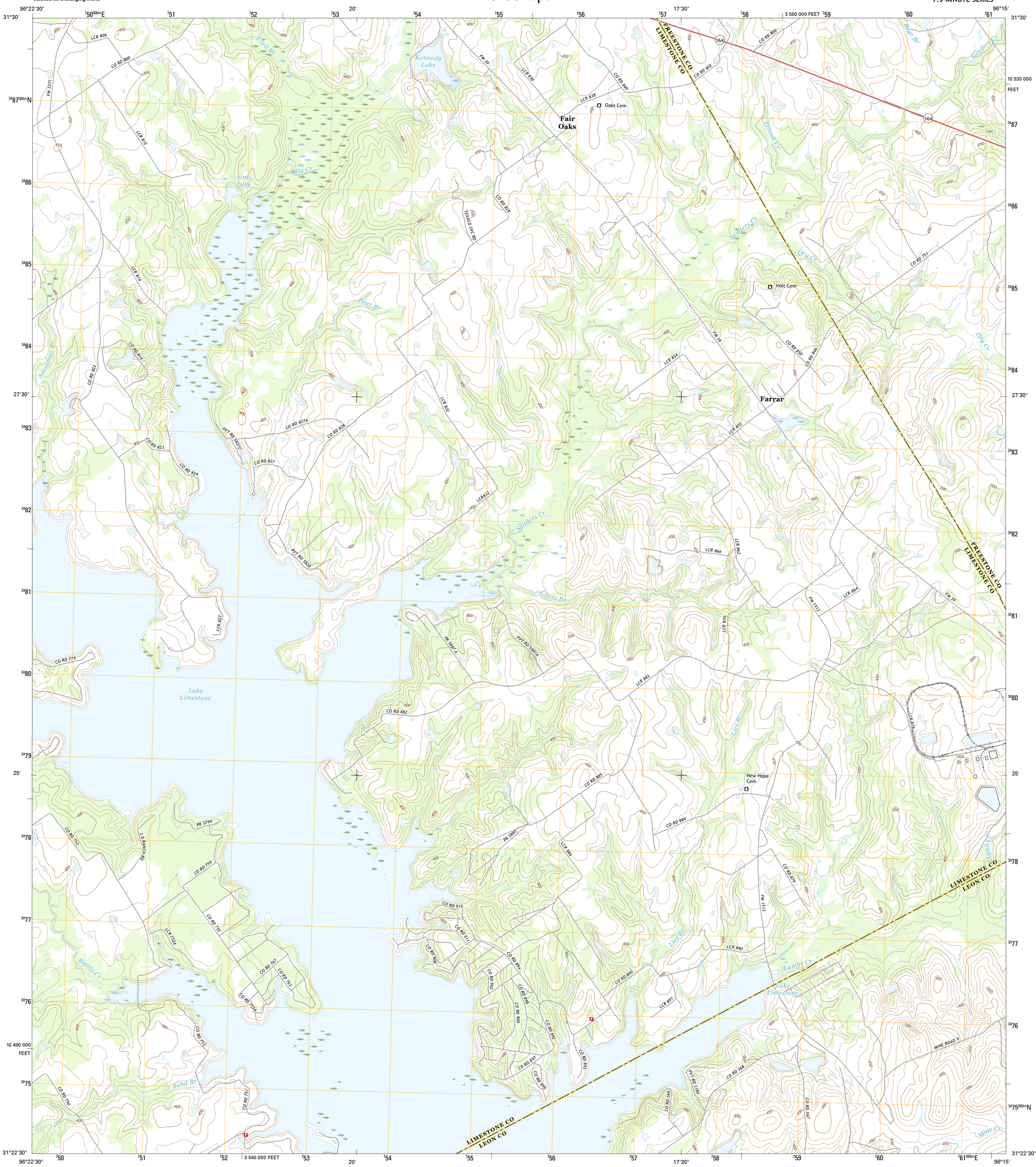
FALLON, TX
2016

NSN 7643016396105
NGA REF NO. USGSX24K14938

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

FARRAR QUADRANGLE
TEXAS
7.5-MINUTE SERIES

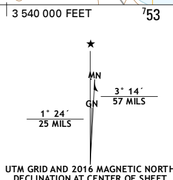


Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014



UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QV
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Fallon
2 Teague South
3 Dew
4 Box Church
5 Donie
6 Oletha
7 Round Prairie
8 Jewett

ADJOINING QUADRANGLES

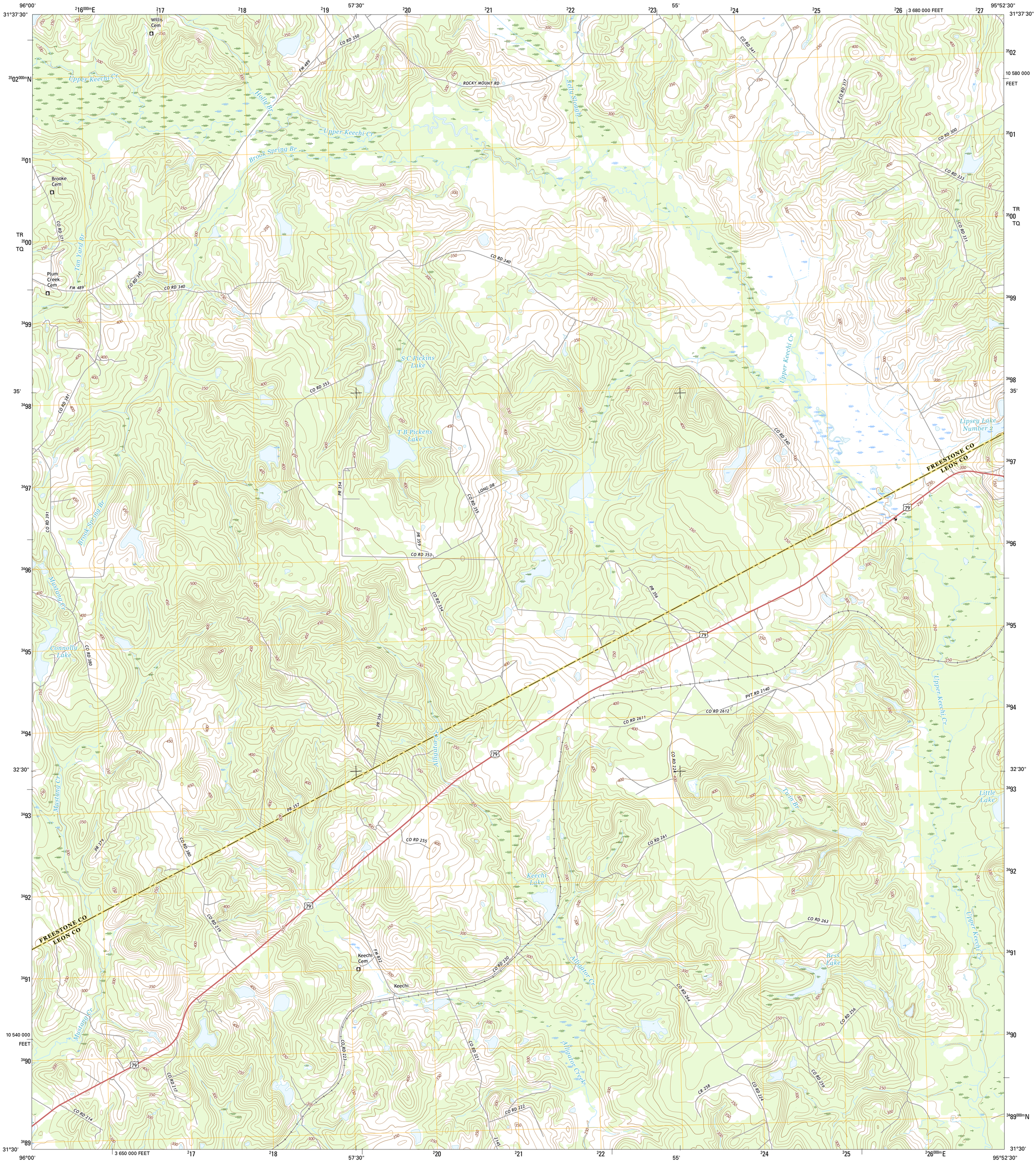
FARRAR, TX
2016

NSN 7643016396116
NGA REF NO. USGSX24K15054

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

KEECHI QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
3° 0´
53 MILS
1° 32´
27 MILS

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR / TQ 35 00
Grid Zone Designation 15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Turlington
2 Butler
3 Long Lake
4 Lanely
5 Oakwood
6 Buffalo
7 Flo
8 Lake Leon

ADJOINING QUADRANGLES

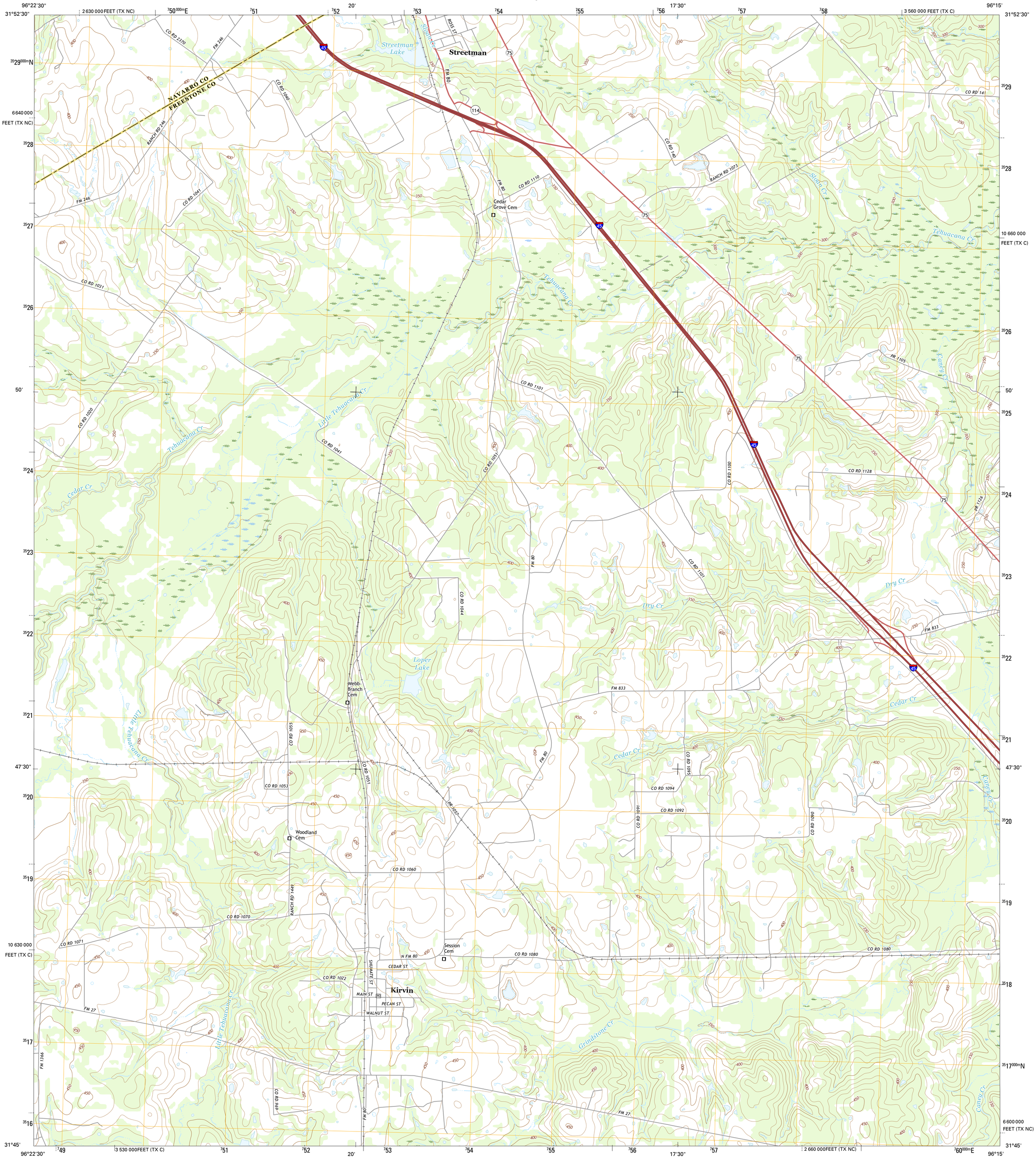
KEECHI, TX
2016

NSN 7643016396760
NGA REF NO. USGSX24K23127

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

KIRVIN QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central and north central zones)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 13' 57 MILS
GN 1° 25' 25 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Richland
2 Streetman
3 Winkler
4 Wortham
5 Stewards Mill
6 Mexia
7 Teague North
8 Fairfield
ADJOINING QUADRANGLES

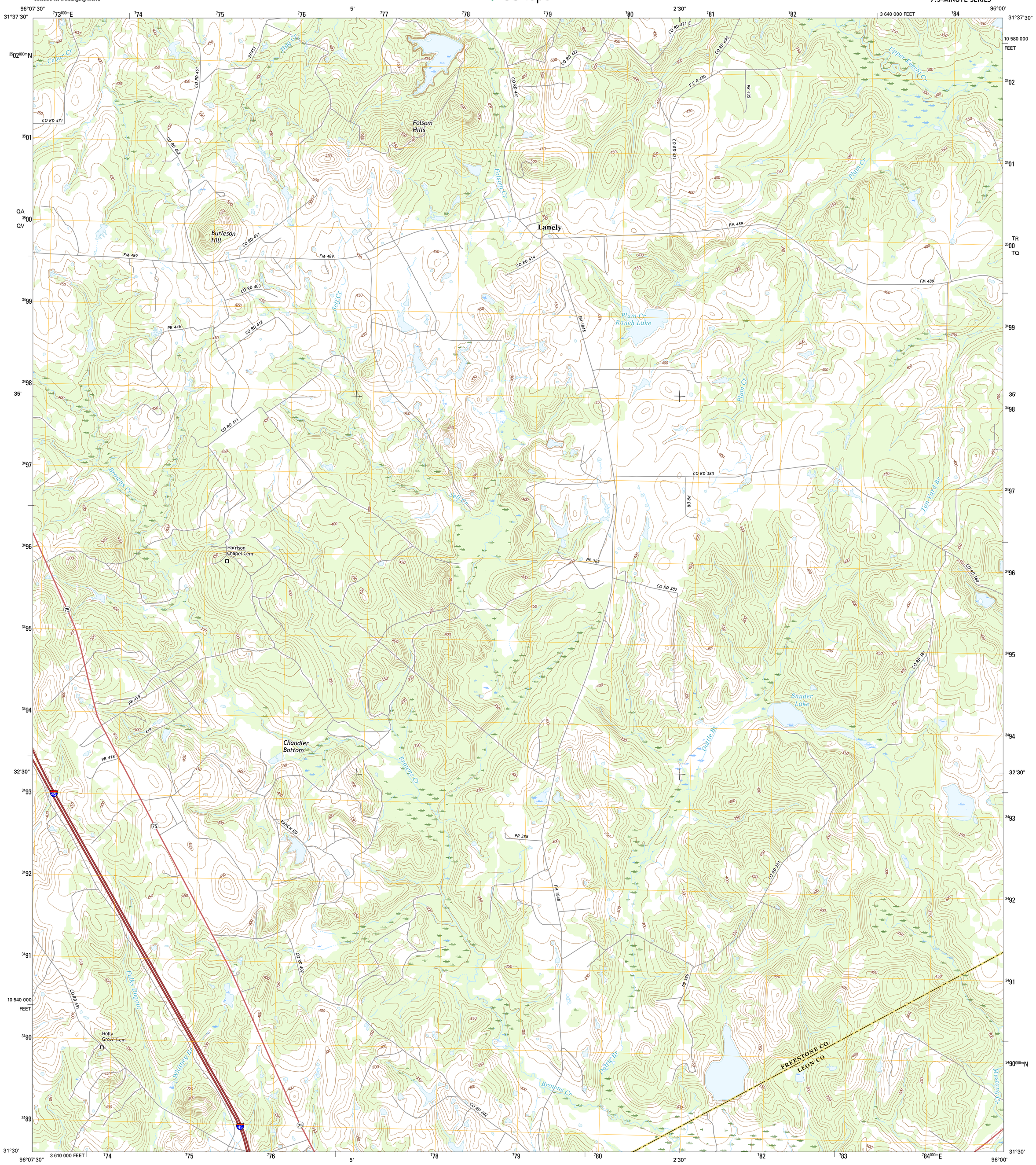
KIRVIN, TX
2016

NSN. 7643016396816
NGA REF NO. USGSX24K23774

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

LANELY QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2007
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

MN 3° 5' 55 MILS
GN 1° 32' 27 MILS

U.S. National Grid
100,000-m Square ID: QA / QV
Grid Zone Designation: 14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Fairfield
2 Turlington
3 Butler
4 Dew
5 Keechi
6 Donie
7 Buffalo
8 Flo

ADJOINING QUADRANGLES

LANELY, TX
2016

NSN 7643016394943
NGA REF NO. USGSX24K24897

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

LONG LAKE QUADRANGLE
TEXAS
7.5-MINUTE SERIES

95°52'30" 95°45' 31°45' 31°37'30" 42'30" 40' 50' 47'30"
3 720 000 FEET 3 690 000 FEET 10 620 000 FEET 10 590 000 FEET
$^{2}28^{000m}$E $^{35}16^{000m}$N $^{35}03^{000m}$N

Clear Lake
Trinity River
ANDERSON CO
FREESTONE CO
Cedar Lake Slough
Cedar Lake
Lake Cr
Big Lake
Cedar Lake Slough
Big Lake
Keechie Cr
Trinity River
ANDERSON CO
FREESTONE CO
Blue Lake
Trinity River
Gaston Br
Suline Br
Lake Port
Smith Lake
Little Lake
Salt Water Slough
Tucker
Long Lake
Mud Lake
Salt Water Slough
Trinity River
FREESTONE CO
ANDERSON CO
Long Lake
Brushy Lake
Little Brushy Lake
Trinity River
LEON CO
ANDERSON CO
Canted Cr

CO RD 2901
CO RD 2903
CO RD 2904
CO RD 2905
CO RD 2906
CO RD 2907
CO RD 2908
CO RD 2909
CO RD 2912
CO RD 2915
CO RD 2916
CO RD 2933
CO RD 2210
ACR 2202
CO RD 280
CO RD 290
CO RD 295
CO RD 296
CO RD 298
CO RD 300
F C O ROAD 318
PR 289
PR 294
PR 303
PR 304
PR 305
PR 322
LAZY LN
FM 3452
FM 320
FM 645
RD 6907
CR 2122
CR 2123
CR 2124
84
79
294

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
2° 55' 52 MILS
1° 29' 26 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR
Grid Zone Designation
15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Yard
2 Tennessee Colony
3 Northwest Palestine
4 Butler
5 Southwest Palestine
6 Keechi
7 Oakwood
8 Indian Lake

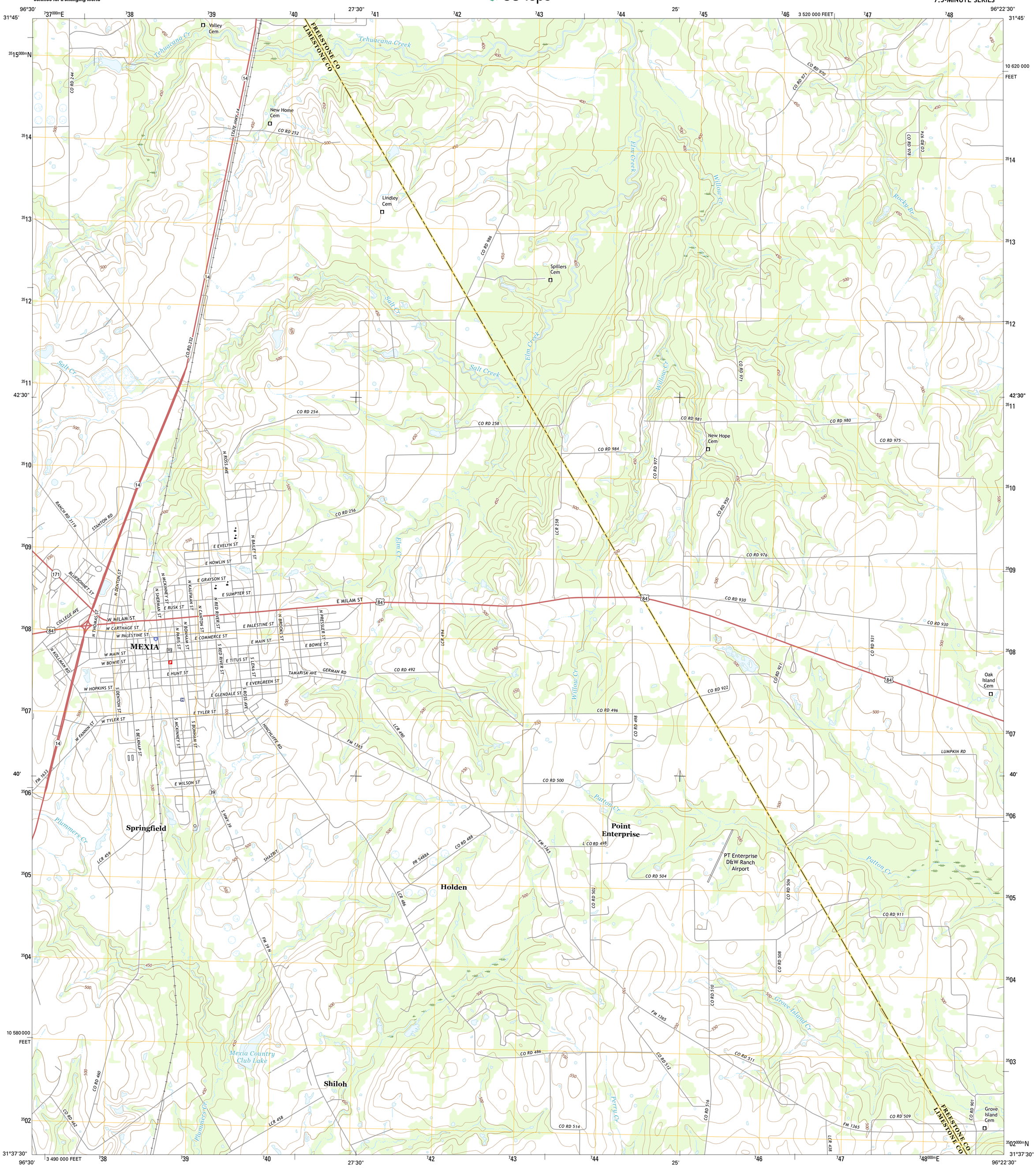
LONG LAKE, TX
2016

NSN. 7643016397083
NGA REF NO. USGSX24K26459

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

MEXIA QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

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Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 18' 59 MILS
GN 1° 21' 24 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Union High
2 Wortham
3 Kirvin
4 Tehuacana
5 Teague North
6 Groesbeck
7 Fallon
8 Teague South

ADJOINING QUADRANGLES

MEXIA, TX
2016

NSN. 7643016391269
NGA REF NO. USGSX24K28995

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

OAKWOOD QUADRANGLE
TEXAS
7.5-MINUTE SERIES

95°52'30" 95°45' 31°37'30" 31°30' 50' 47'30" 35' 32'30"
3 720 000 FEET 10 580 000 FEET 3 690 000 FEET 10 540 000 FEET
TR TQ

Oakwood
Mount Pisgah Cem
Carter Ranch Airport
McCrakin Lake
Glaze Lake
Criminal Lake
Lipsey Lake Number 2
Little Lake
Gilliam Lake
Caney Cr
Toms Cr
Trinity River
Northwest Fork Boons Cr
Boons Cr
Ringgold Cr
Tuescott Cr
Eads Cr
Birch Cr
Red Br

FREESTONE CO
LEON CO
ANDERSON CO

CO RD 200
OAKWOOD RD
CO RD 311
CO RD 321
CO RD 333
CO RD 331
WOW RD
ROCKY MOUNT RD
FCK 301
PR 315
PR 304
CO RD 239
CO RD 241
SHADY LN
KNOWLES ST
HOLLY ST
LOVE ST
COLLEGE ST
MOODY ST
PALMER ST
COMMERCE ST
FRONT ST
KIMBELL ST
MAIN ST
MAYES ST
W WALSTON ST
WALSTON ST
BROAD ST
CO RD 237
CO RD 2391
FM 542
CO RD 234
CO RD 2421
FM 831
PVT RD 2125
CO RD 260
CO RD 242
CATTLE RANCH RD
CO RD 244
CO RD 243
CO RD 2431
CO RD 246
CO RD 245
CO RD 2461
CO RD 252
79

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

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Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 2° 56' 52 MILS
GN 1° 28' 26 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR / TQ
Grid Zone Designation 15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Butler
2 Long Lake
3 Southwest Palestine
4 Keechi
5 Indian Lake
6 Flo
7 Lake Leon
8 Stanmire Lake
ADJOINING QUADRANGLES

OAKWOOD, TX
2016

MSN 7643016397498
NGA REF NO. USGSX24K22816

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

ROUSTABOUT CAMP QUADRANGLE
TEXAS
7.5-MINUTE SERIES

96°07'30" 32°00' 5' 2'30" 96°00'

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and
1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central and north central zones)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET
MN 3° 4' 55 MILS
GN 1° 33' 28 MILS

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the
National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Goodlow Park
2 Creslenn Ranch
3 Cross Roads
4 Winkler
5 Cayuga
6 Stewards Mill
7 Young
8 Yard

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

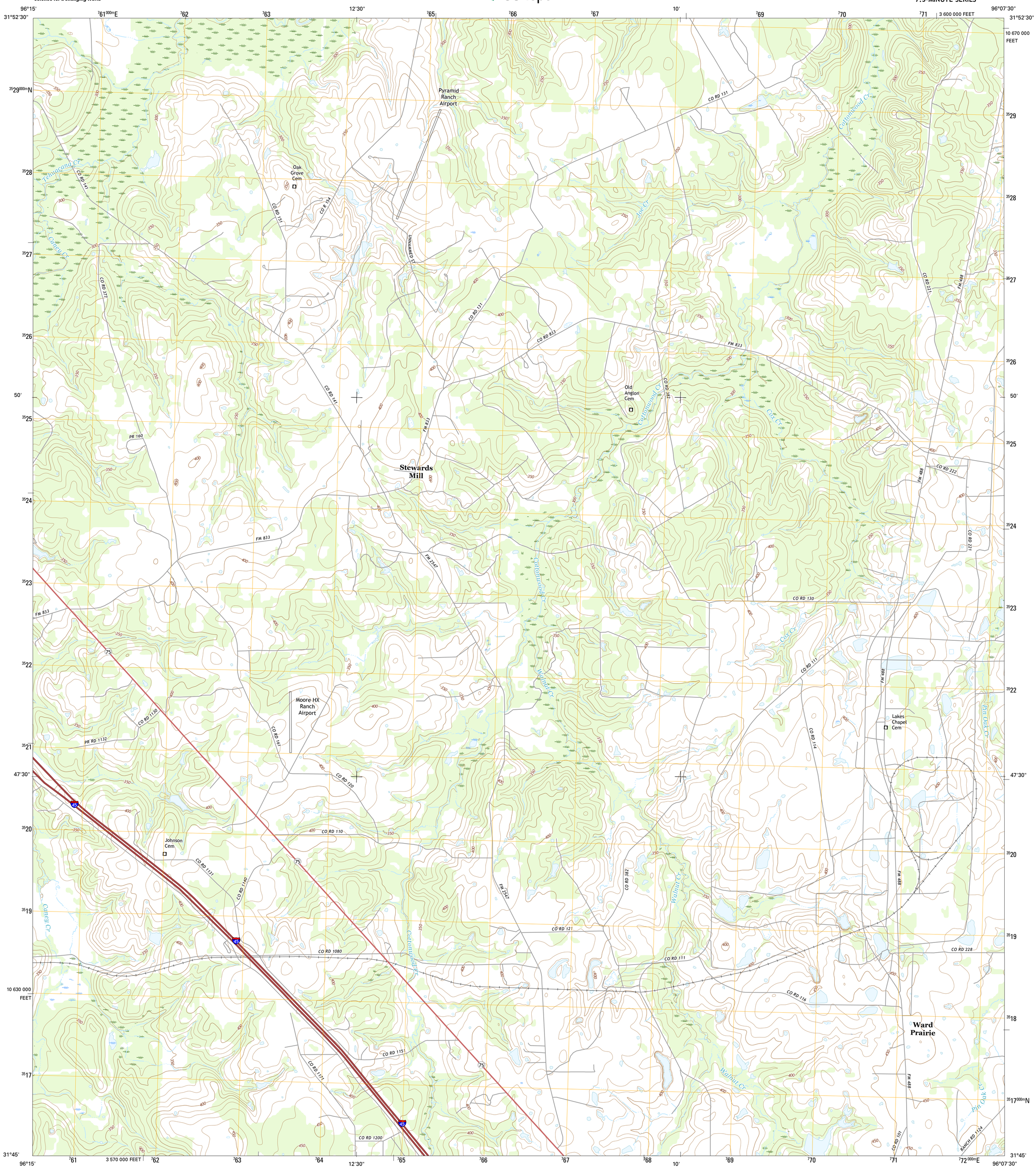
ROUSTABOUT CAMP, TX
2016

NSN. 7643016398024
NGA REF NO. USGSX24K38788

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

STEWARDS MILL QUADRANGLE
TEXAS-FREESTONE CO.
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
1° 29'
26 MILS
3° 9'
56 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Streetman
2 Winkler
3 Roustabout Camp
4 Kirvin
5 Young
6 Teague North
7 Fairfield
8 Turlington
ADJOINING QUADRANGLES

STEWARDS MILL, TX
2016

NSN, NGA REF NO. USGSX24K4316964

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

STREETMAN QUADRANGLE
TEXAS
7.5-MINUTE SERIES

96°22'30" | 7490000mE | 3530000 FEET (TX C) | 20' | 7753 | 7754 | 7755 | 17'30" | 7757 | 2 660 000 FEET (TX NC) | 96°15'

32°00' | 35 43 000mN | 10 710 000 FEET (TX C) | 35 42 | 35 41 | 35 40 | 35 39 | 57'30" | 35 38 | 35 37 | 35 36 | 35 35 | 55' | 35 34 | 35 33 | 35 32 | 6 650 000 FEET (TX NC) | 35 31 | 35 30 | 31°52'30"

Navarro
Cheneyboro
Hopewell Cem
Handcock Cem
Gunn Cem
Eureka
Johnson Cem
Birdston Cem
Streetman

Crab Cr
Grape Cr
Richland Chambers Reservoir
Texas Industries Lake
Spring Br
Emerson Lake Number 1
Emerson Lake Number 2

RE 57
S E 1090
FM 3243
NAVARRO SLAB
SE CO RD 2120
SECR 2130
SE 2130
SE 2140
SE 2145
SE CO RD 2135
SE CO RD 2175
SE 2215
CURFY LN
SE 2210
VANNICE RD
SECR 2220
SE CO RD 2247
CO HWY 2240
SE CO RD 2260
JONES BRANCH RD
LA BOTA
SE CO RD 2270
SE 2240
SE CO RD 2230
SUNDANCE
SE CO RD 2230K
TONKAWA TRL
EL SUENO
SE CO RD 2160
CO HWY 2170
CO RD 2140
SE CO RD 1090
S E 1020
S E 2160
SE 1090
SE 1095 RD
SE 1098
SE 1270 RD
SE 1230 RD
SE CO RD 1200
R 1200
SE 1210 RD
SE 1220 RD
SE 1260
SE 1280
SE 2310 RD
SE CO RD 2359
RUDYS WAY
LOPER RD
LOPER DR
SE 2360
WHITLEY DR
SE CO RD 2336
SE 2330 RD
R 2355
RANCH RD 416
FM 416
2340
2370
2310
CO RD 2320
RANCH RD 3059
CO RD 171
CO RD 141
CO RD 143
SE 1240
SE 1230
FM 416
S I- 45
N SVC RD I- 45
FRONTAGE RD
SW CO RD 2360
CO RD 2360
OLD STREETMAN RD
ROSS ST
MAIN ST
262
CO RD 150
NAVARRO CO
FREESTONE CO

2 630 000 FEET (NX NC) | 7750 | 7751 | 7752 | 20' | 7753 | 7754 | 7755 | 7756 | 17'30" | 7757 | 7758 000mE | 3560000 FEET (TX C) | 96°15'

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (north central and central zones)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
3° 13' 57 MILS
1° 25' 25 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway	Local Connector
Secondary Hwy	Local Road
Ramp	4WD
Interstate Route	US Route
State Route	

1	2	3
4		5
6	7	8

1 Corsicana
2 Powell
3 Goodlow Park
4 Richland
5 Winkler
6 Wortham
7 Kirvin
8 Stewards Mill

ADJOINING QUADRANGLES

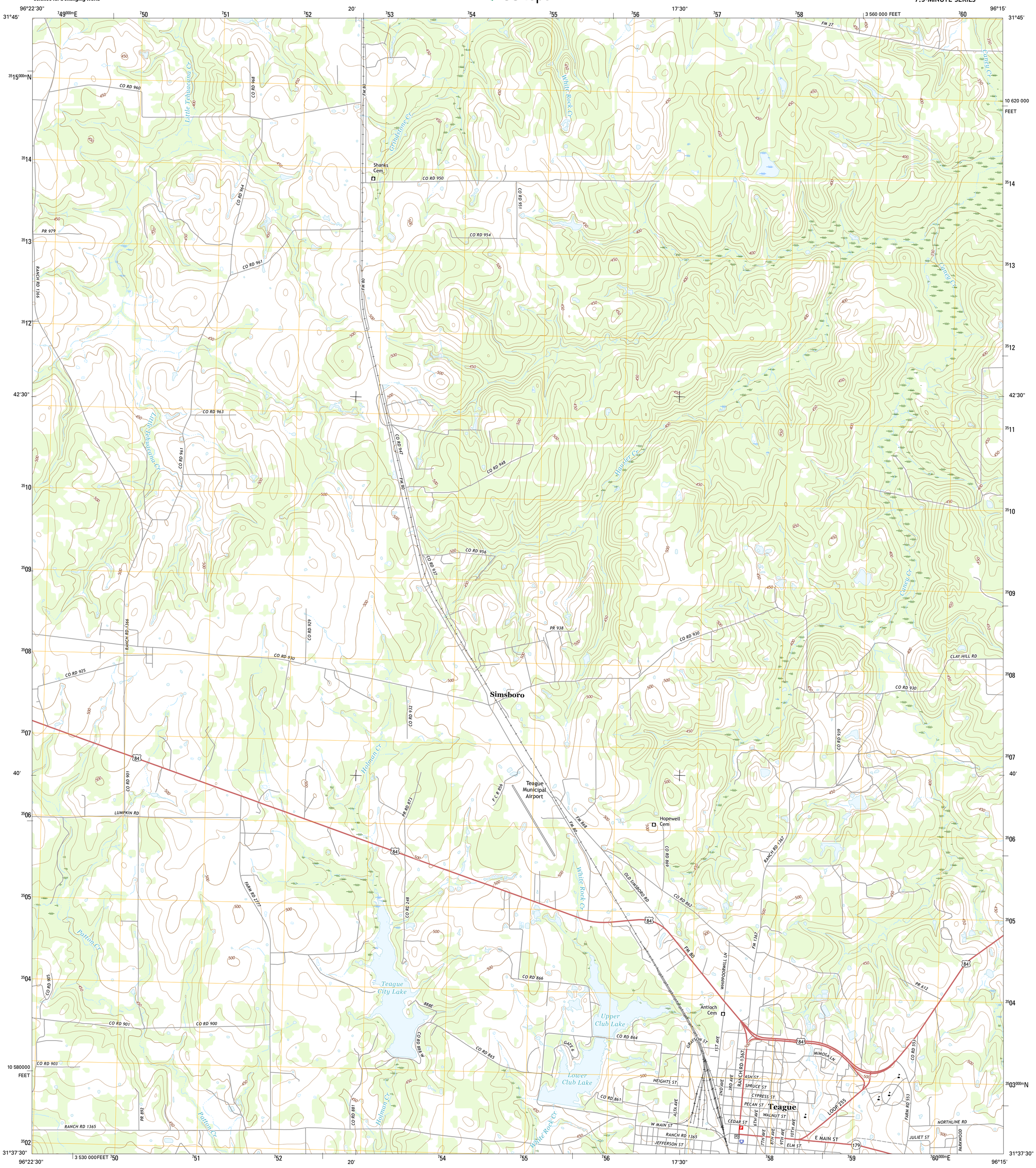
STREETMAN, TX
2016

NSN. 7643016398485
NGA REF NO. USGSX24K43491

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

TEAGUE NORTH QUADRANGLE
TEXAS-FREESTONE CO.
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 14' 57 MILS
GN 1° 25' 25 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Wortham
2 Kirvin
3 Stewards Mill
4 Mexia
5 Fairfield
6 Fallon
7 Teague South
8 Dew

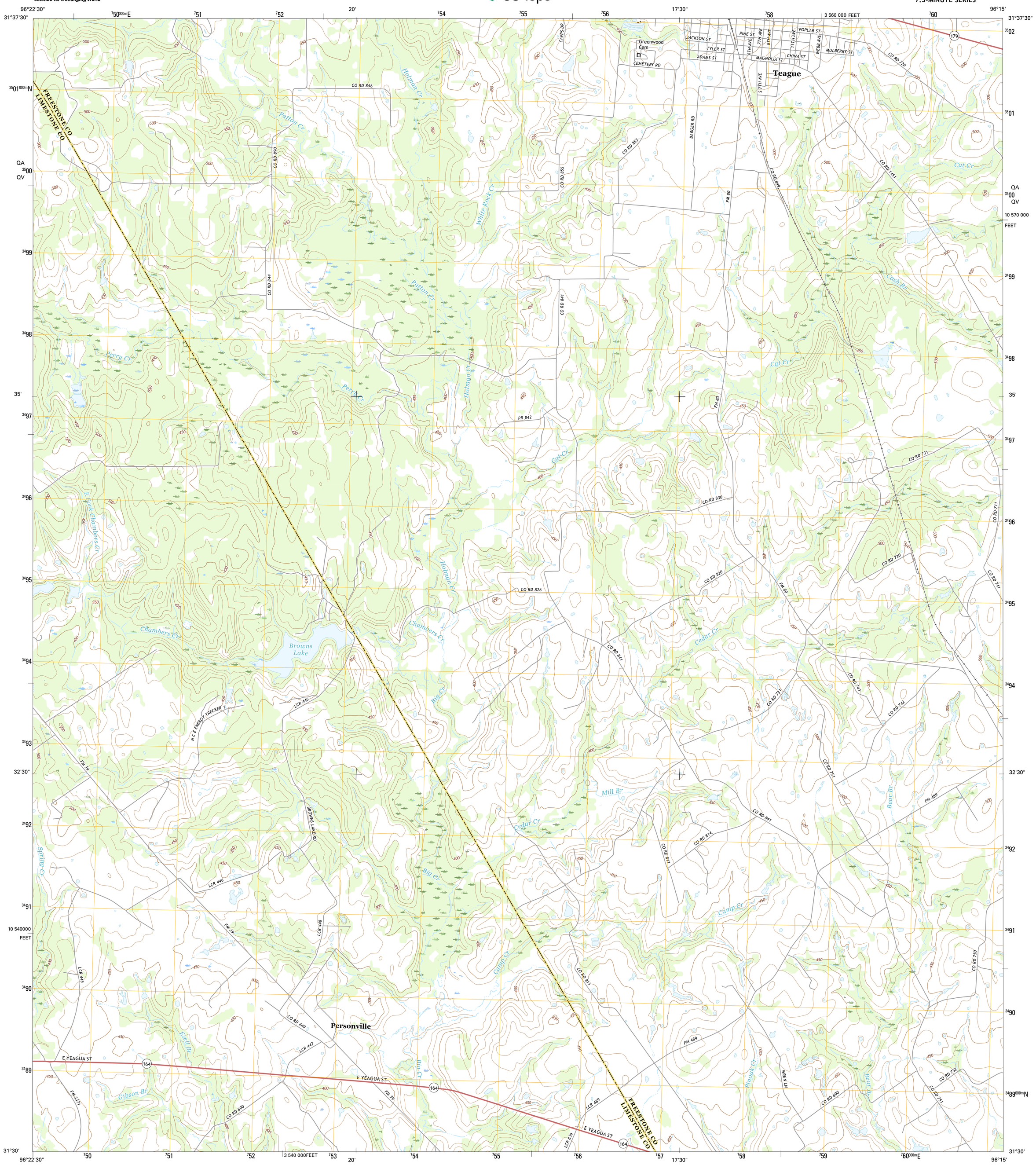
TEAGUE NORTH, TX
2016

NSN. 7643016398543
NGA REF NO. USGSX24K44420

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

TEAGUE SOUTH QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, October 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 14' 57 MILS
GN 1° 24' 25 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA / QV
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Mexia
2 Teague North
3 Fairfield
4 Fallon
5 Dew
6 Box Church
7 Farrar
8 Donie
ADJOINING QUADRANGLES

TEAGUE SOUTH, TX
2016

NSN 7643016398546
NGA REF NO. USGSX24K44421

USGS
science for a changing world

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map
US Topo

TENNESSEE COLONY QUADRANGLE
TEXAS
7.5-MINUTE SERIES

95°52'30" 2290000mE 30 31 50' 33 34 35 47'30" 37 3 710 000 FEET 38 39 95°45'
31°52'30" 3530000mN 29 28 27 26 50' 25 24 23 22 47'30" 21 20 19 18 10 630 000 FEET 17000mN 31°45'
10 670 000 FEET

Beaver Creek
Beaver Cr
Spring Cr
Lake Cr
Keechie Cr
Tennessee Colony
Massey Lake
Jimmison Quarter Cem
Soil Conservation Service Site 1 Reservoir
Soil Conservation Service Site 2 Reservoir
Spring Lake
Cedar Lake Slough
Clear Lake
Harmony
ANDERSON CO
Trinity River

FM 860
CO RD 2401
CO RD 2402
CO RD 2410
CO RD 2409
CO RD 2405
CO RD 321
CO RD 6203
CO RD 2303
CO RD 2310
CO RD 2307
CO RD 2403
CO RD 2306
CO RD 2305
CO RD 2501
FM 645
CO RD 2301
CO RD 2304
CO RD 2311
ACR 2231
CO RD 2223
CO RD 2224
CO RD 2220
CO RD 2302
WILDCAT FERRY RD
FM 2054
CO RD 2802
CO RD 2803
CO RD 2218
CO RD 2217
CO RD 2219
CO RD 2810
CO RD 2809
CO RD 2212
CO RD 2216
CO RD 2812
CO RD 2811
CO RD 2215
CO RD 2213
CO RD 2214
FM 3328
FM 320
FM 3452
287
324
IPO
F

Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

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Imagery.....NAIP, July 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN
GN
1° 29′ 26 MILS
2° 55′ 52 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR
Grid Zone Designation
15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Cayuga
2 Blackfoot
3 Wilkerson Mountain
4 Yard
5 Northwest Palestine
6 Butler
7 Long Lake
8 Southwest Palestine

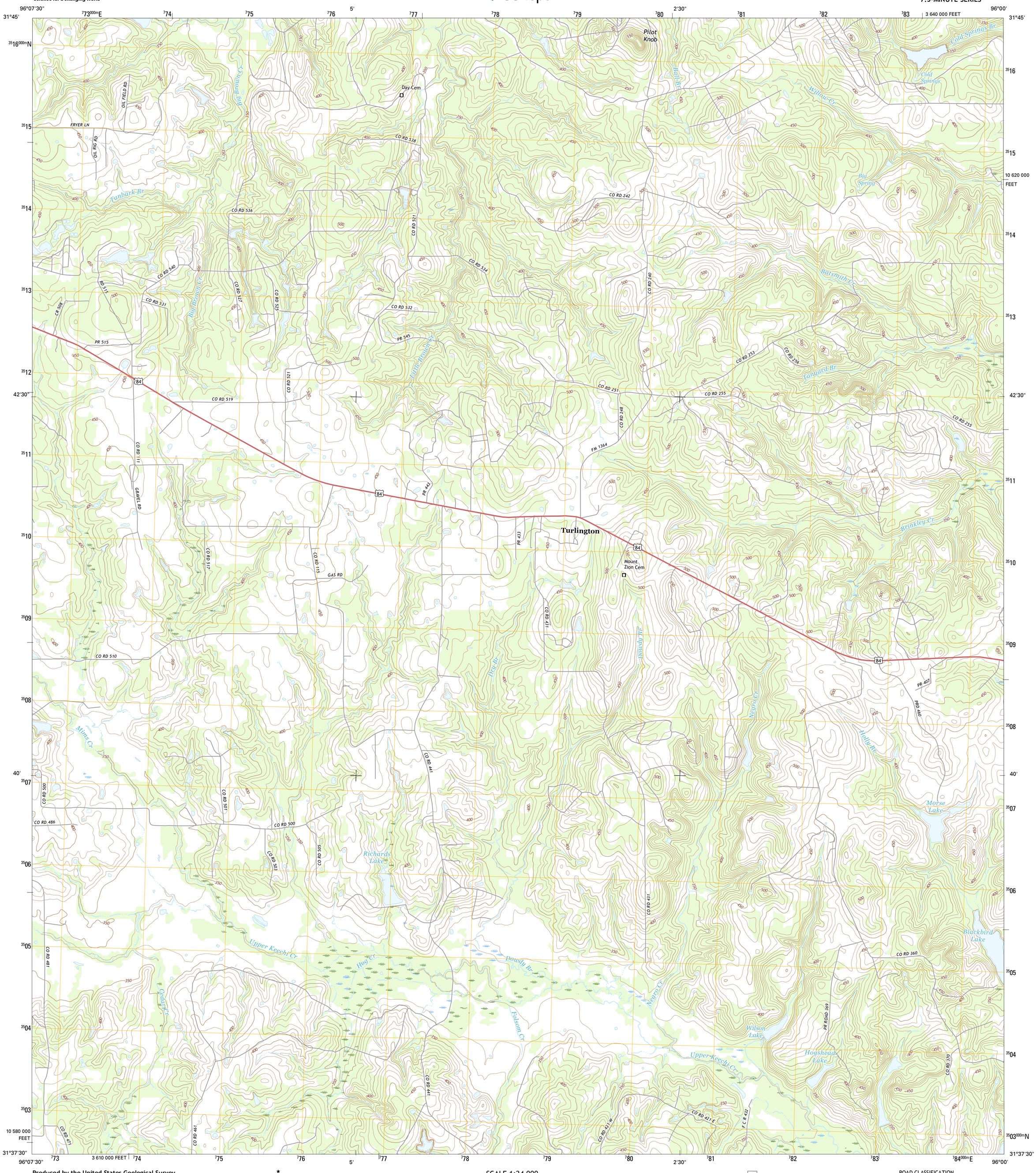
TENNESSEE COLONY, TX
2016

NSN. 7643016398557
NGA REF NO. USGSX24K44569

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

TURLINGTON QUADRANGLE
TEXAS-FREESTONE CO.
7.5-MINUTE SERIES



Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

MN 3° 5' 55 MILS
GN 1° 32' 27 MILS

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway	Local Connector
Secondary Hwy	Local Road
Ramp	4WD

Interstate Route · US Route · State Route

1	2	3
4		5
6	7	8

1 Stewards Mill
2 Young
3 Yard
4 Fairfield
5 Butler
6 Dew
7 Lanely
8 Keechi

ADJOINING QUADRANGLES

TURLINGTON, TX
2016

NSN 7643016398690
NGA REF NO. USGSX24K45967

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

WINKLER QUADRANGLE
TEXAS
7.5-MINUTE SERIES

96°15' | 12'30" | 10' | 96°07'30"
32°00' | 57'30" | 55' | 31°52'30"

6 690 000 FEET (TX NC) | 2 680 000 FEET (TX NC) | 3 600 000 FEET (TX C) | 10 710 000 FEET (TX C)
6 650 000 FEET (TX NC) | 3 570 000 FEET (TX C) | 2 700 000 FEET (TX NC)

Winkler
Sweatman - Blackmon Cem
Lee Cem
Greenbriar Cem
Petersburg Cem
Lake Corsicana
Texas Industries Lake
NAVARRO CO
FREESTONE CO
FM 416
RANCH RD 416
OLD HWY 287
STATE HWY 309
THE SHORES DR
LAKE VIEW LNDG
DEEP WATER CV
HATTERAS DR
CANAL ST
WATERS EDGE DR
SAILBOAT DR
CLUBHOUSE DR
SEC R1170
Long Arm Bay
SE CO RD 3190
SE CO RD 3310
SE CO RD 4271
NELSONS LOOP
SAPPHIRE LN
AZURE LN
PLET DR
PELICAN PT
PLEASANT HALL CEMETERY RD
E CO RD 2385
SE CO RD 2385
SE 2370
SE CO RD 2390
SE 2380
SE 2390
CO RD 2390
CO RD 2380
MIDWAY RD
MIDWAY LANDING RD
APPOMATTOX
CHELSEA DR
SOUTHERN OAKS DR
SANDY LN
TEAKWOOD AVE
SEPTEMBER DR
APRIL COVE
190 SPR
SHENANDOAH DR
STONEWALL DR
BULL RUN DR
KINGSWOOD DR
KNIGHTS CIR
SUSAN GROVE BLVD
WALKER TRCE
RANCH RD 416
CO RD 192
CO RD 196
CO RD 190
CO RD 185
CO RD 193
CO RD 181
CO RD 771
CO RD 131
CO RD 221
R 188
R 186
R 171
Greenbriar Cr
Spring Br
Tehuacana Creek
Little Sandy Cr
Sandy Cr
Tehuacana Creek
Cottonwood Cr
Campbell Slough
Cottonwood Cr

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and
1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central and north central zones)

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Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 9' 56 MILS
GN 1° 29' 26 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Powell
2 Goodlow Park
3 Creslenn Ranch
4 Streetman
5 Roustabout Camp
6 Kirvin
7 Stewards Mill
8 Young
ADJOINING QUADRANGLES

WINKLER, TX
2016

NSN. 7643016398943
NGA REF NO. USGSX24K49490

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

WORTHAM QUADRANGLE
TEXAS
7.5-MINUTE SERIES

Currie
Wortham
Long Bottom Cem
Miller Cem
Red Cem
Little Pin Oak Cr
Mesquite Cr
Cedar Cr
New Wortham Lake
Wortham Lake Number 1
Wortham Lake Number 2
Wortham Lake
Jackson Br
Tehuacana Creek
Wolf Cr
Elm Creek
Rocky Br
NAVARRO CO
FREESTONE CO
LIMESTONE CO
CO RD 2190
CO RD 2210
CO RD 2200
RANCH RD 641
FM 1394
SW 2307
SW 2300
FARM RD 641
CO RD 2230
CO RD 2240
CO RD 2400
SE CO RD 2390
SW CO RD 2380
SW 2383
SW 2380
SW CO RD 2386
CO RD 2410
CO RD 2380
SE 2380 RD
SW 2430 RD
SW 2410 RD
SW 2420 RD
CO RD 1021
CO RD 1020
FM 246
CO RD 1031
CO RD 1001
CO RD 1081
CO RD 990
RANCH RD 246
PR 994
CO RD 1011
CO RD 1010
SW 2330
MARTIN LUTHER KING AVE
N 3RD ST
W CANADIAN AVE
N AVE D
CO RD 1000
CO RD 995
N 6TH ST
W SAN SABA AVE
W CONCH AVE
E SAN SABA AVE
E MAIN ST
S 5TH ST
S 4TH ST
S 2ND ST
S 1ST ST
S AVE B
S AVE C
THIN CIR
W BRAZOS AVE
S 3RD ST
S AVE D
FM 27
RANCH RD 27
LCR 244
CO RD 992
CO RD 2244
CO RD 228
CO RD 246
CO RD 983
CO RD 248
CO RD 234
CO RD 250
CO RD 991
CO RD 996
CO RD 970
CO RD 971
14

Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and
1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central and north central zones)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2004
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 18′ 59 MILS
GN 1° 21′ 24 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the
National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

TEXAS
QUADRANGLE LOCATION

ROAD CLASSIFICATION
Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

1 Purdon
2 Richland
3 Streetman
4 Union High
5 Kirvin
6 Tehuacana
7 Mexia
8 Teague North
ADJOINING QUADRANGLES

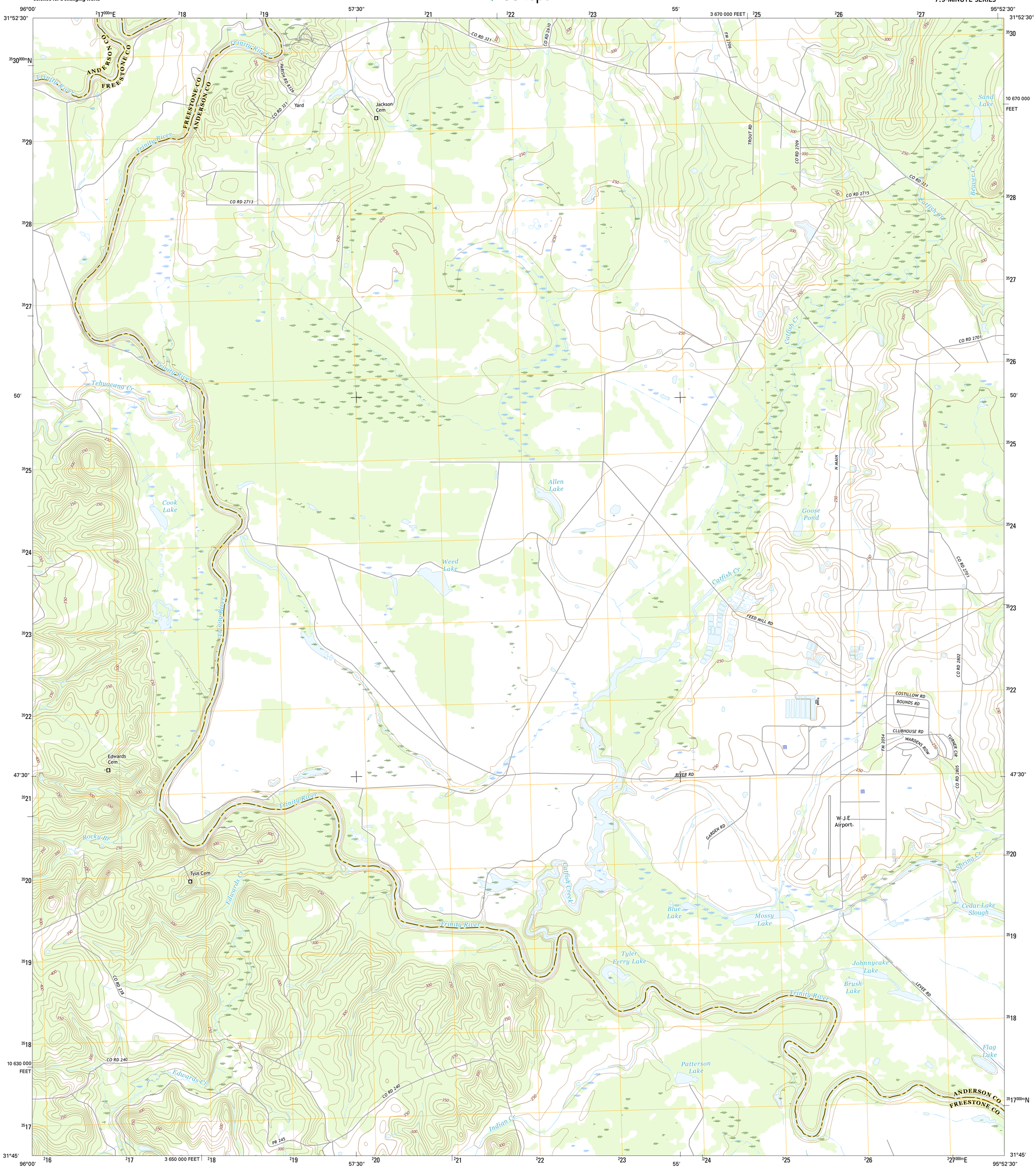
WORTHAM, TX
2016

NSN. 7643016398969
NGA REF NO. USGSX24K49888

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

YARD QUADRANGLE
TEXAS
7.5-MINUTE SERIES



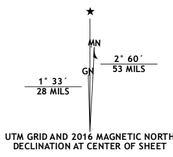
Produced by the United States Geological Survey

North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 15R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, July 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014



UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
TR
Grid Zone Designation
15R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
NORTH AMERICAN VERTICAL DATUM OF 1988

This map was produced to conform with the National Geospatial Program US Topo Product Standard, 2011.
A metadata file associated with this product is draft version 0.6.19

QUADRANGLE LOCATION

ROAD CLASSIFICATION

Expressway
Secondary Hwy
Ramp
Local Connector
Local Road
4WD
Interstate Route
US Route
State Route

1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES

1 Roustabout Camp
2 Cayuga
3 Blackfoot
4 Young
5 Tennessee Colony
6 Turlington
7 Butler
8 Long Lake

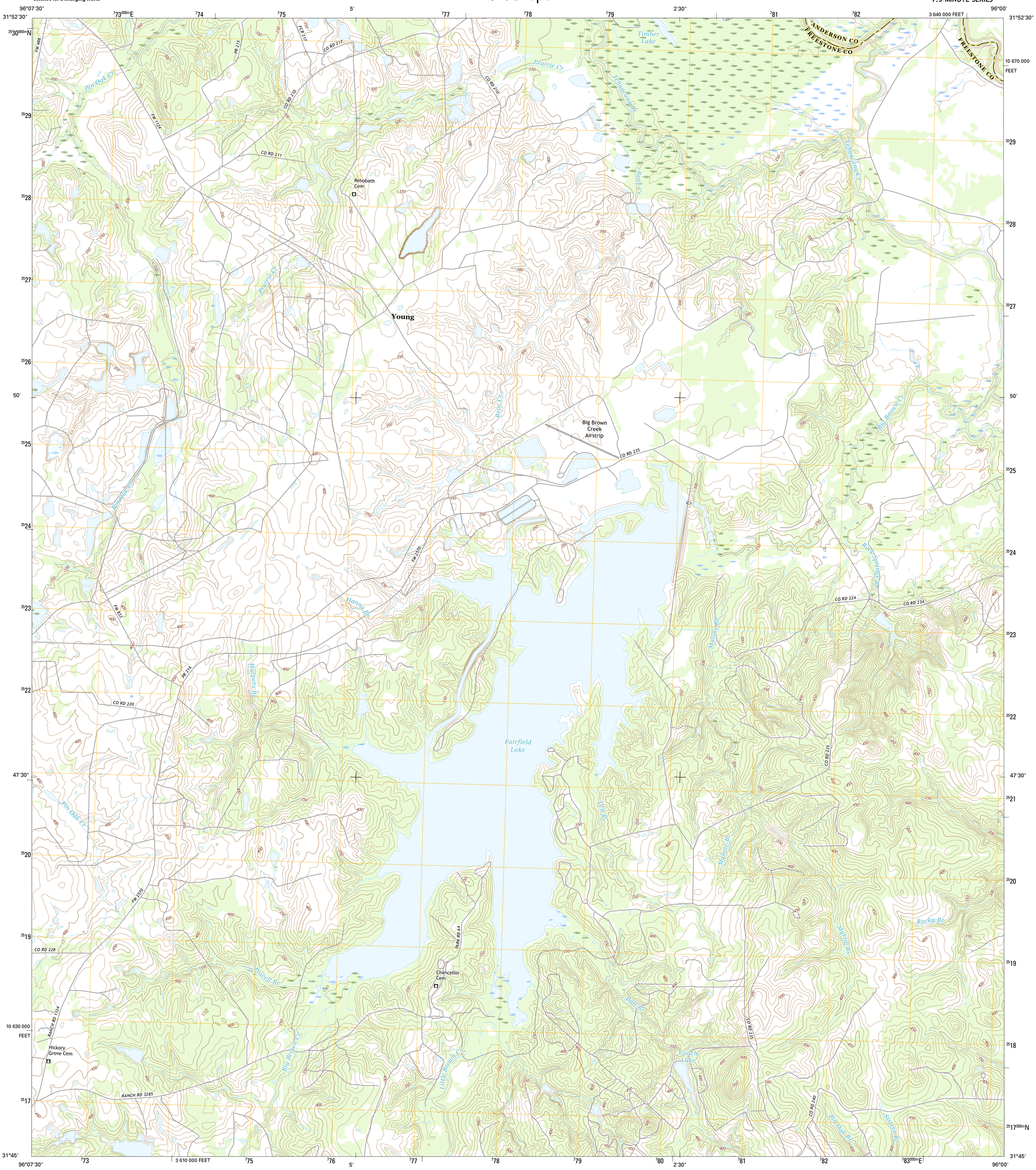
YARD, TX
2016

NSN. 7643016398978
NGA REF NO. USGSX24K50044

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

The National Map US Topo

YOUNG QUADRANGLE
TEXAS
7.5-MINUTE SERIES



Produced by the United States Geological Survey
North American Datum of 1983 (NAD83)
World Geodetic System of 1984 (WGS84). Projection and 1 000-meter grid: Universal Transverse Mercator, Zone 14R
10 000-foot ticks: Texas Coordinate System of 1983 (central zone)

This map is not a legal document. Boundaries may be generalized for this map scale. Private lands within government reservations may not be shown. Obtain permission before entering private lands.

Imagery.....NAIP, August 2014
Roads..... U.S. Census Bureau, 2014 - 2015
Names.....GNIS, 2015
Hydrography.....National Hydrography Dataset, 2014
Contours.....National Elevation Dataset, 2003
Boundaries.....Multiple sources; see metadata file 1972 - 2015

Wetlands.....FWS National Wetlands Inventory 1977 - 2014

MN 3° 4' 55 MILS
GN 1° 33' 28 MILS
UTM GRID AND 2016 MAGNETIC NORTH DECLINATION AT CENTER OF SHEET

U.S. National Grid
100,000-m Square ID
QA
Grid Zone Designation
14R

SCALE 1:24 000

1 0.5 0 KILOMETERS 1 2
1000 500 0 METERS 1000 2000
1 0.5 0 1
MILES
1000 0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000
FEET

CONTOUR INTERVAL 10 FEET
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1	2	3
4		5
6	7	8

ADJOINING QUADRANGLES
1 Winkler
2 Roustabout Camp
3 Cayuga
4 Stewards Mill
5 Yard
6 Fairfield
7 Turlington
8 Butler

YOUNG, TX
2016

NSN 7643016398987
NGA REF NO. USGSX24K50151